



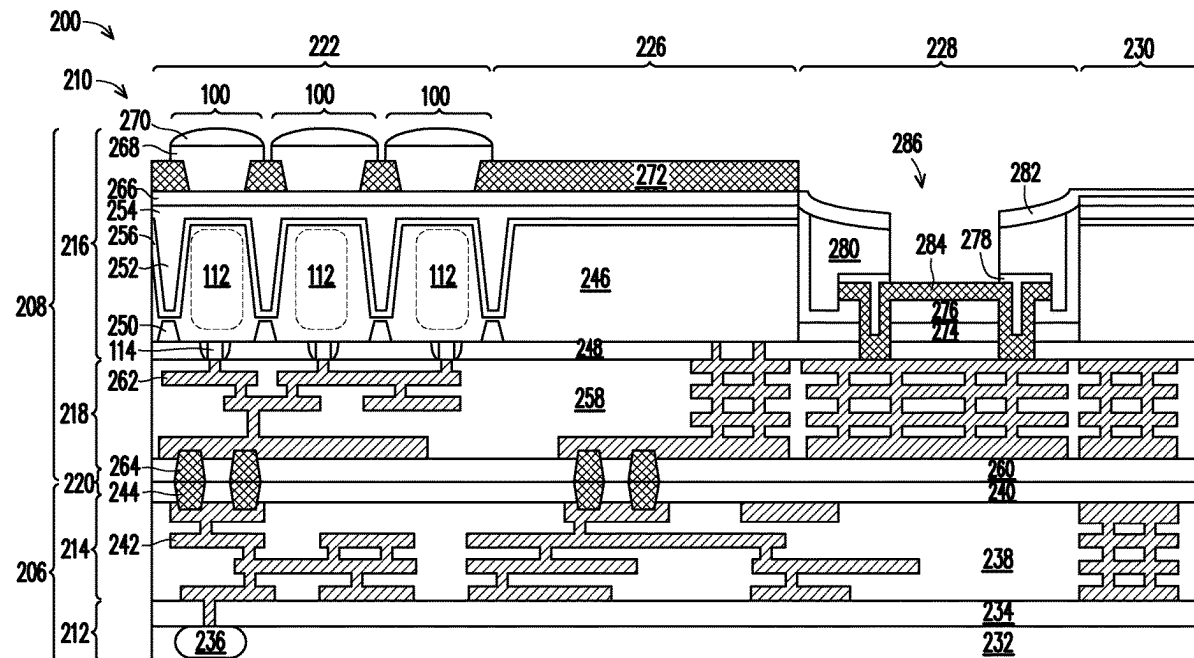
US 20250275271A1

(19) **United States**(12) **Patent Application Publication**
LU et al.(10) **Pub. No.: US 2025/0275271 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **PIXEL SENSOR ARRAYS AND METHODS OF FORMATION**(52) **U.S. Cl.**CPC *H10F 39/8023* (2025.01); *H10F 39/028* (2025.01); *H10F 39/806* (2025.01)(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**,
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ABSTRACT(72) Inventors: **Chun-Liang LU**, Tainan City (TW);
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An image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

(21) Appl. No.: **18/585,042**(22) Filed: **Feb. 22, 2024****Publication Classification**(51) **Int. Cl.***H01L 27/146* (2006.01)

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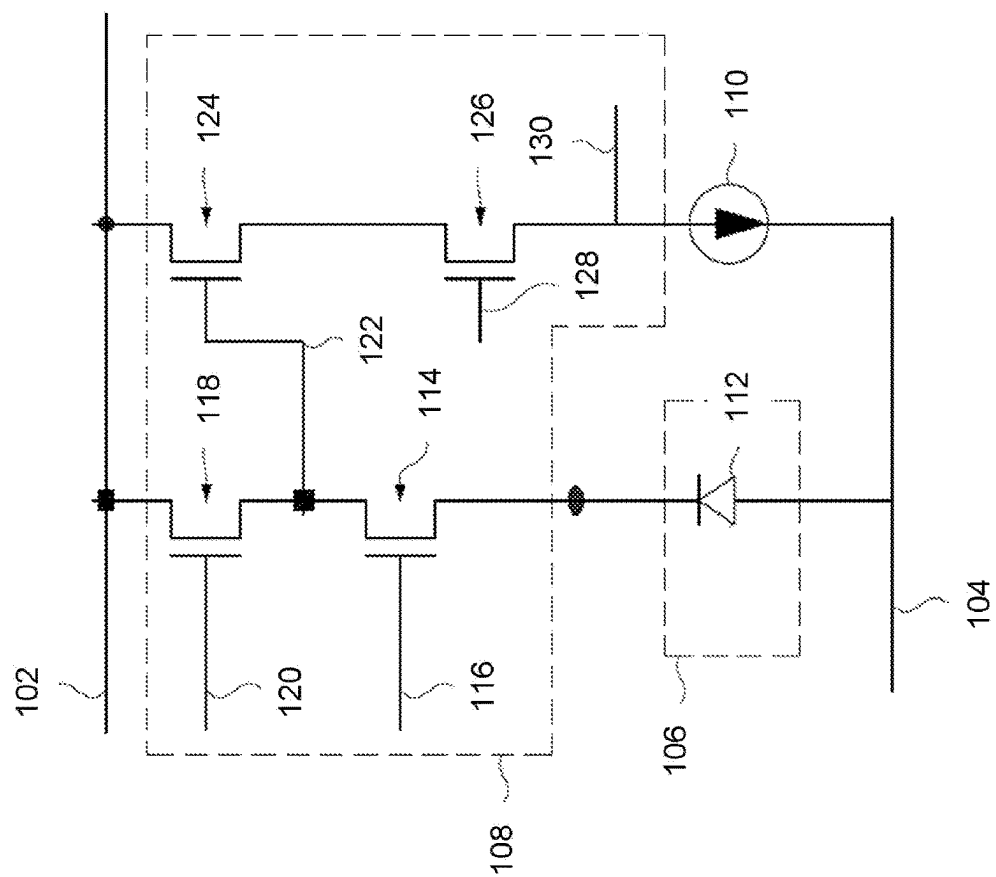


FIG. 1

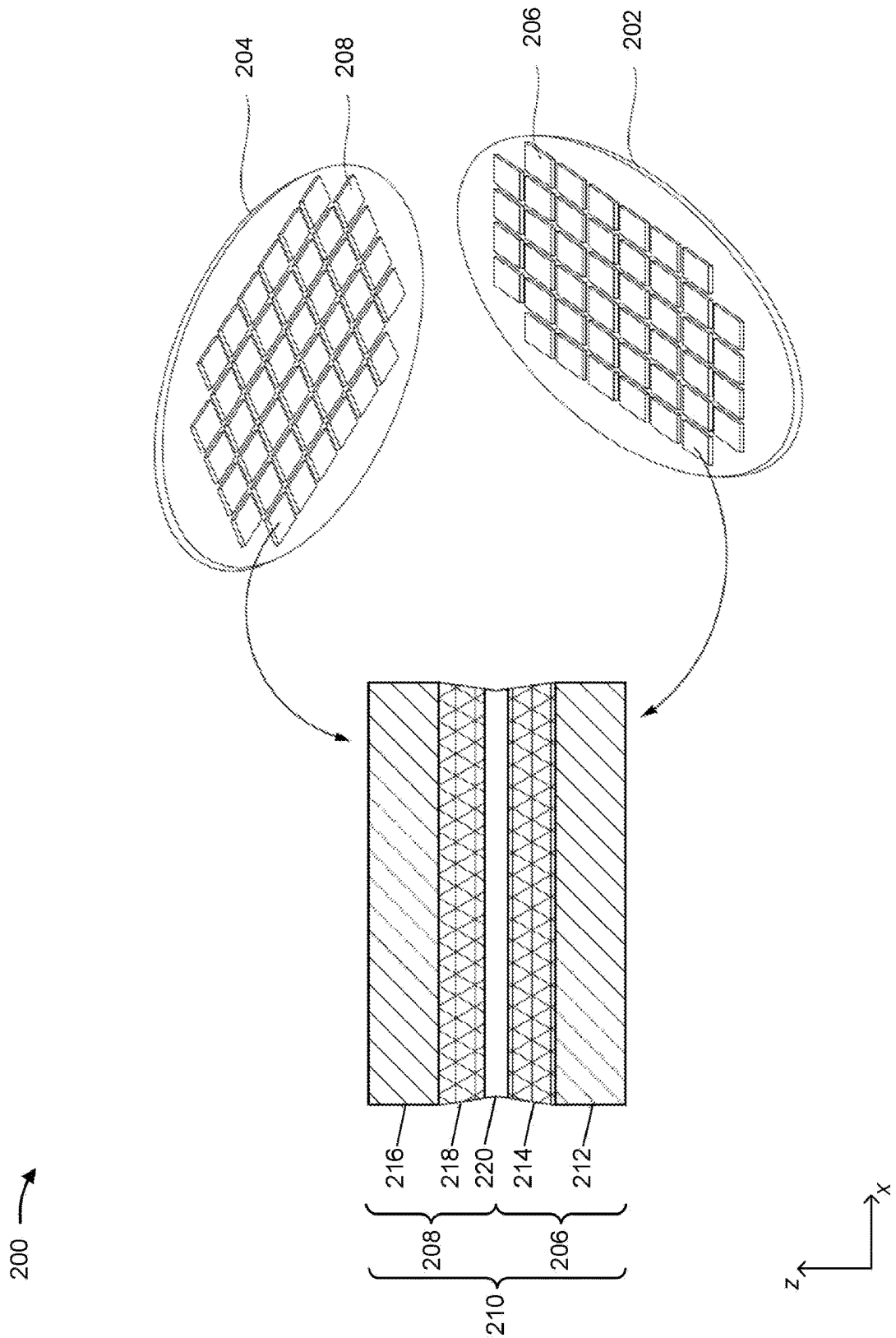


FIG. 2A

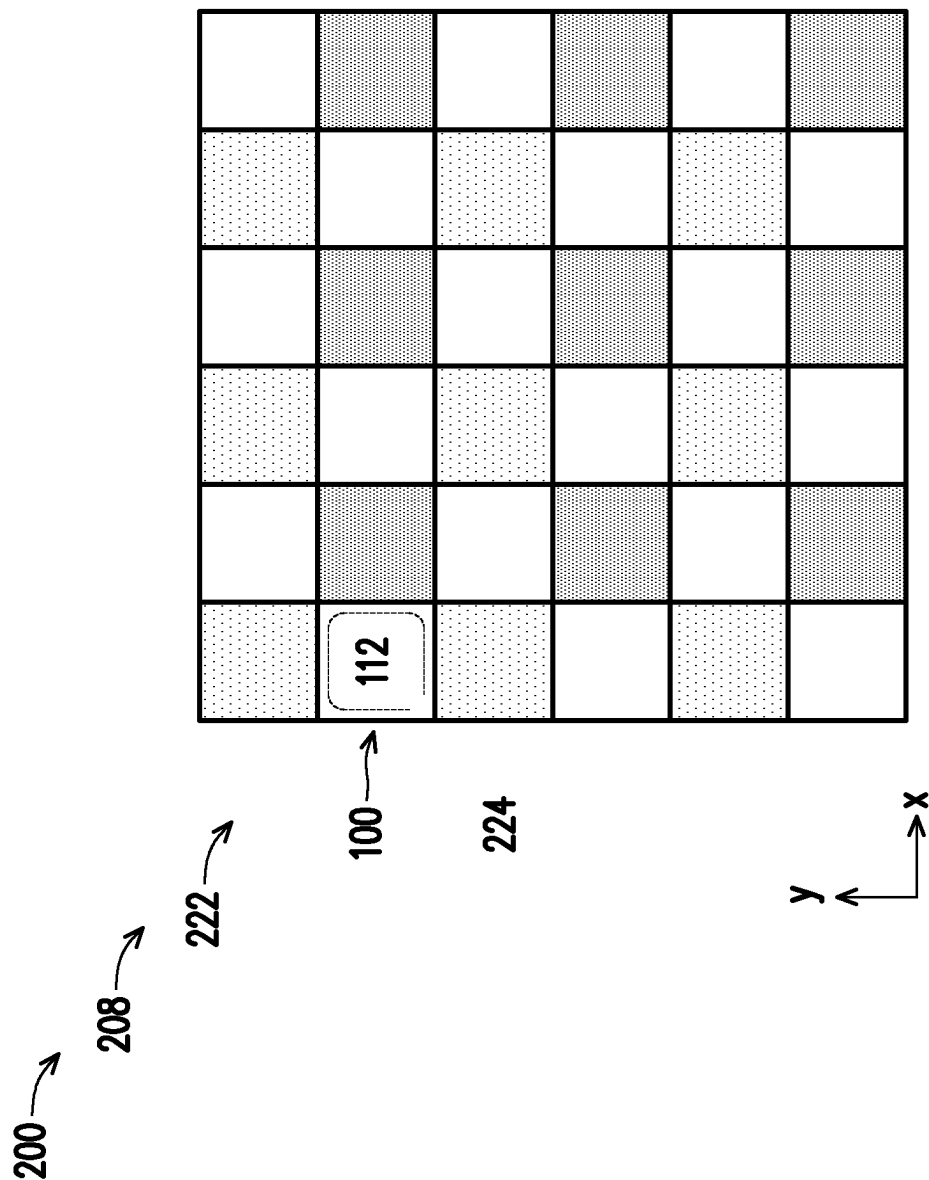


FIG. 2B

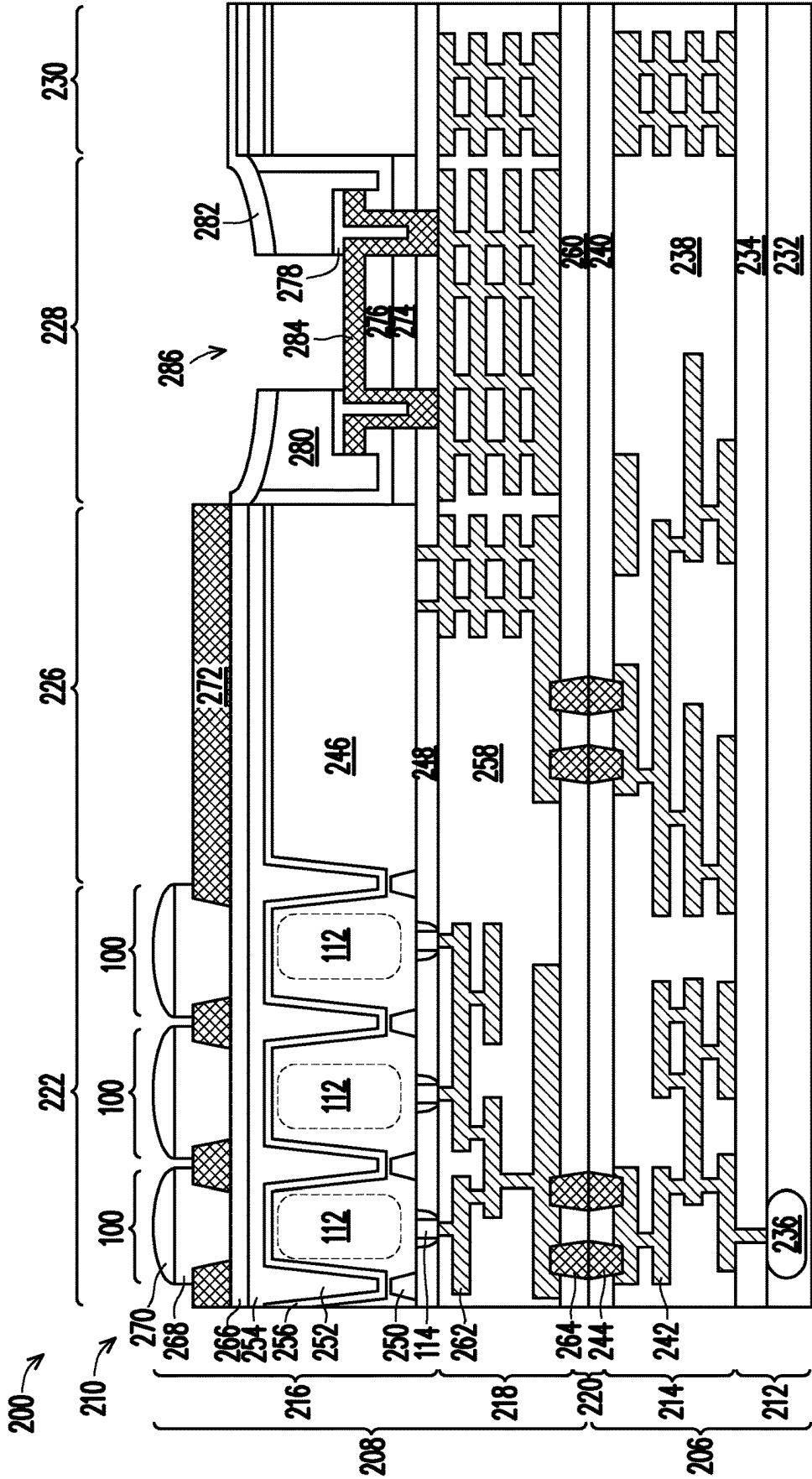


FIG. 2C

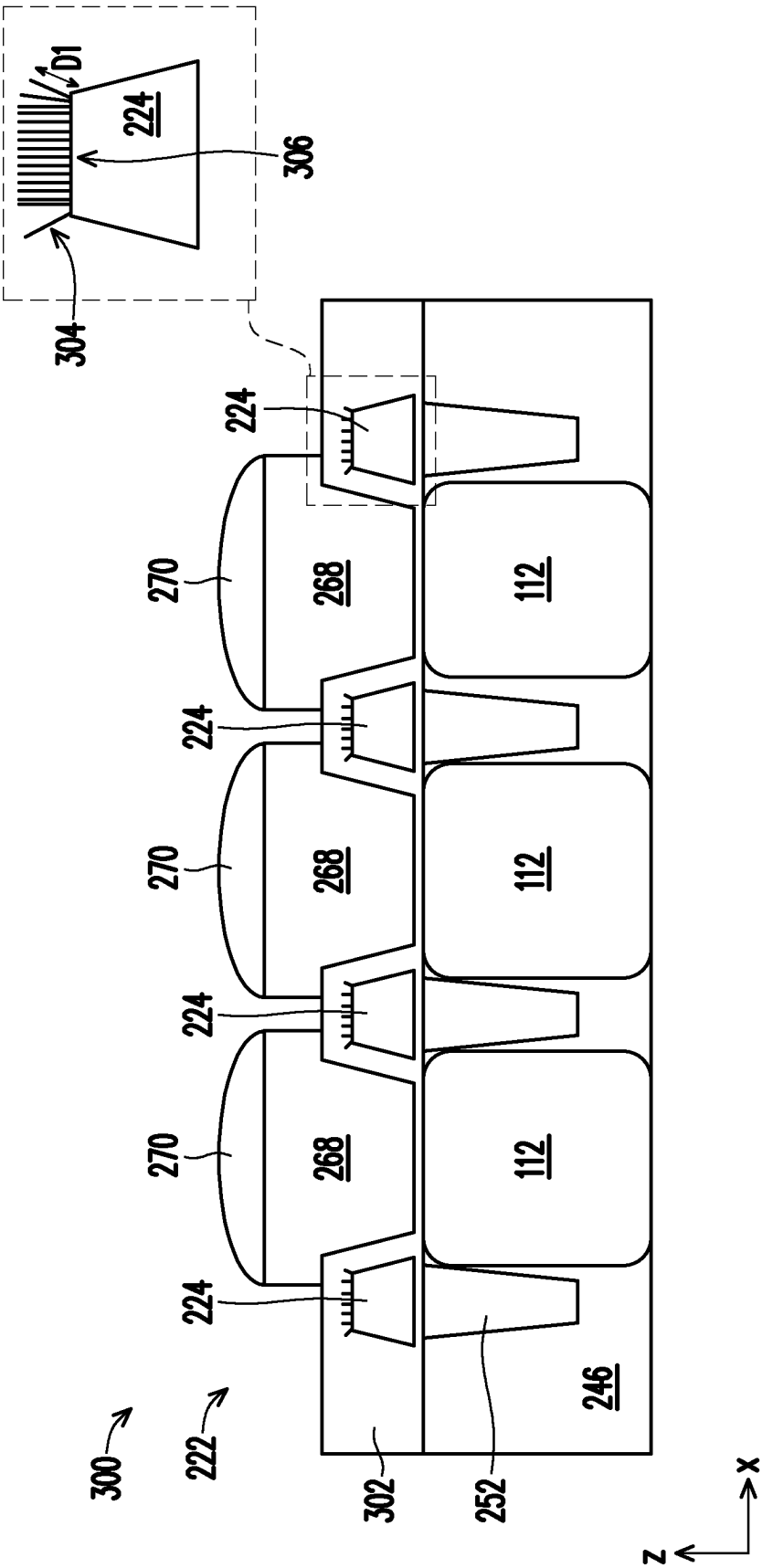


FIG. 3A

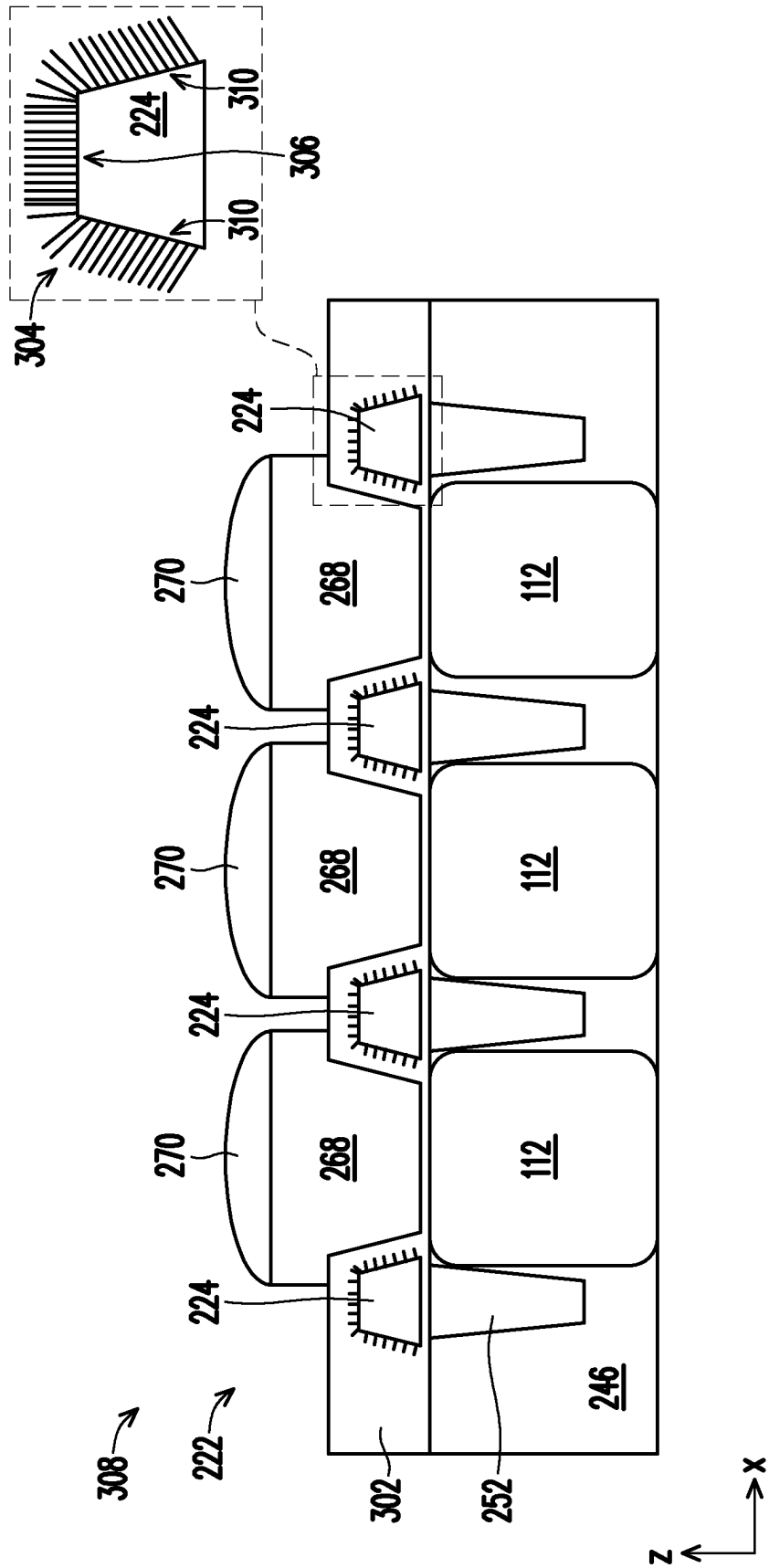


FIG. 3B

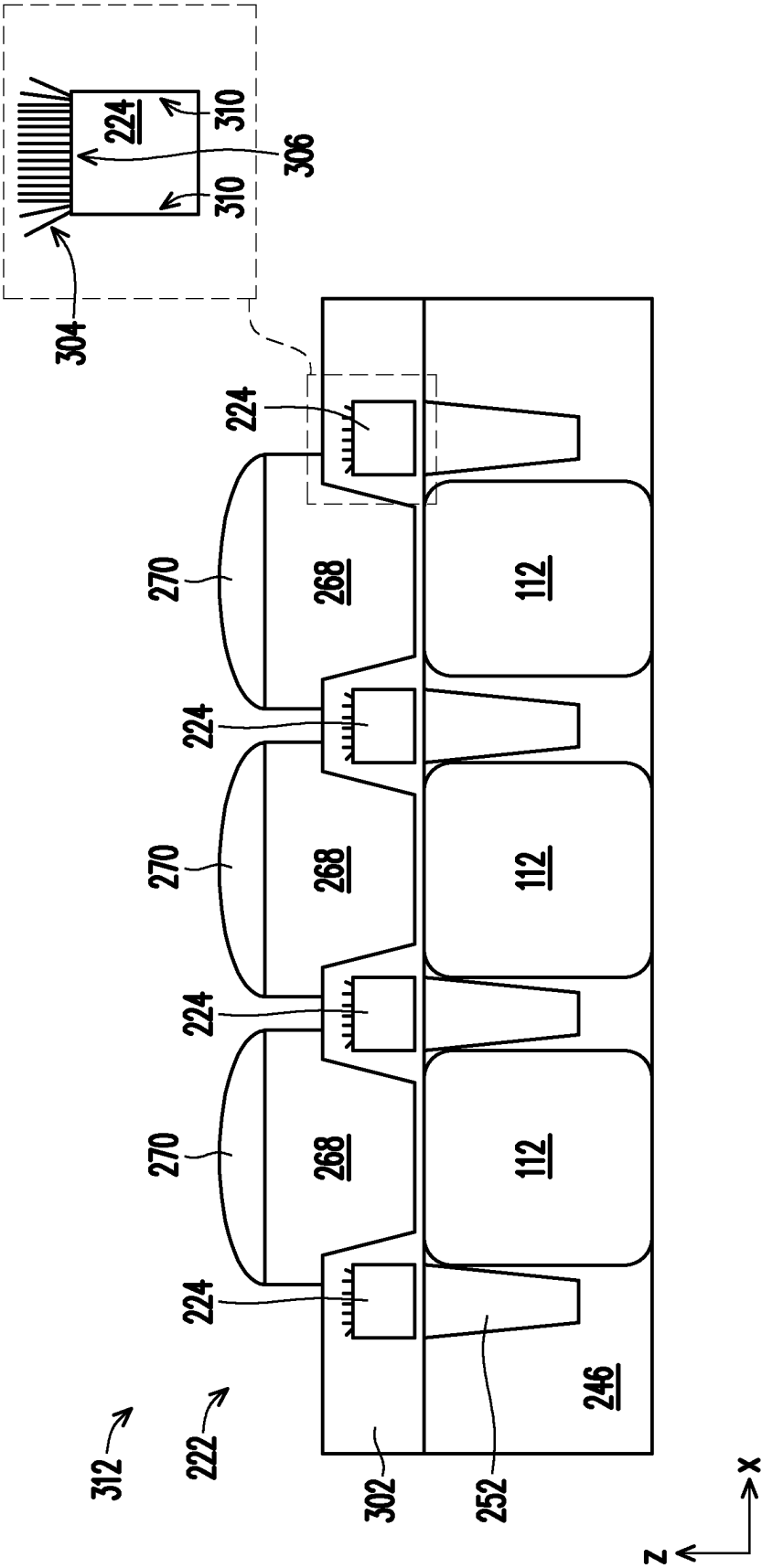


FIG. 3C

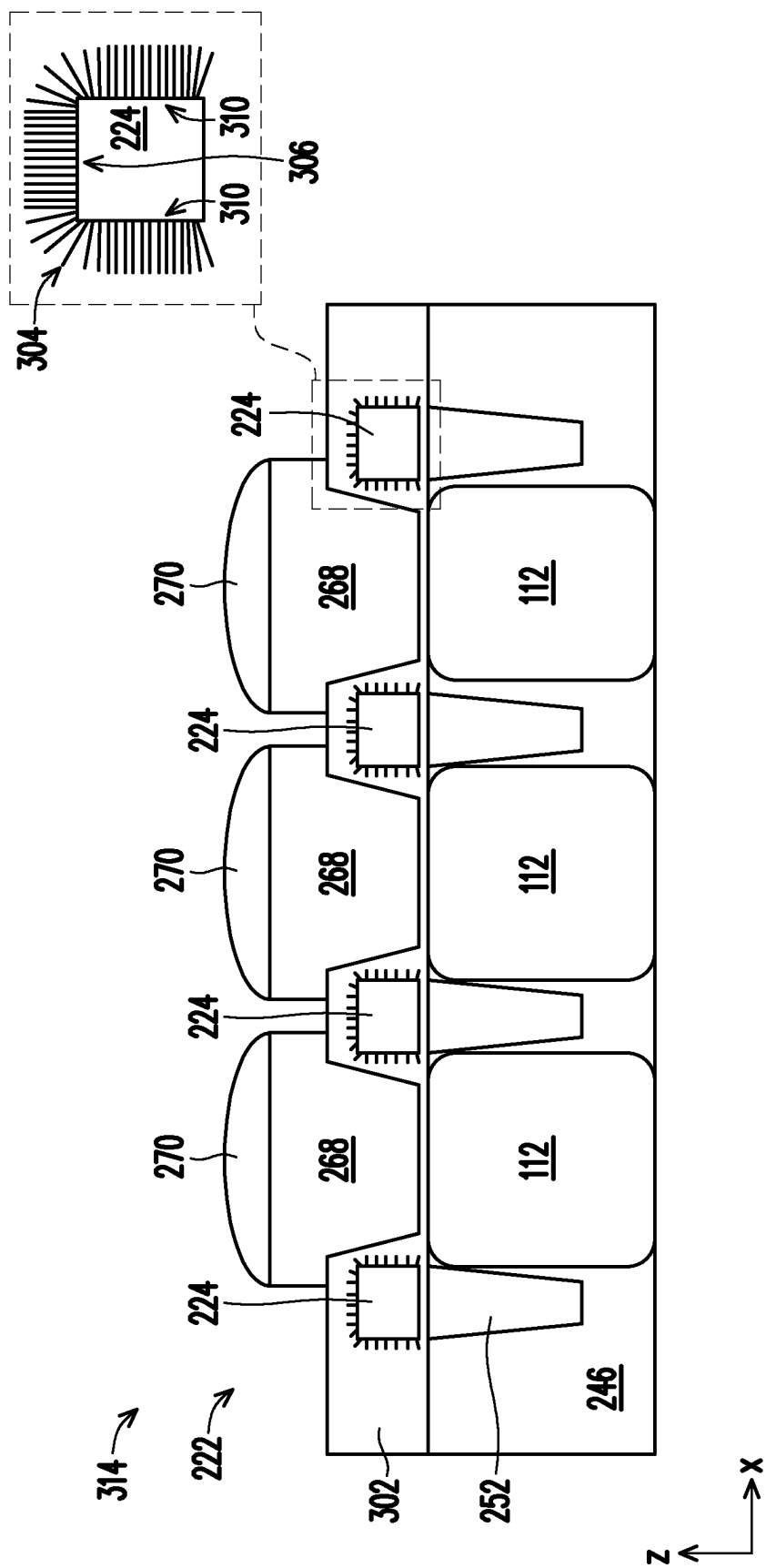
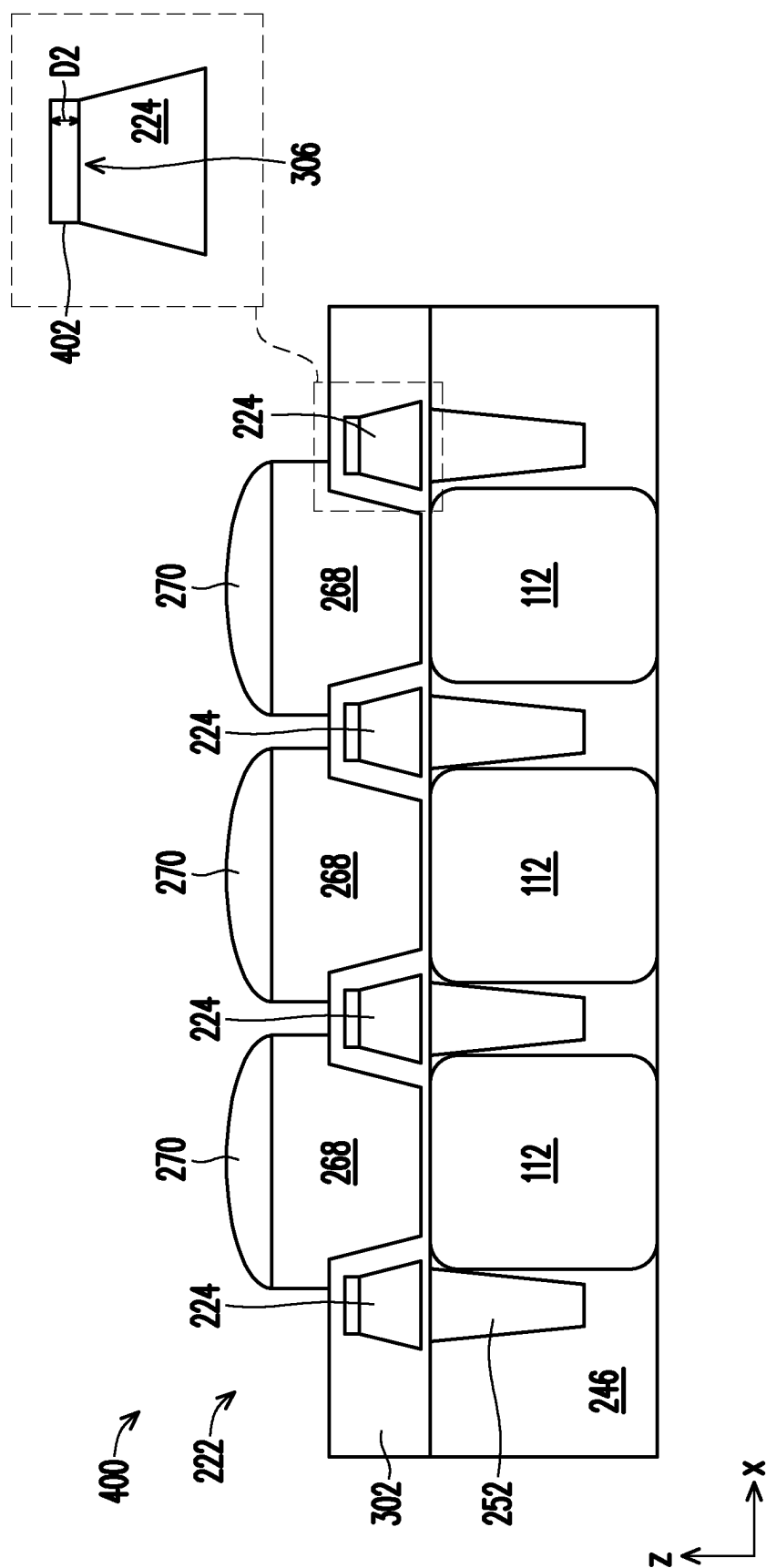
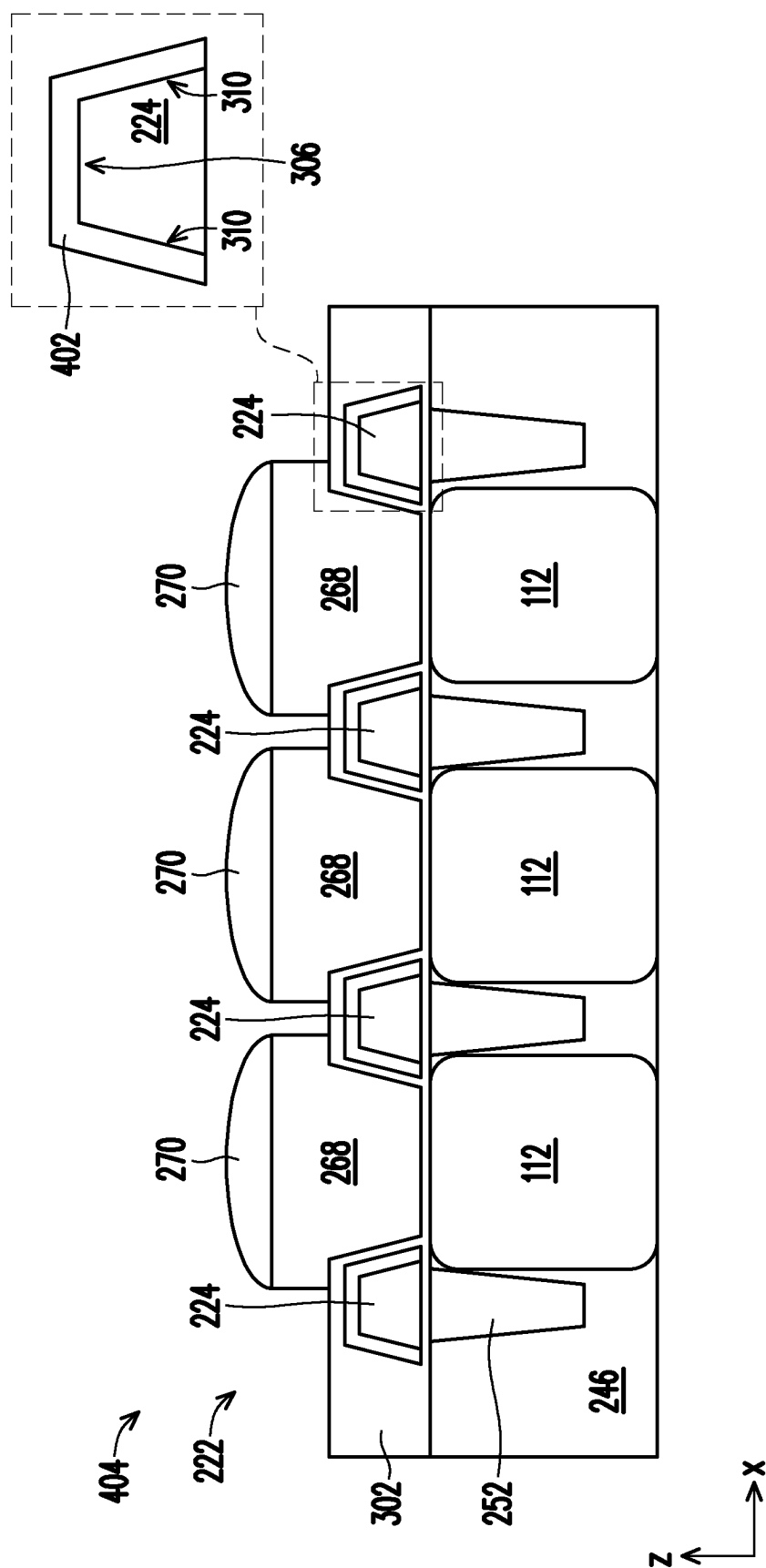


FIG. 3D





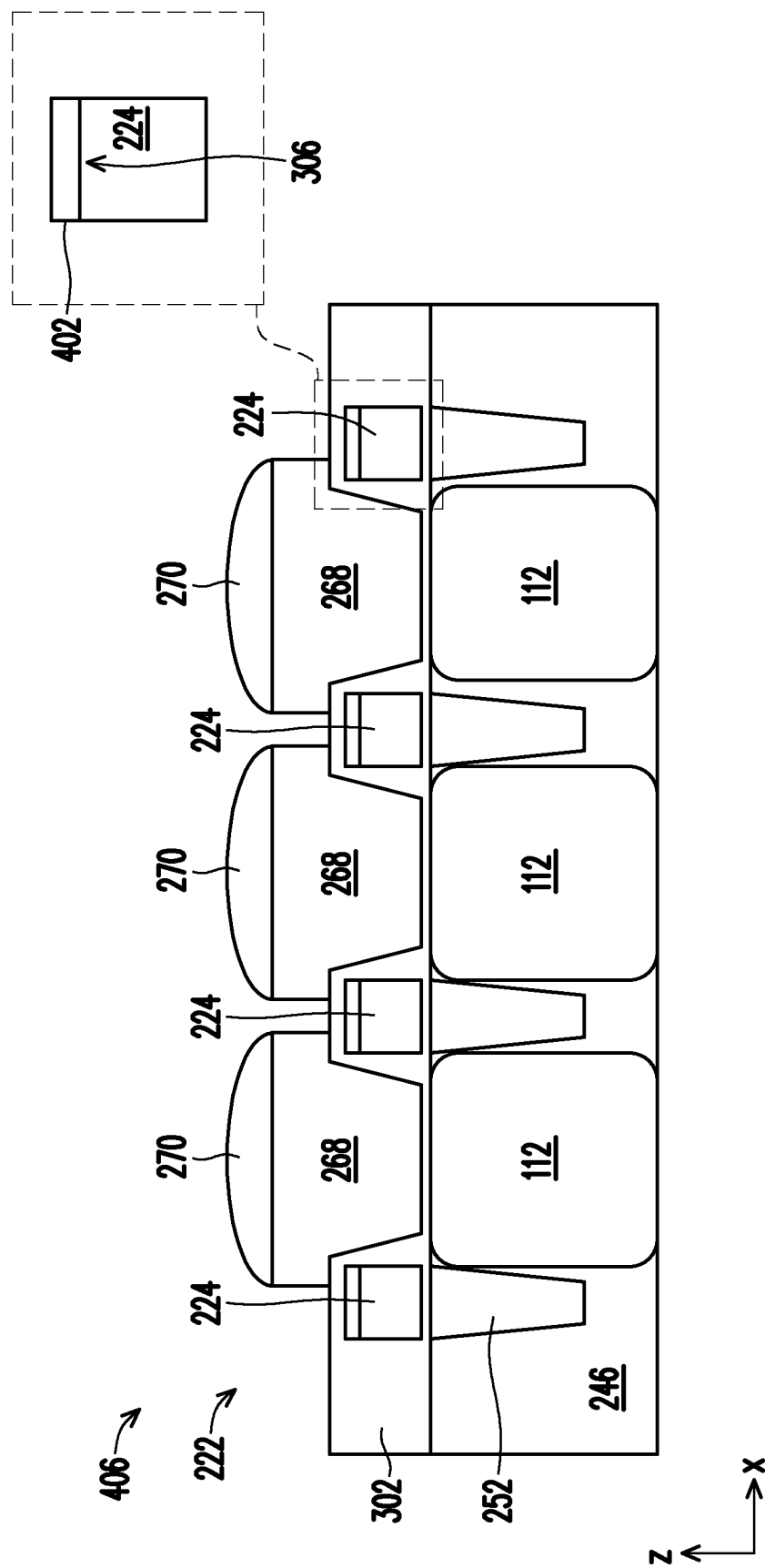
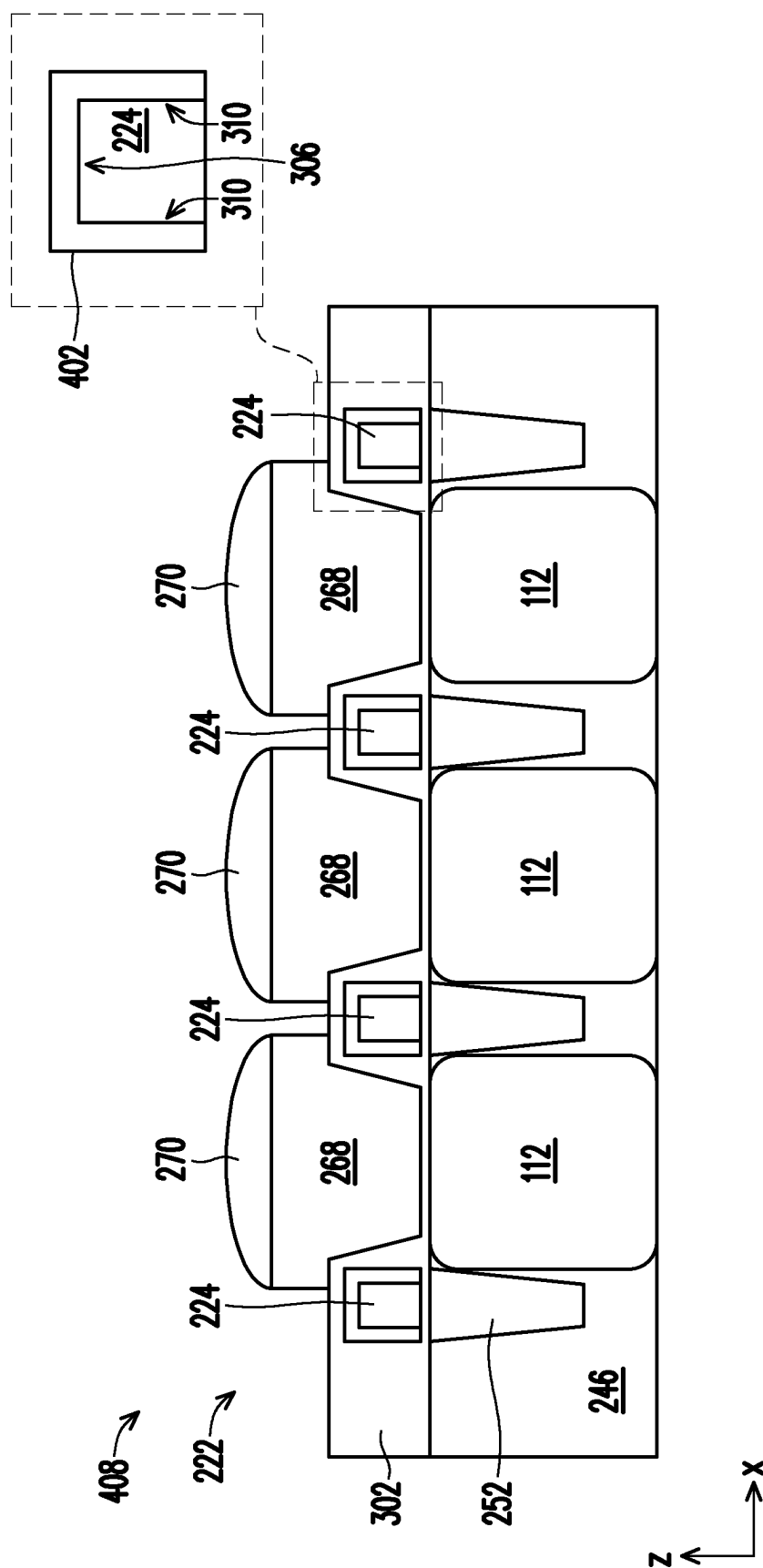


FIG. 4C



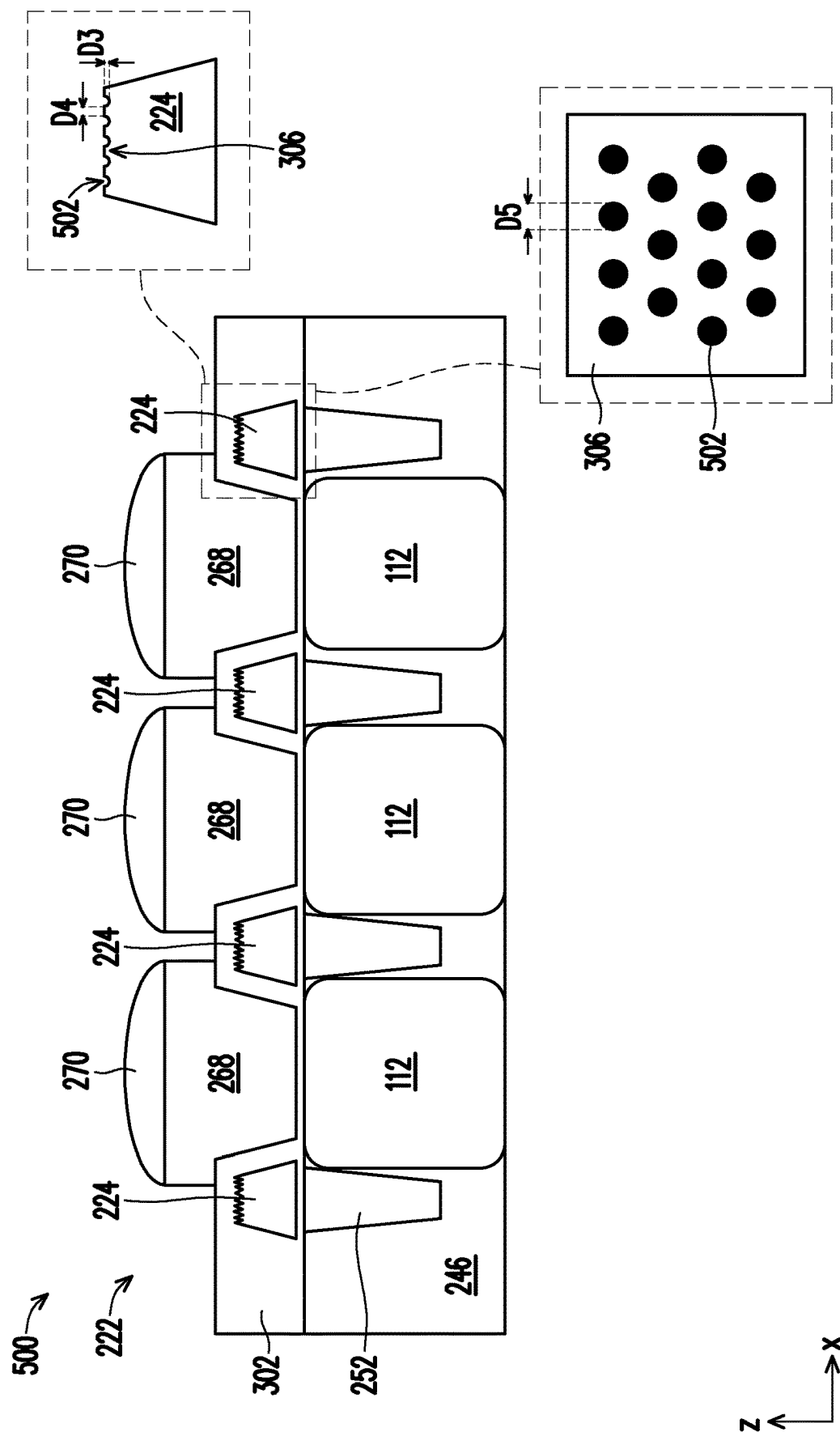


FIG. 5A

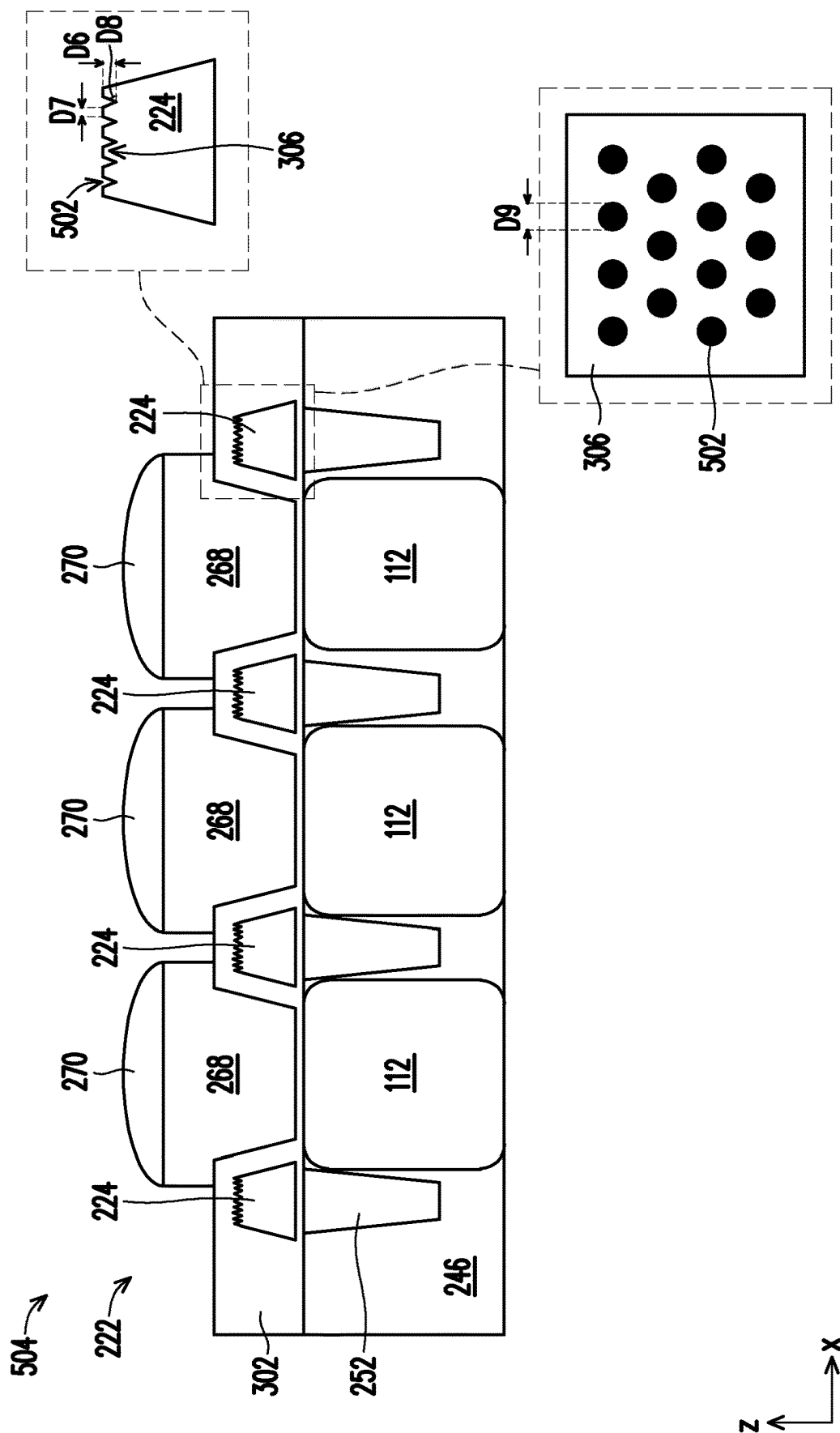


FIG. 5B

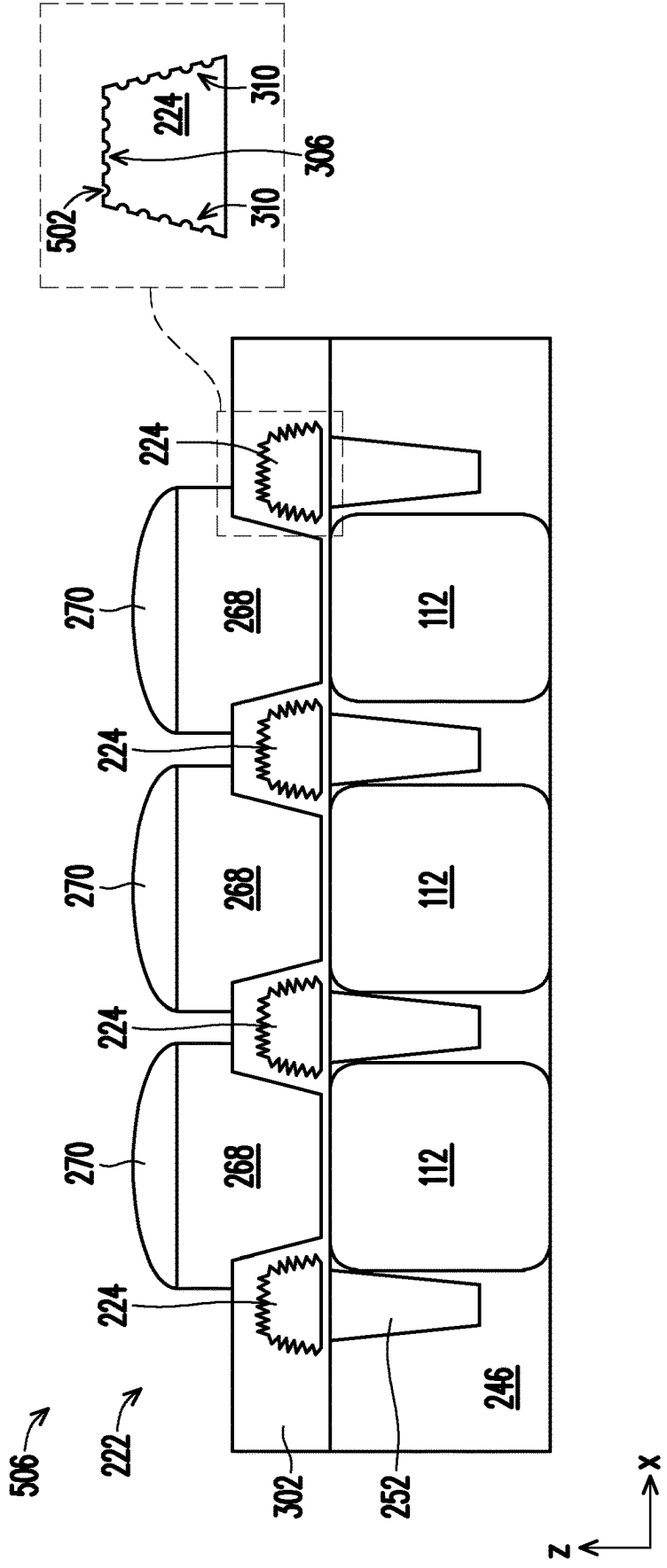


FIG. 5C

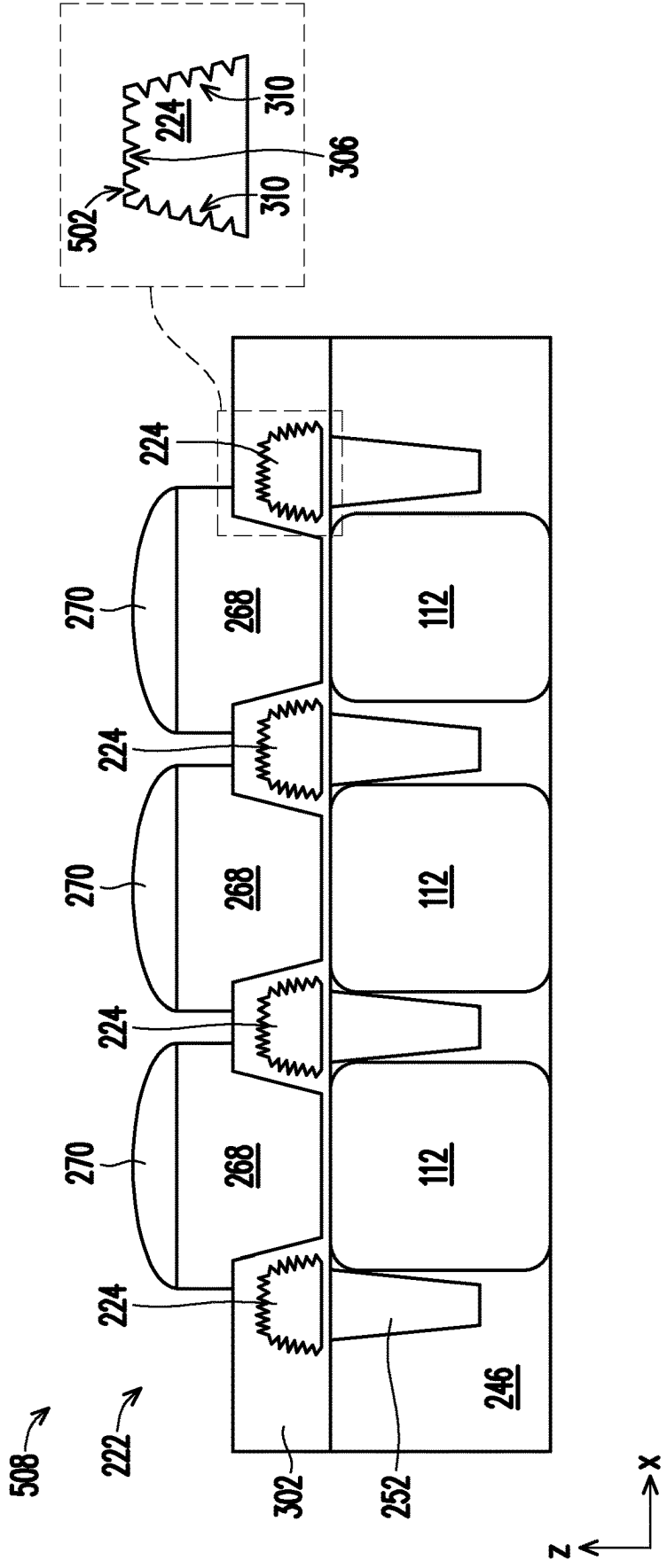


FIG. 5D

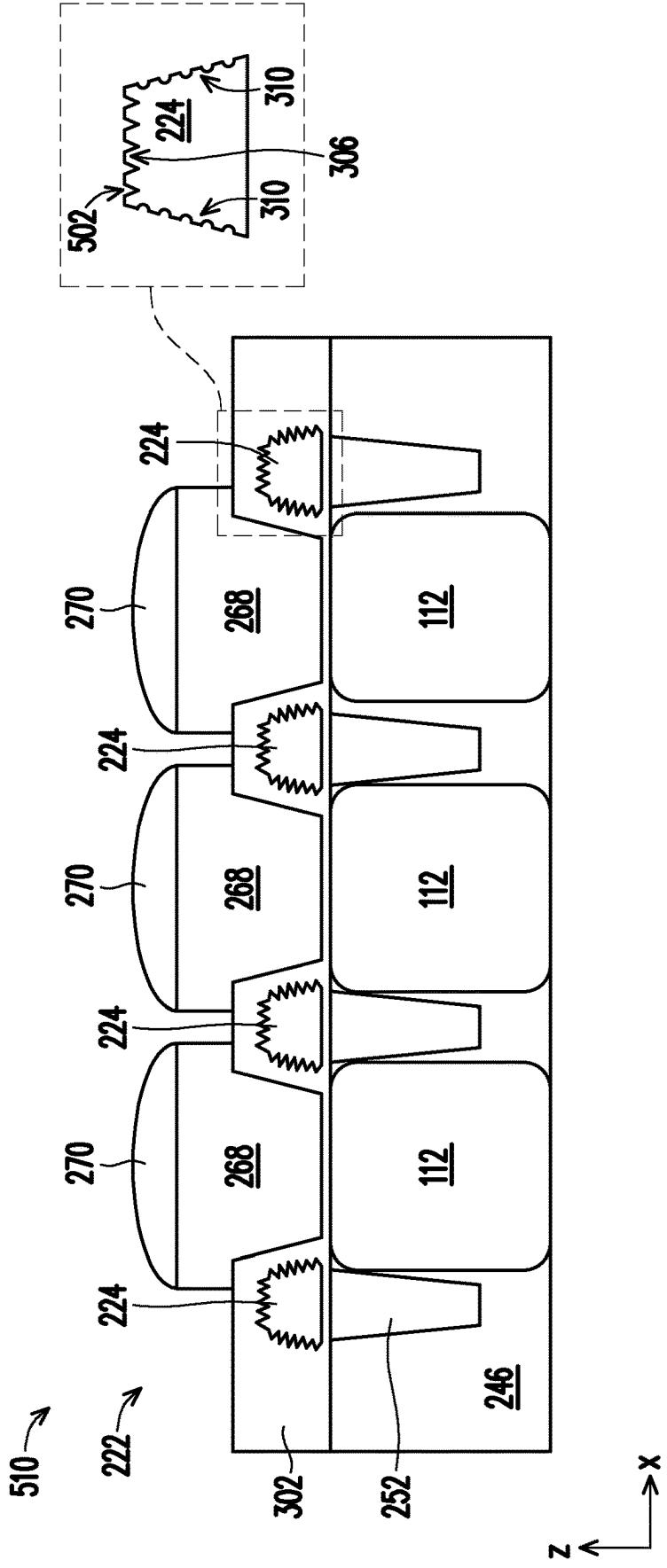
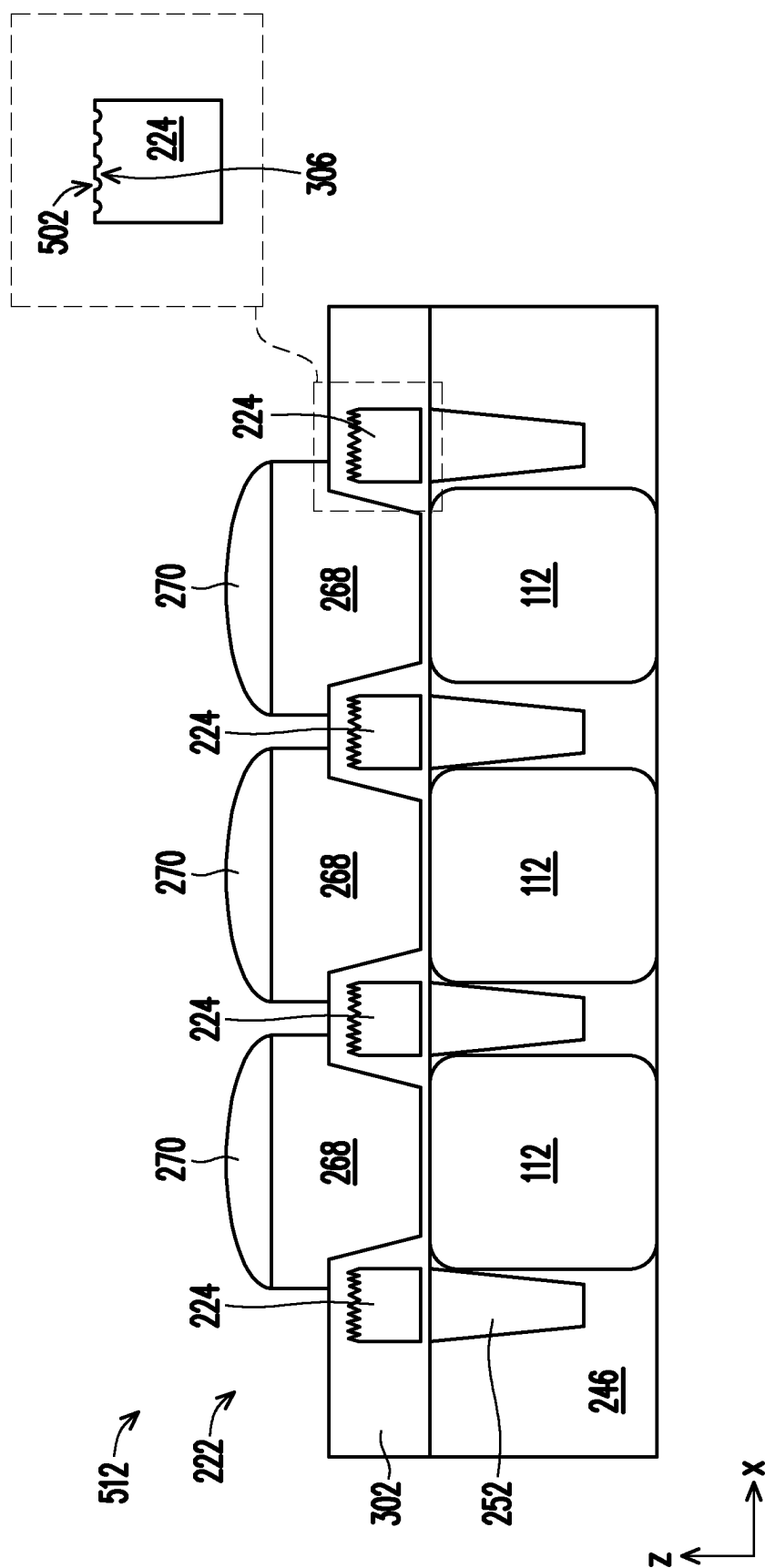


FIG. 5E



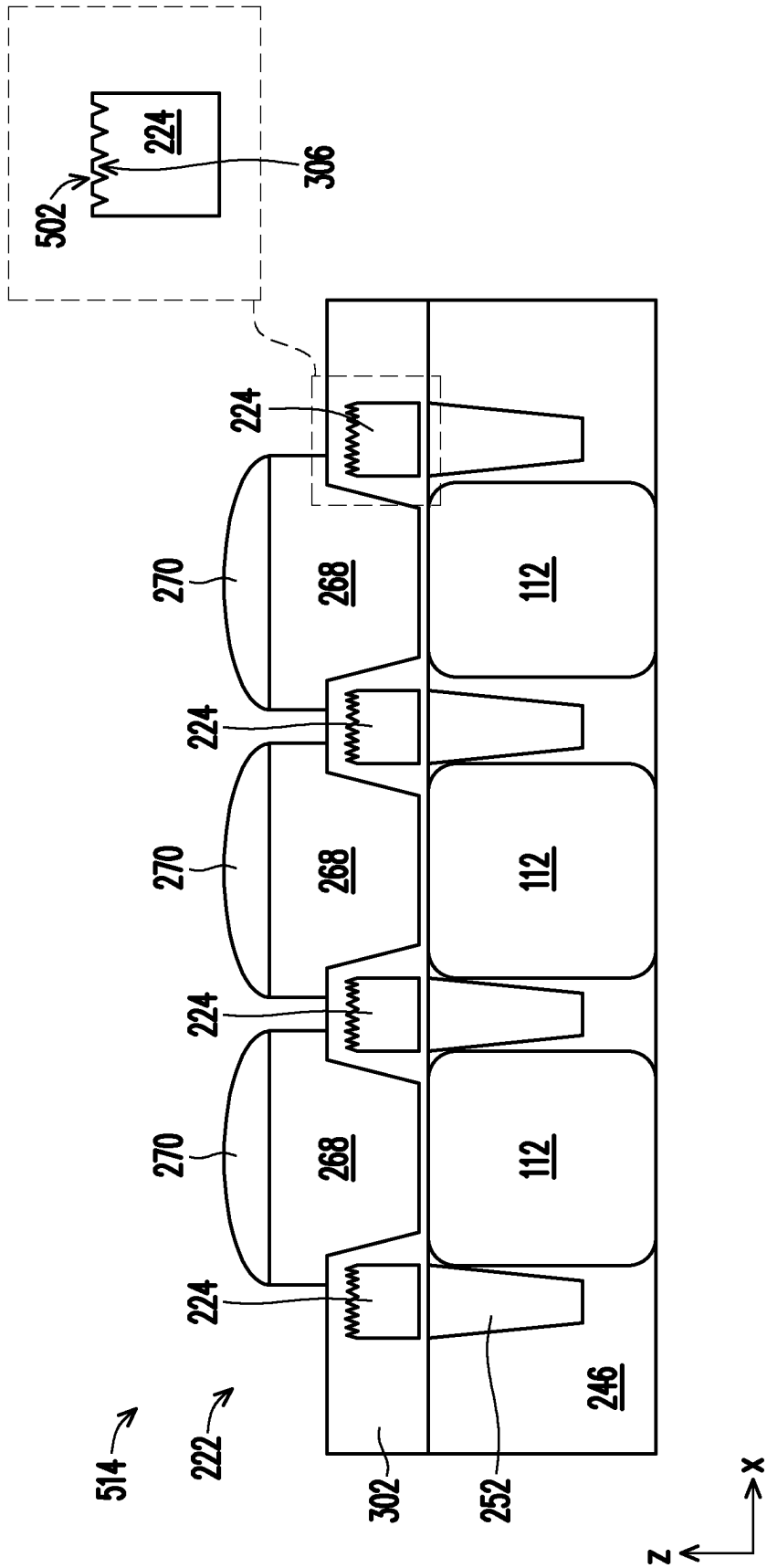


FIG. 5G

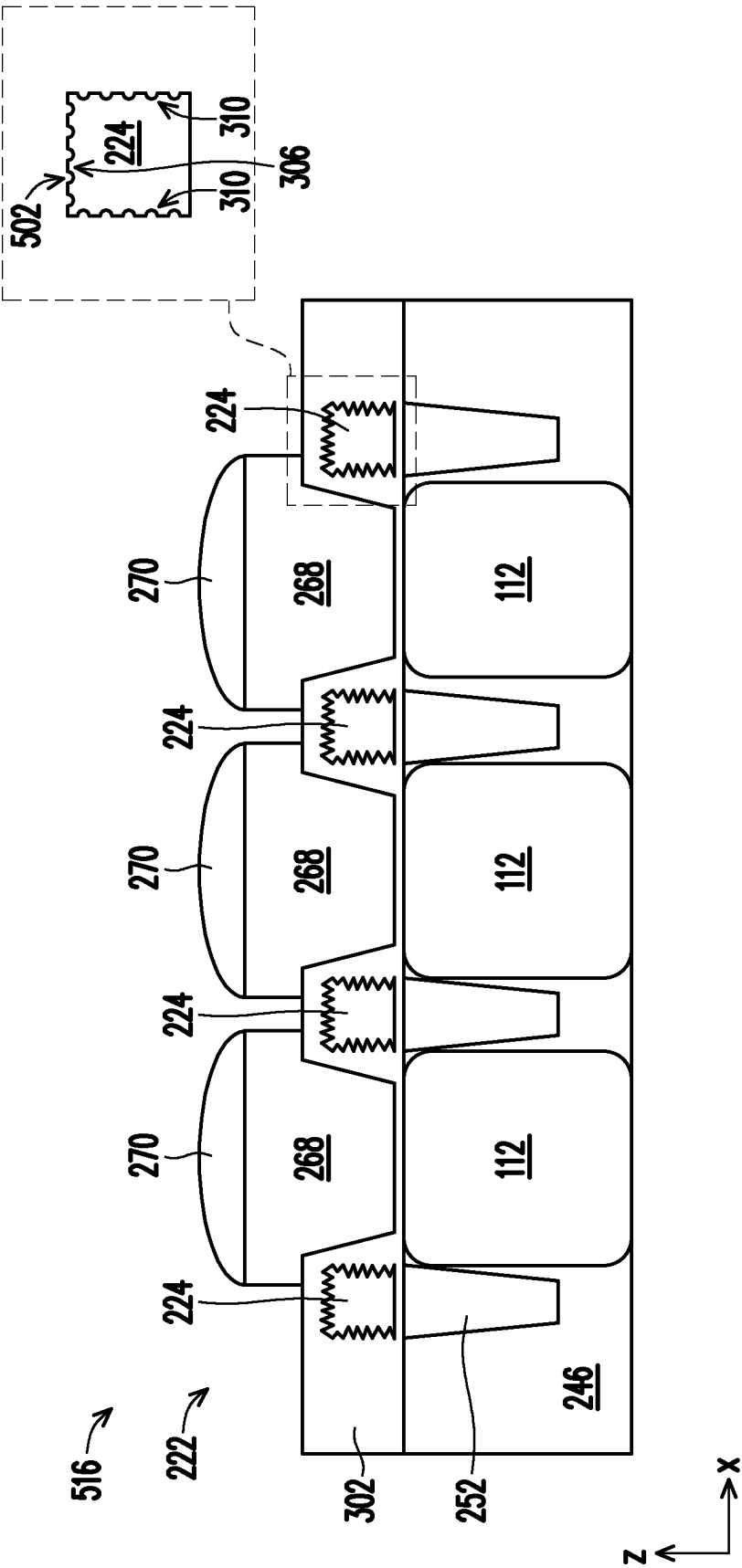


FIG. 5H

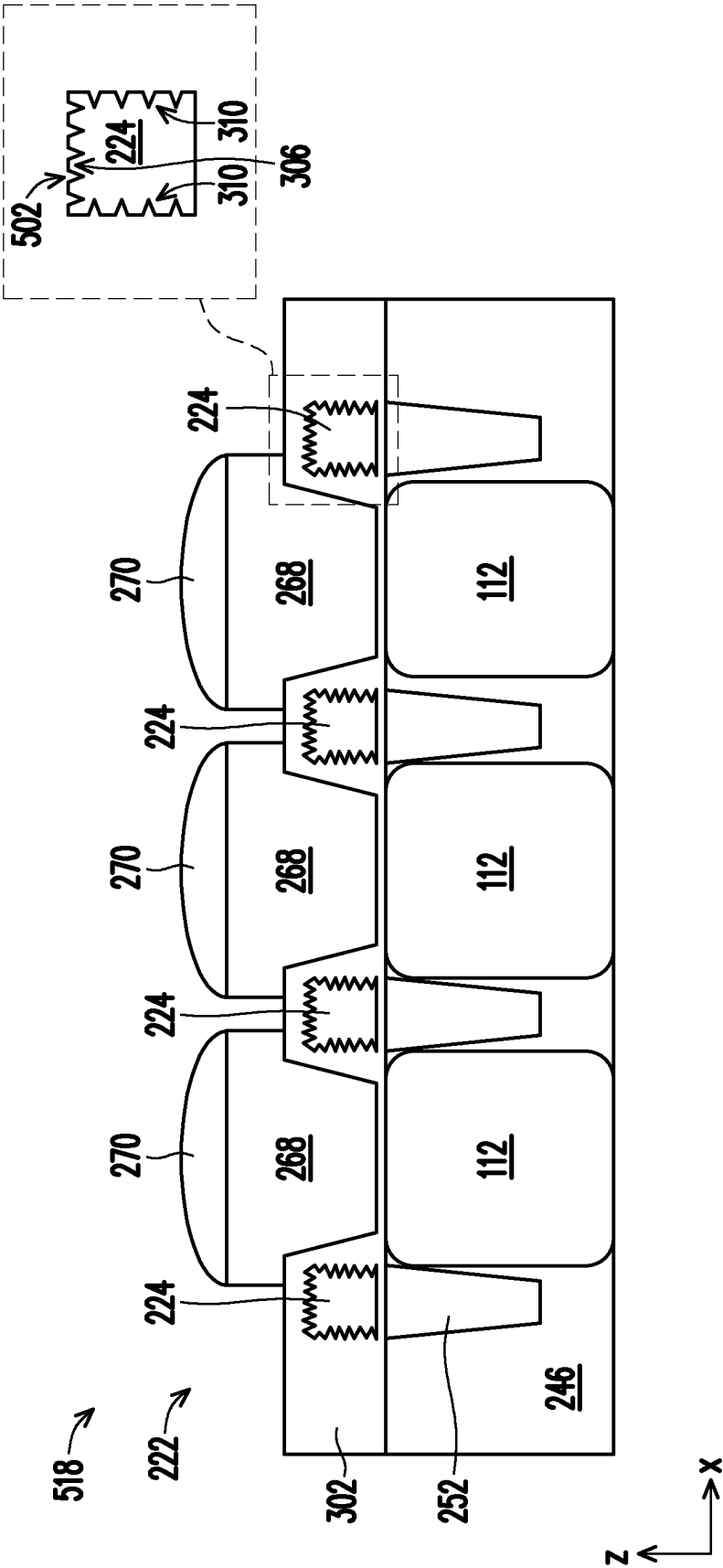


FIG. 5I

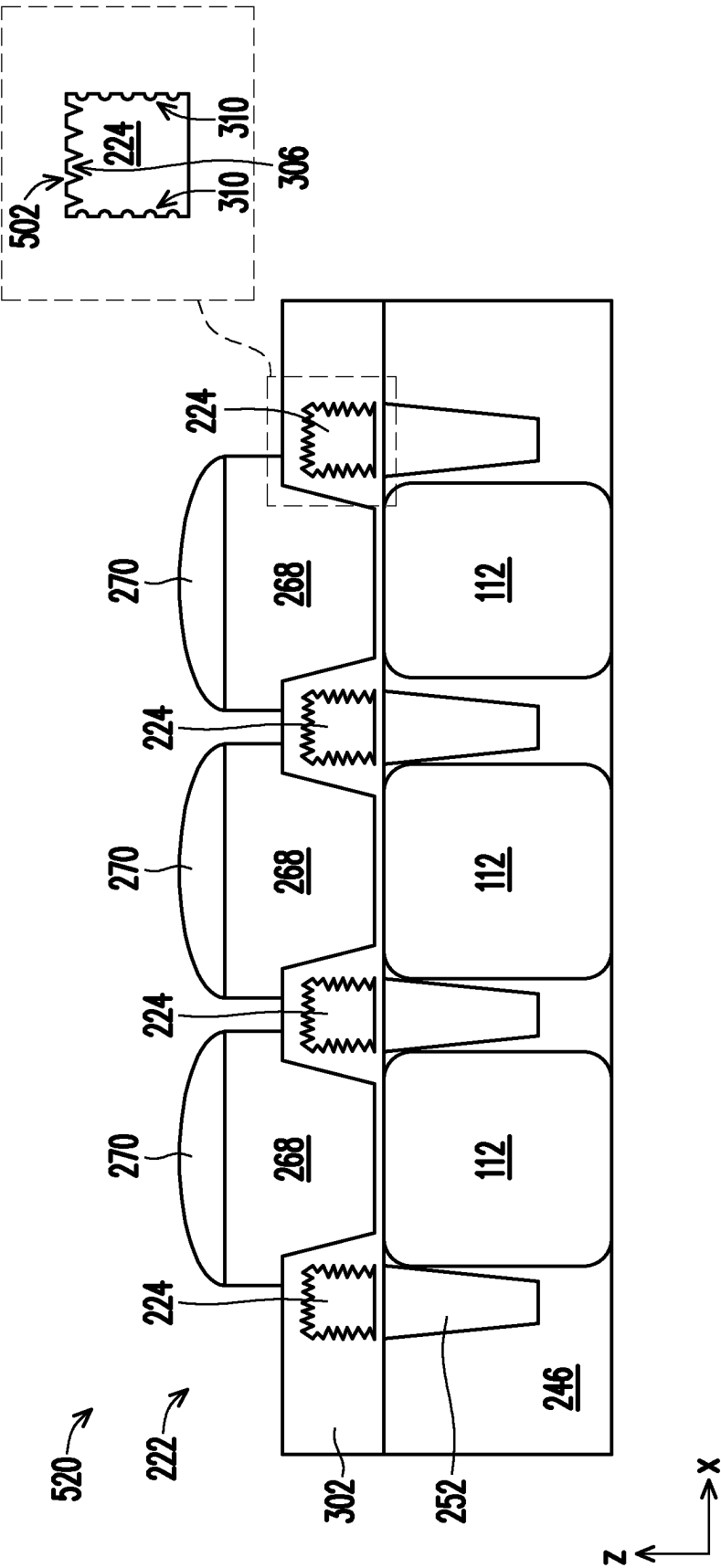


FIG. 5J

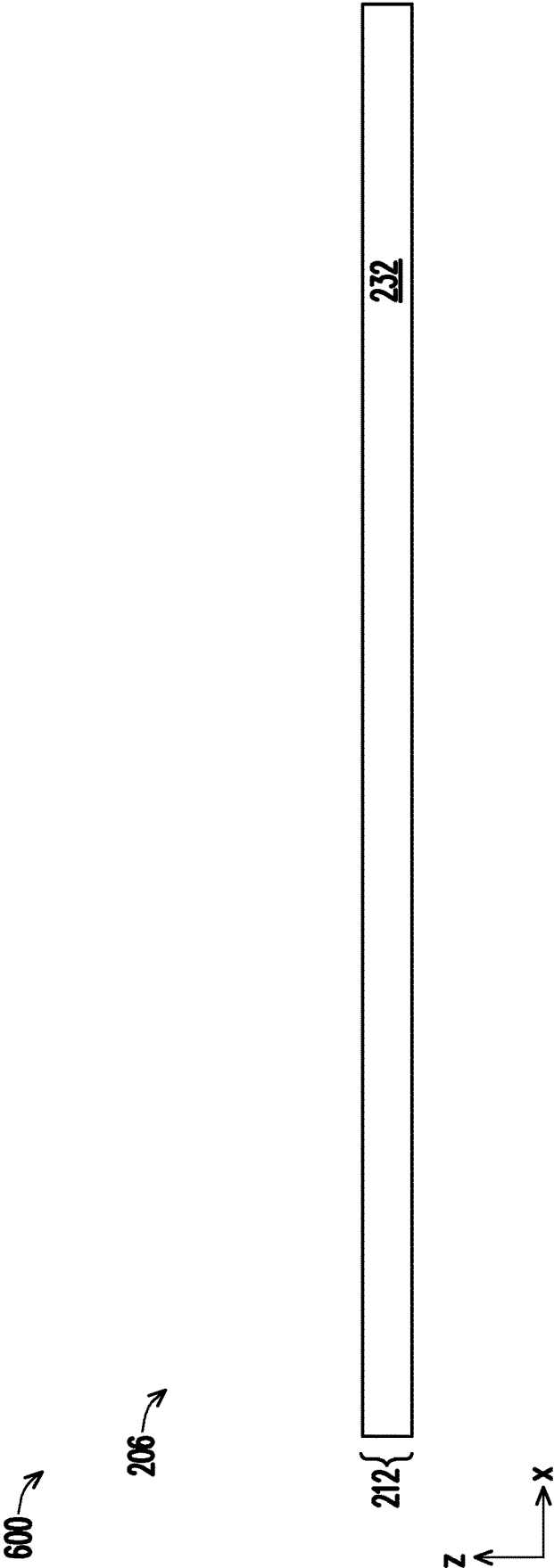


FIG. 6A

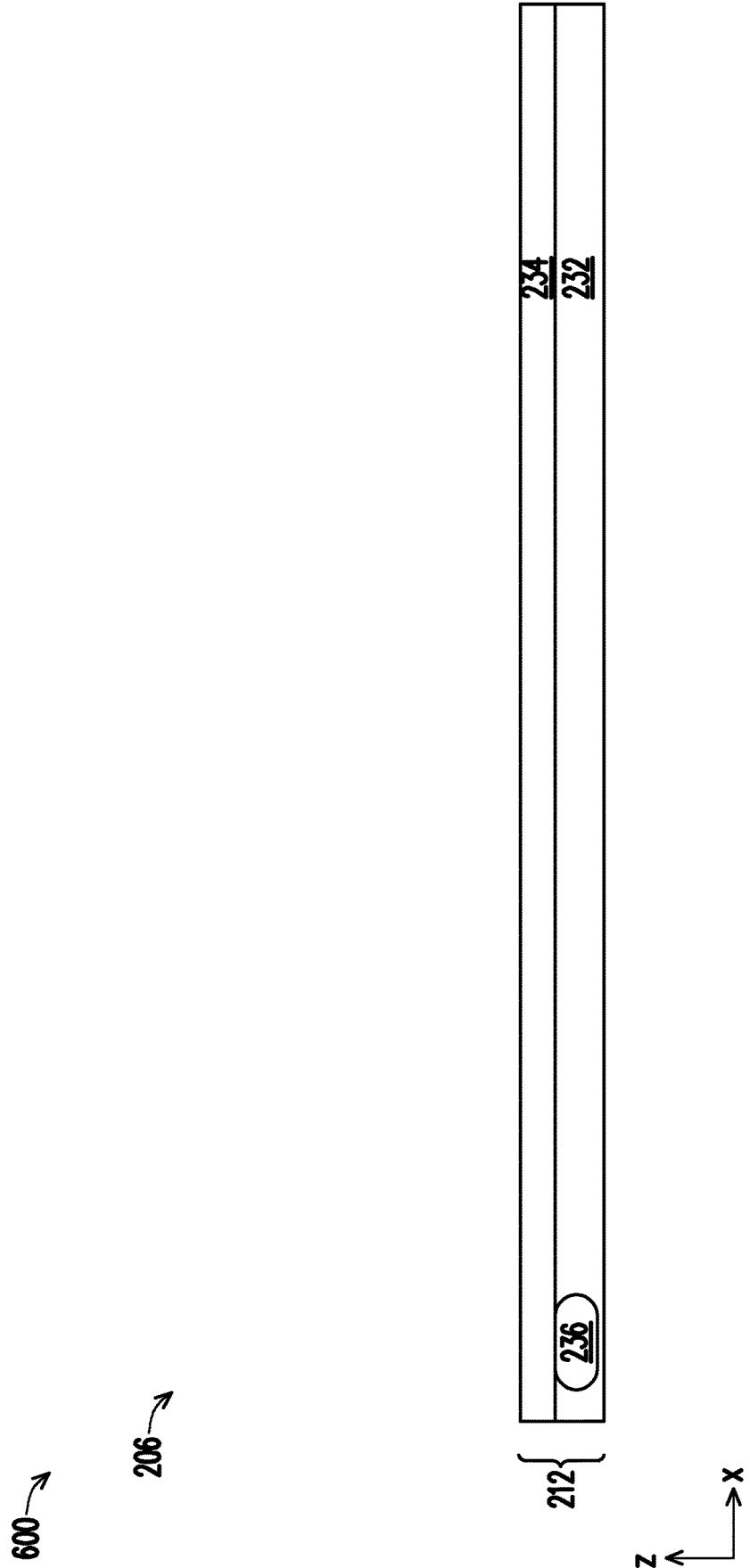


FIG. 6B

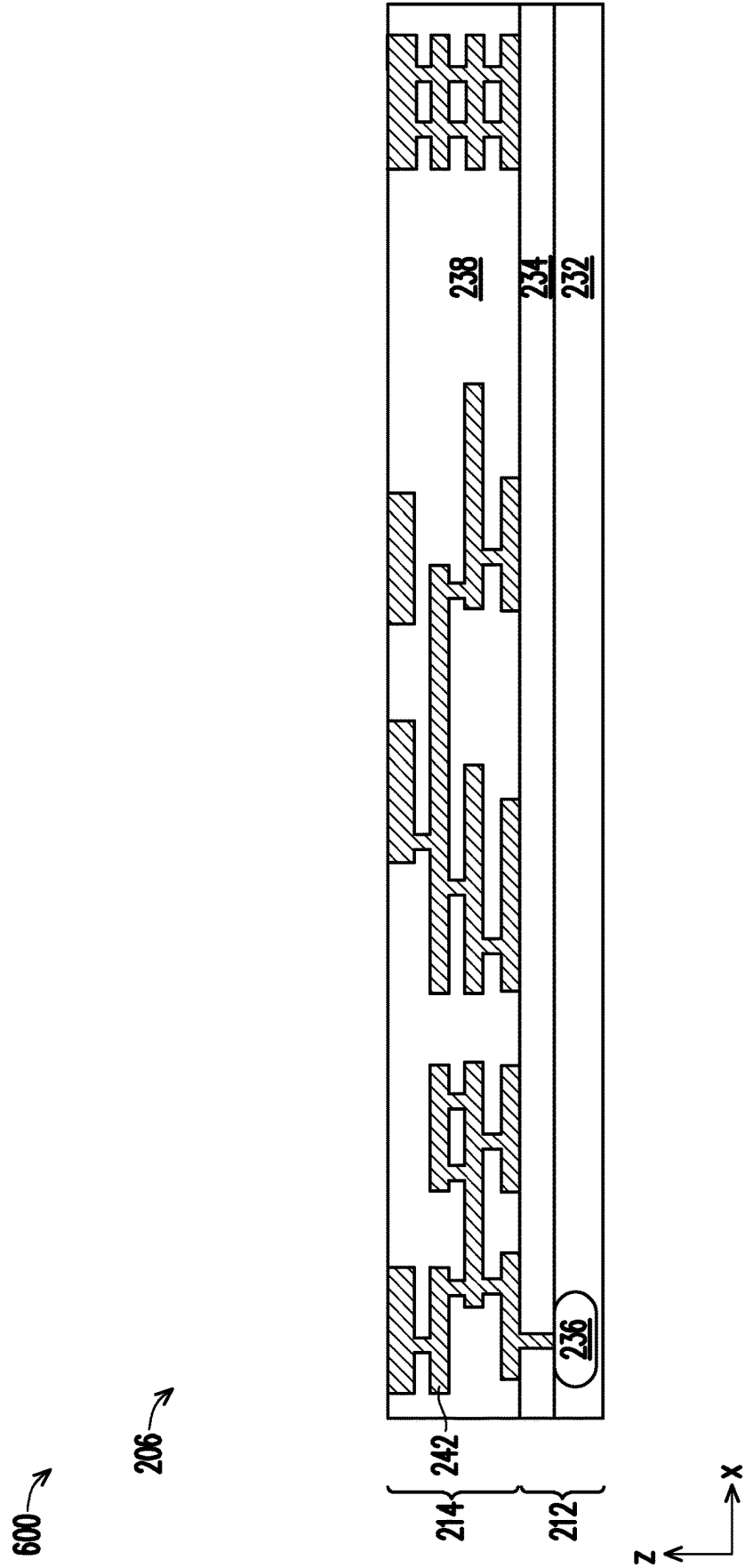


FIG. 6C

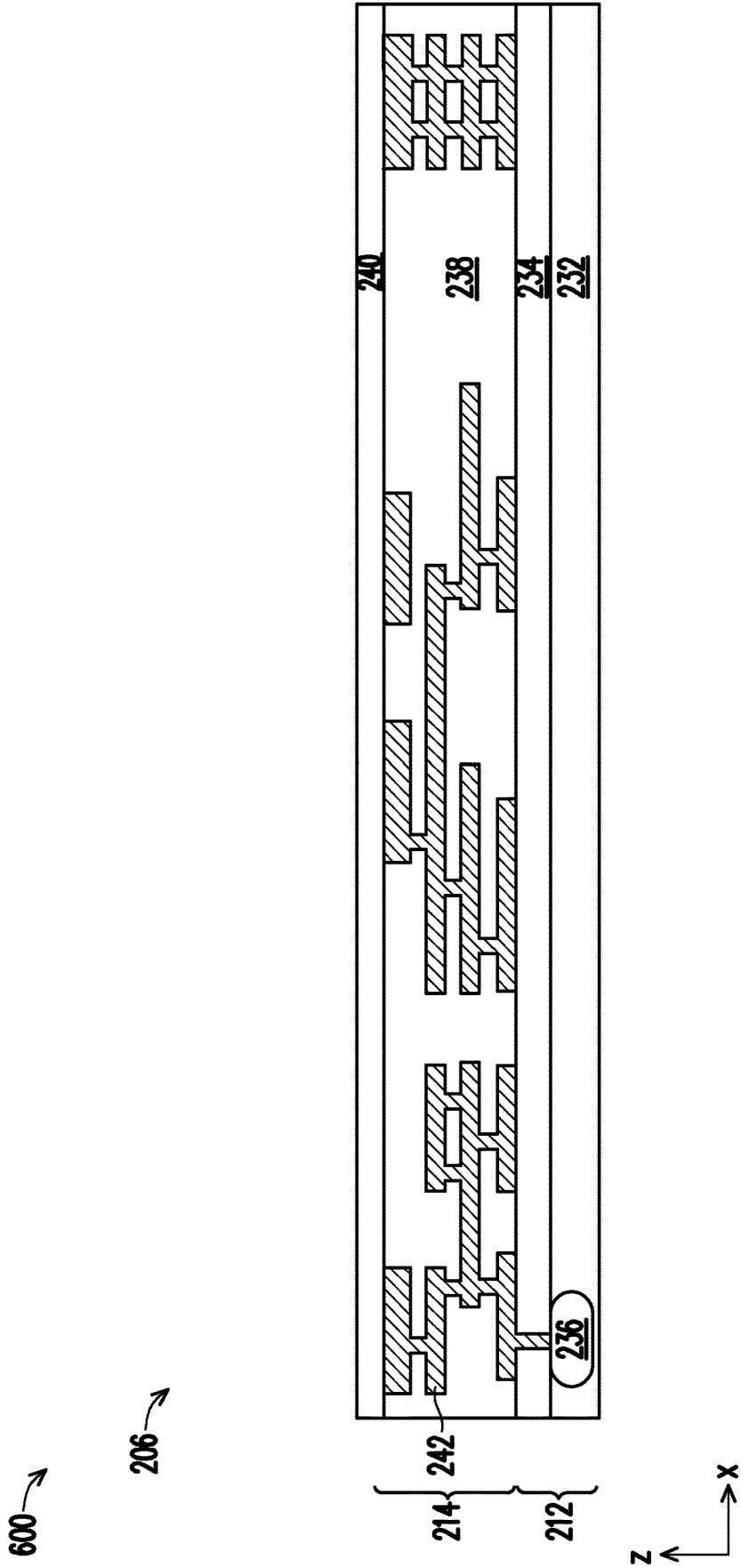


FIG. 6D

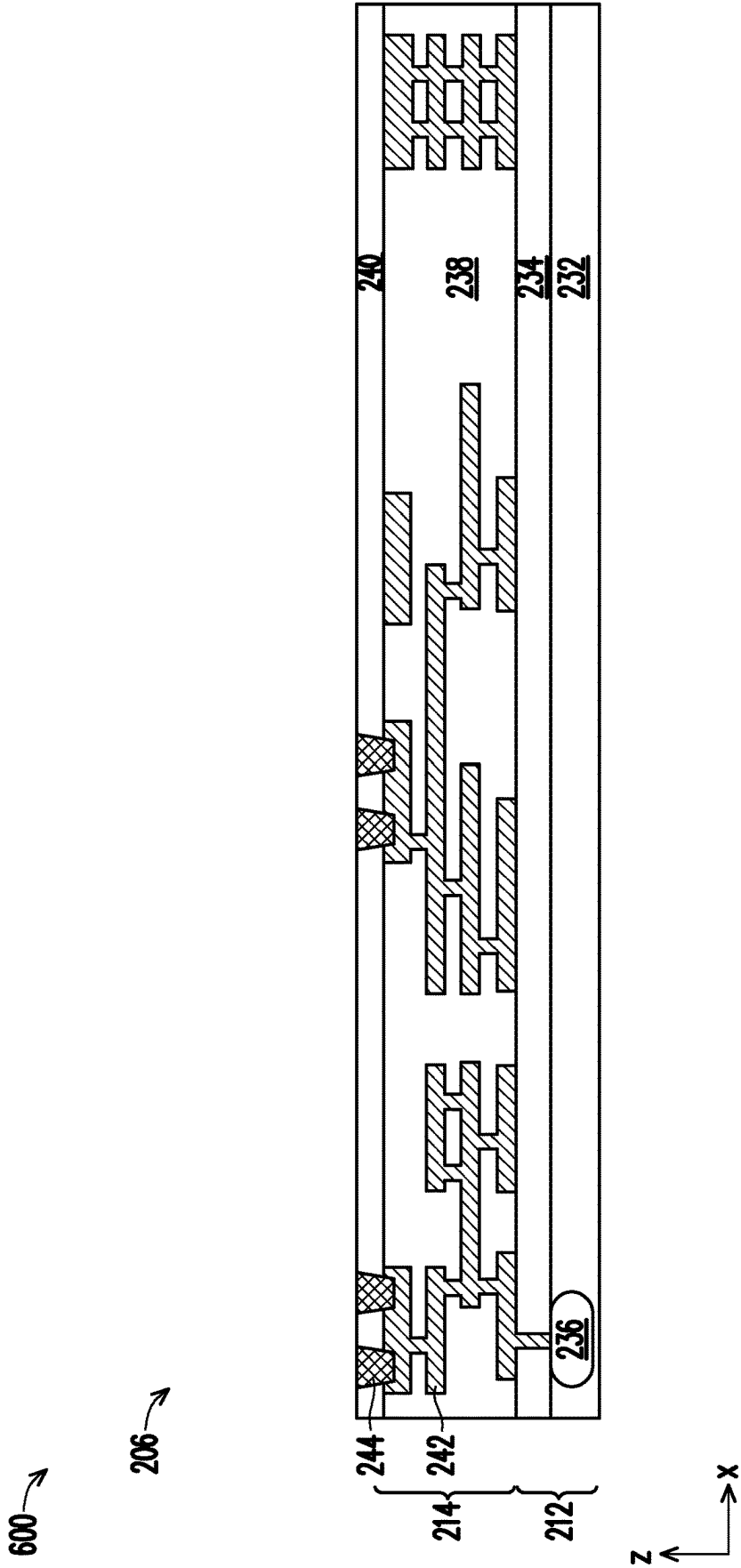
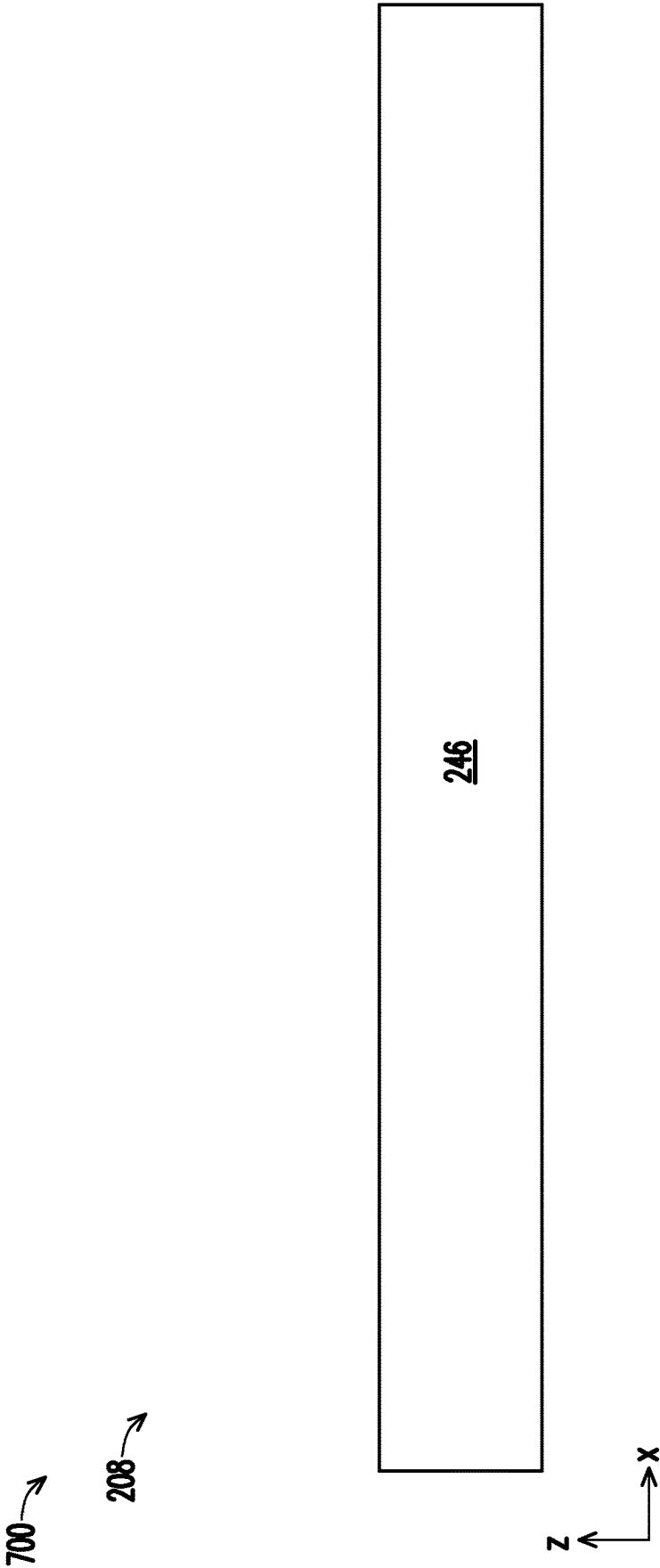


FIG. 6E



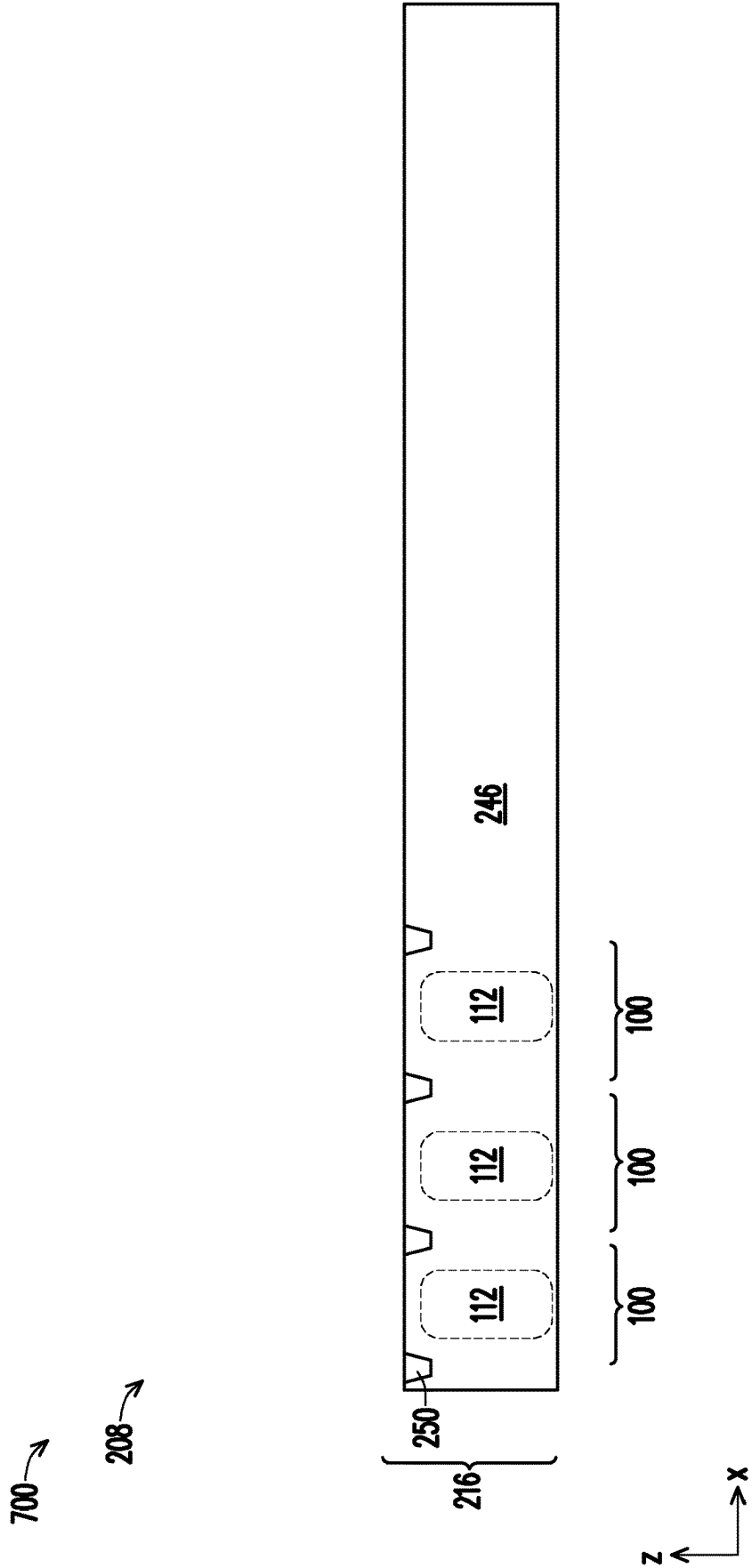


FIG. 7B

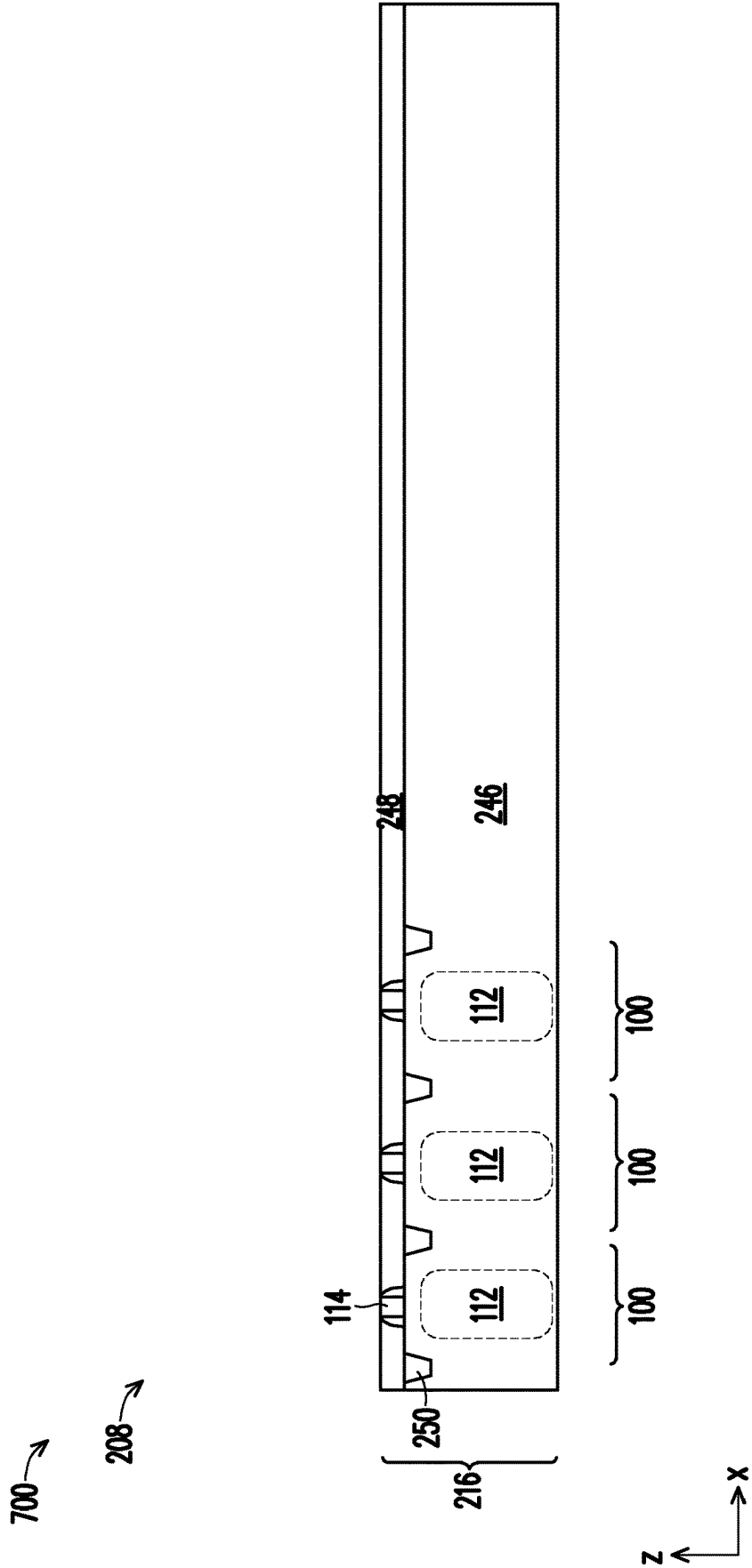


FIG. 7C

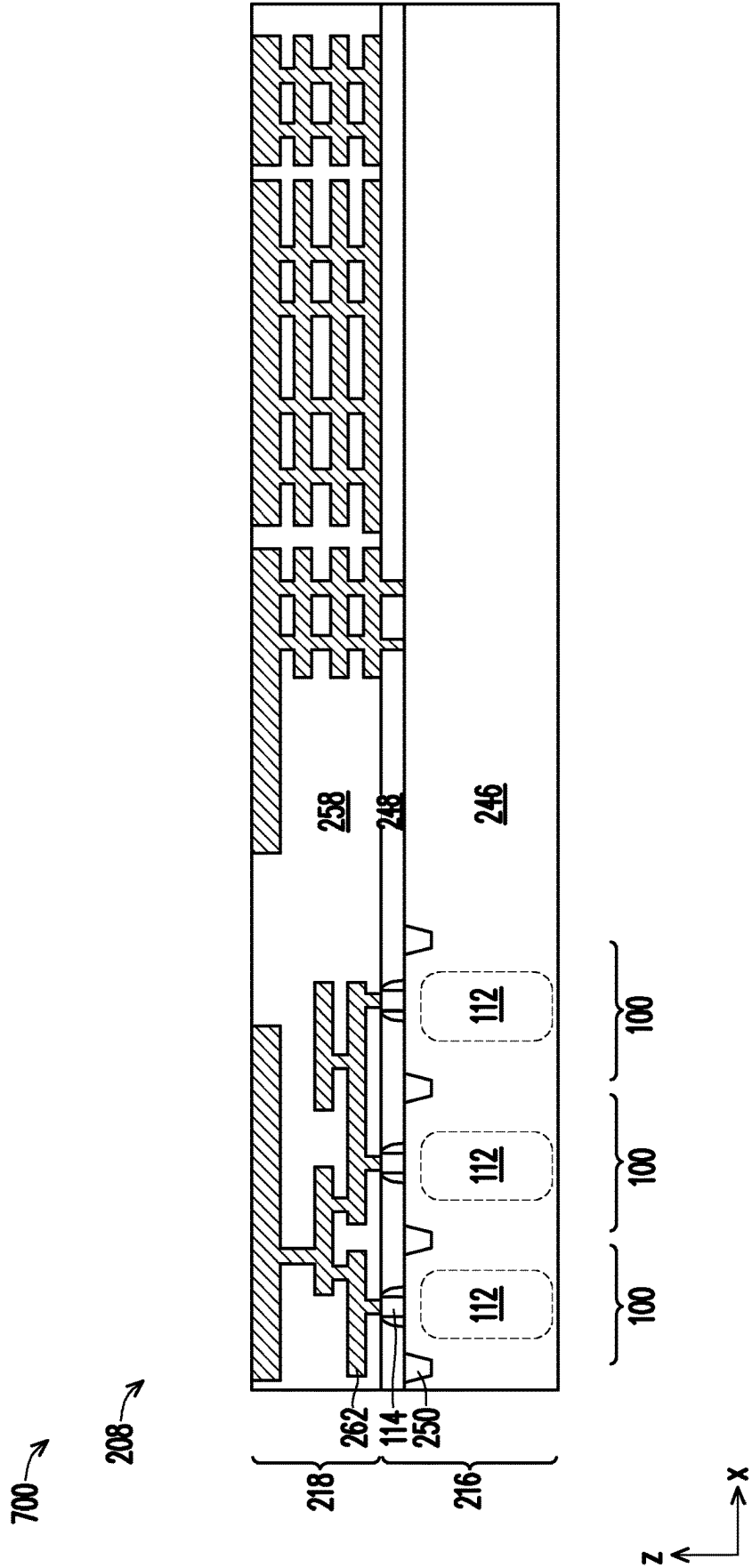


FIG. 7D

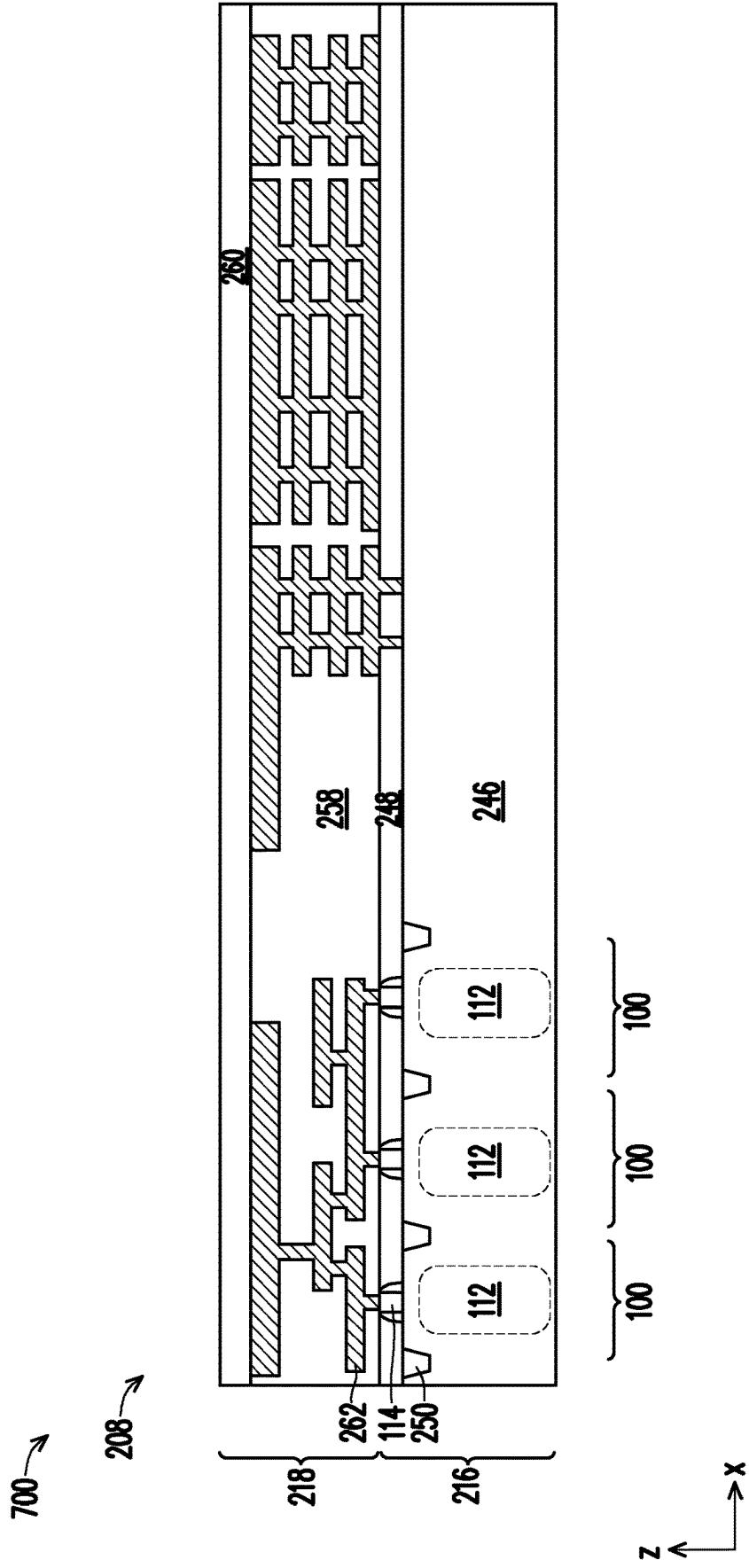


FIG. 7E

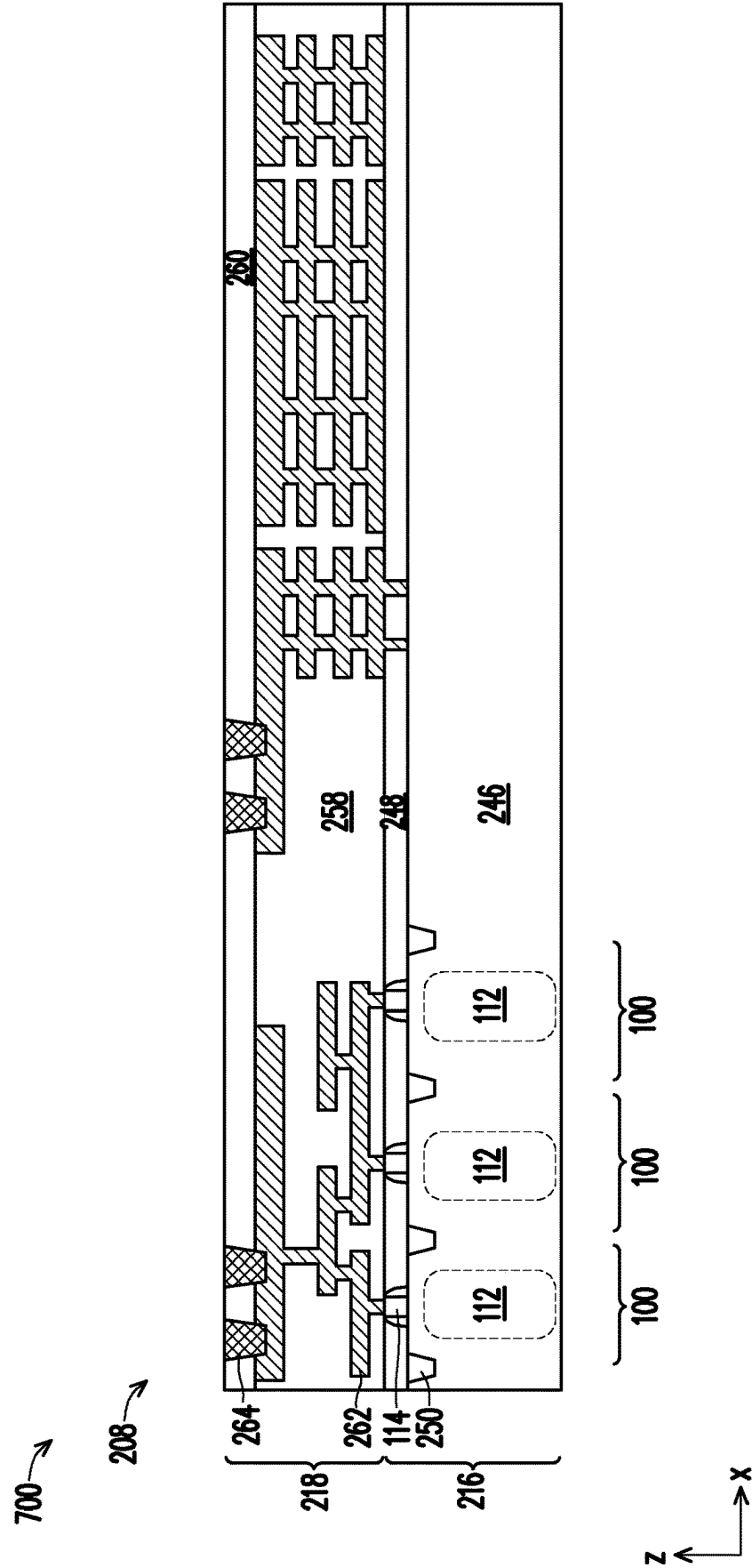
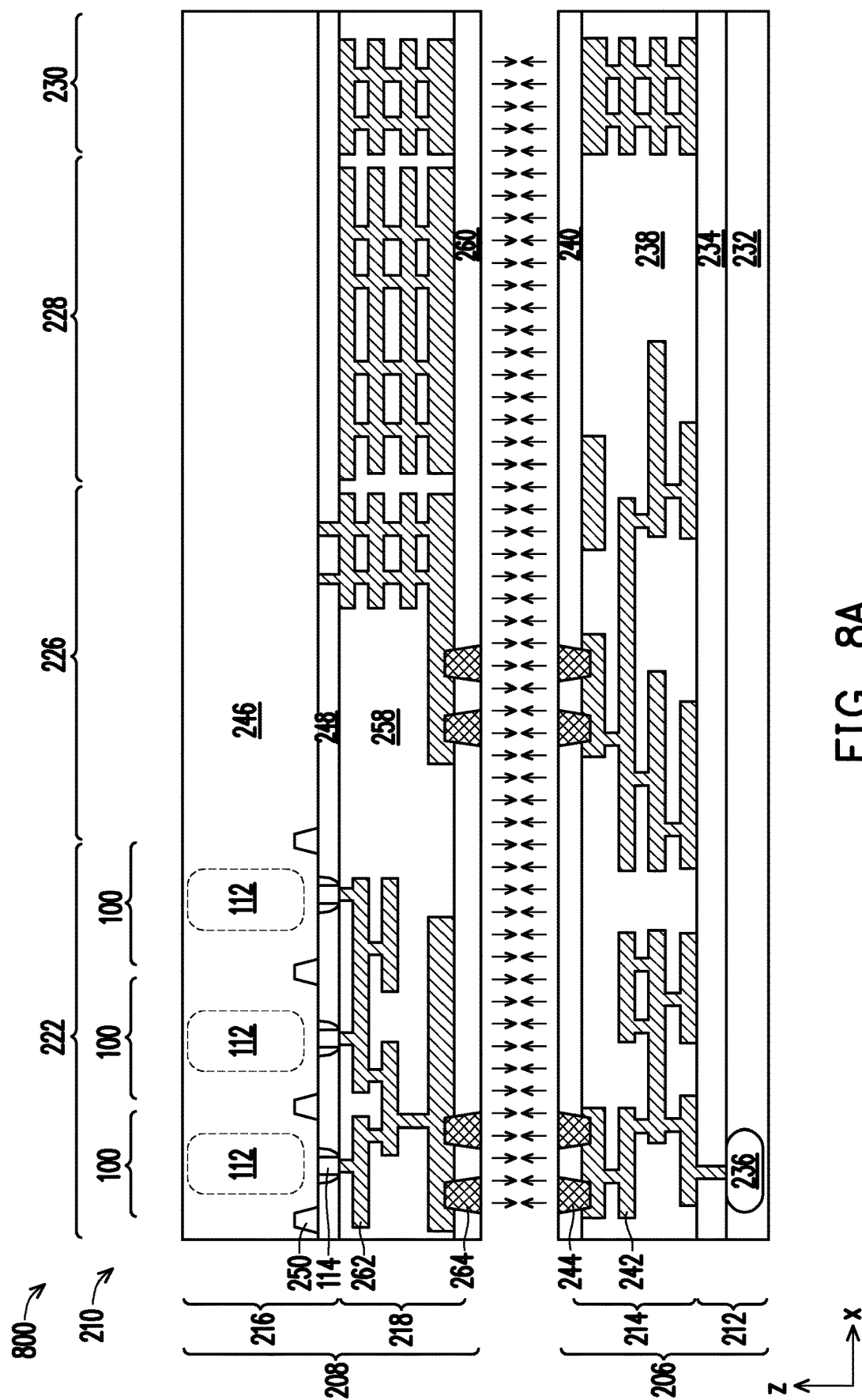


FIG. 7F



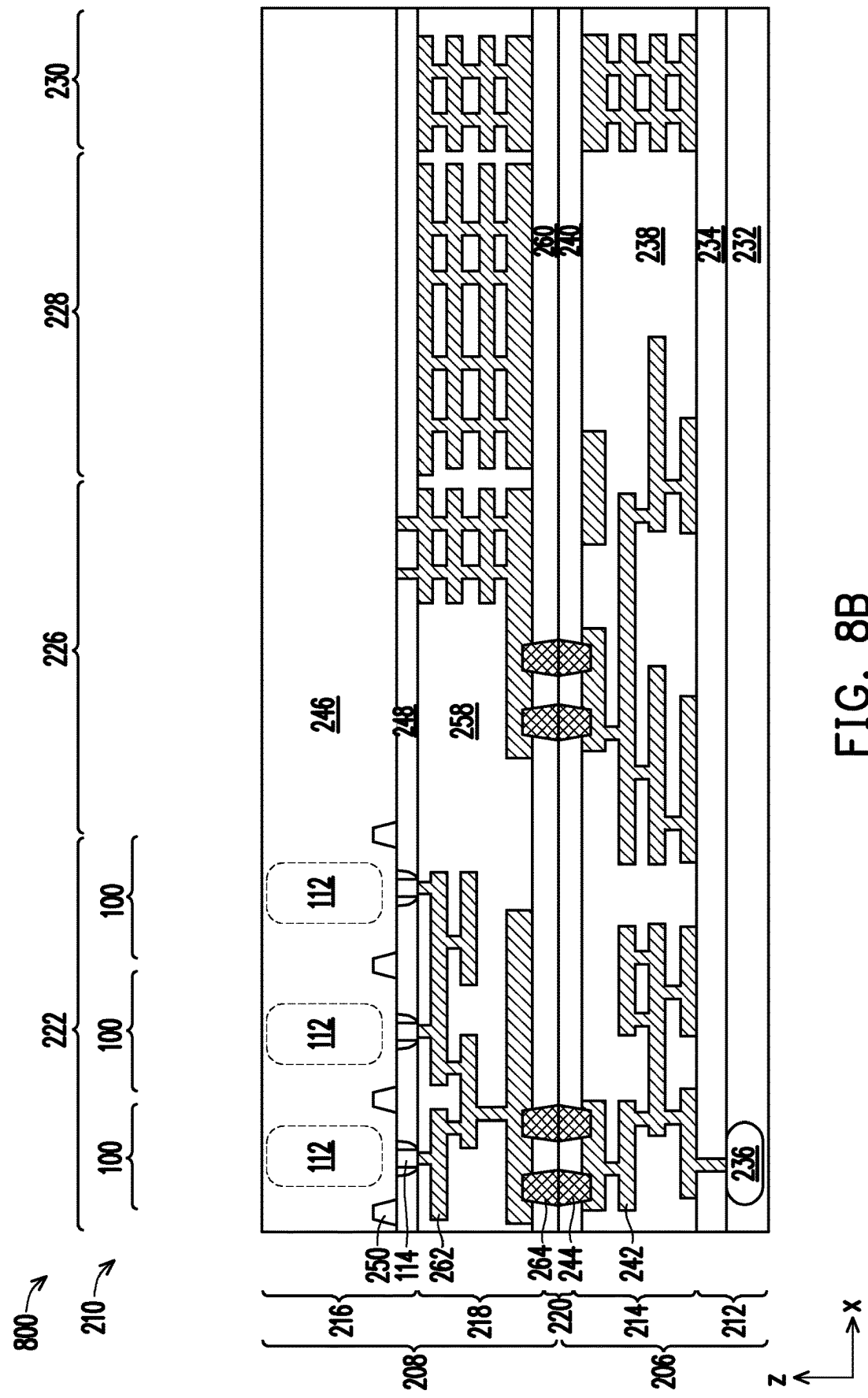


FIG. 8B

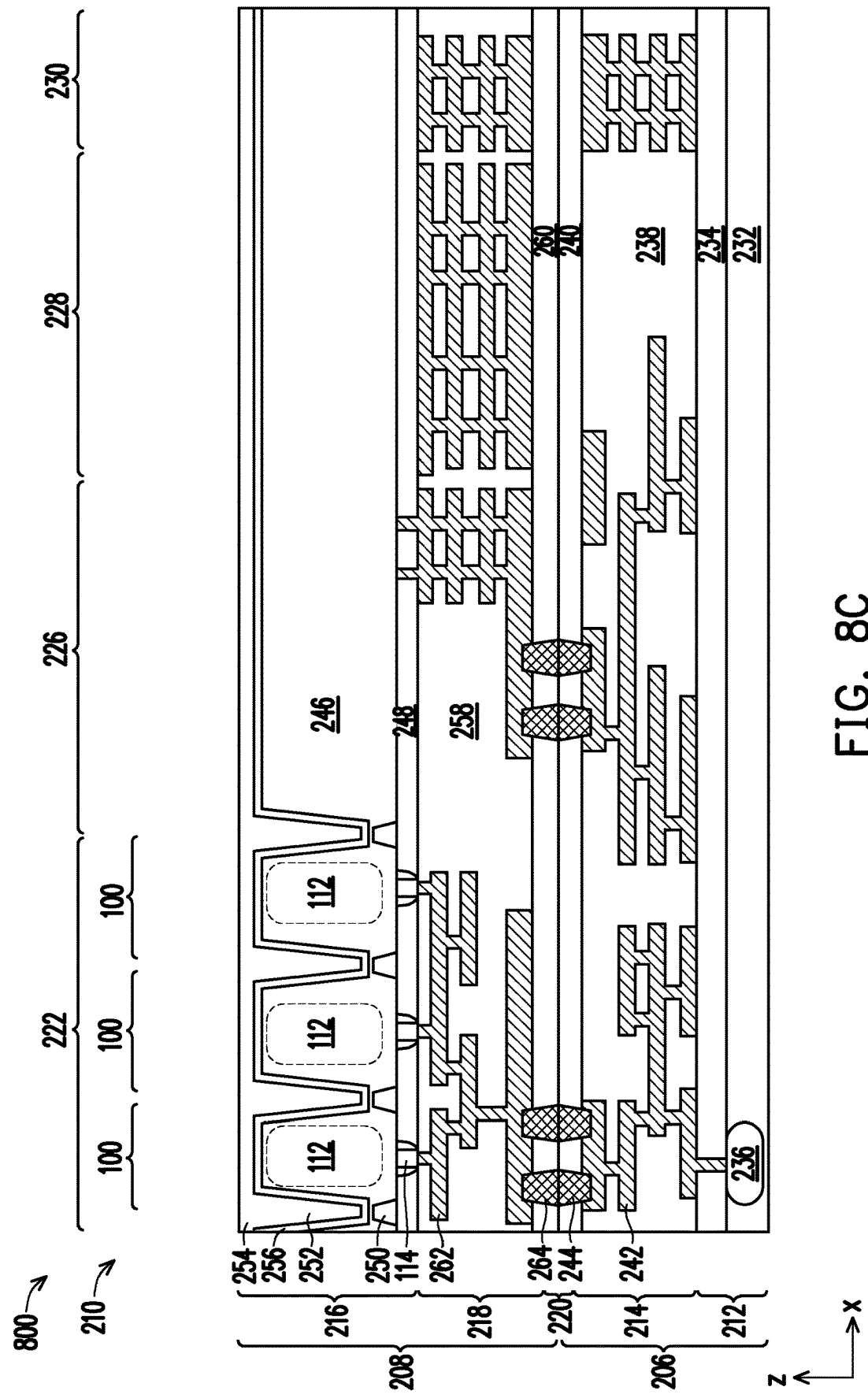
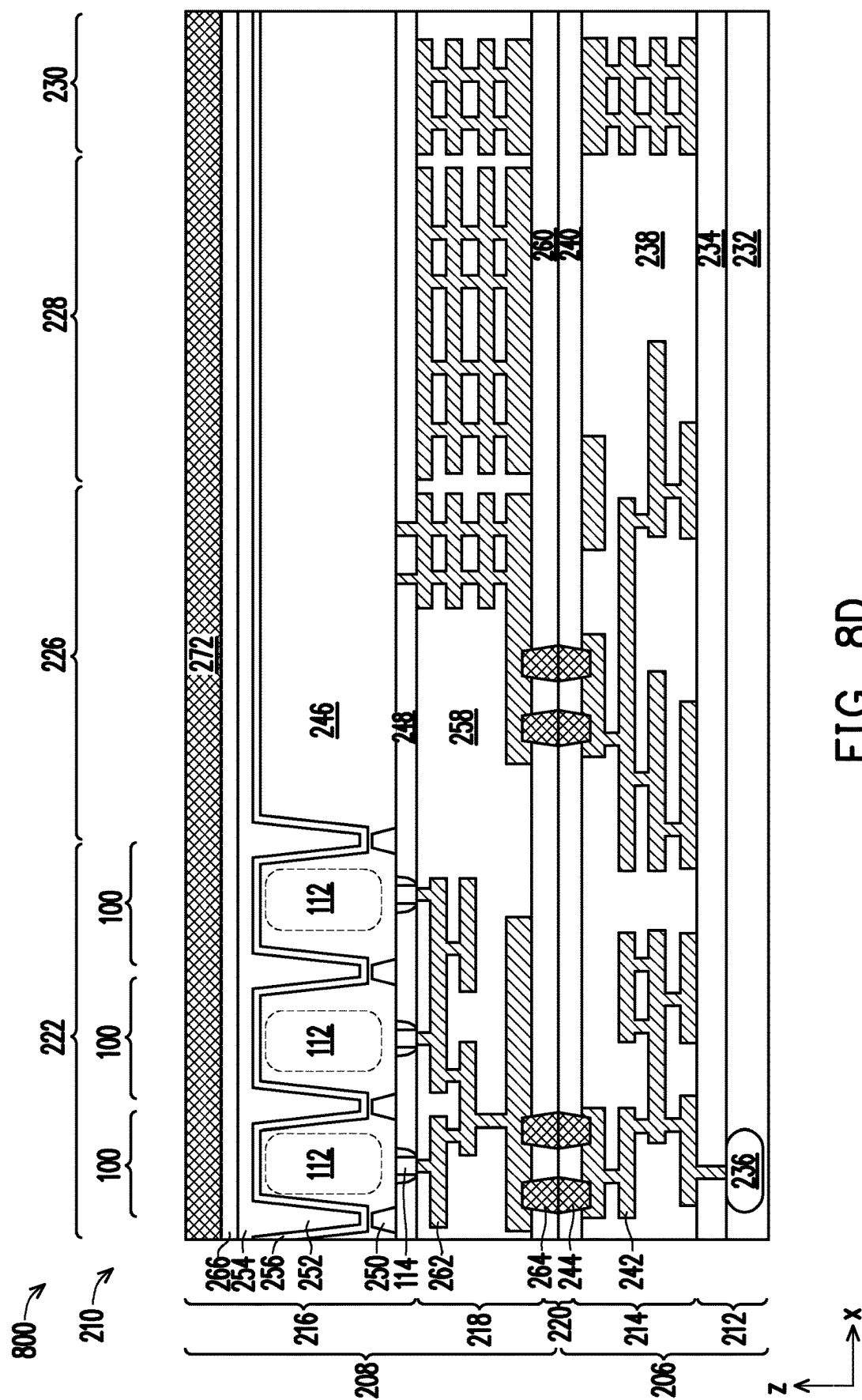
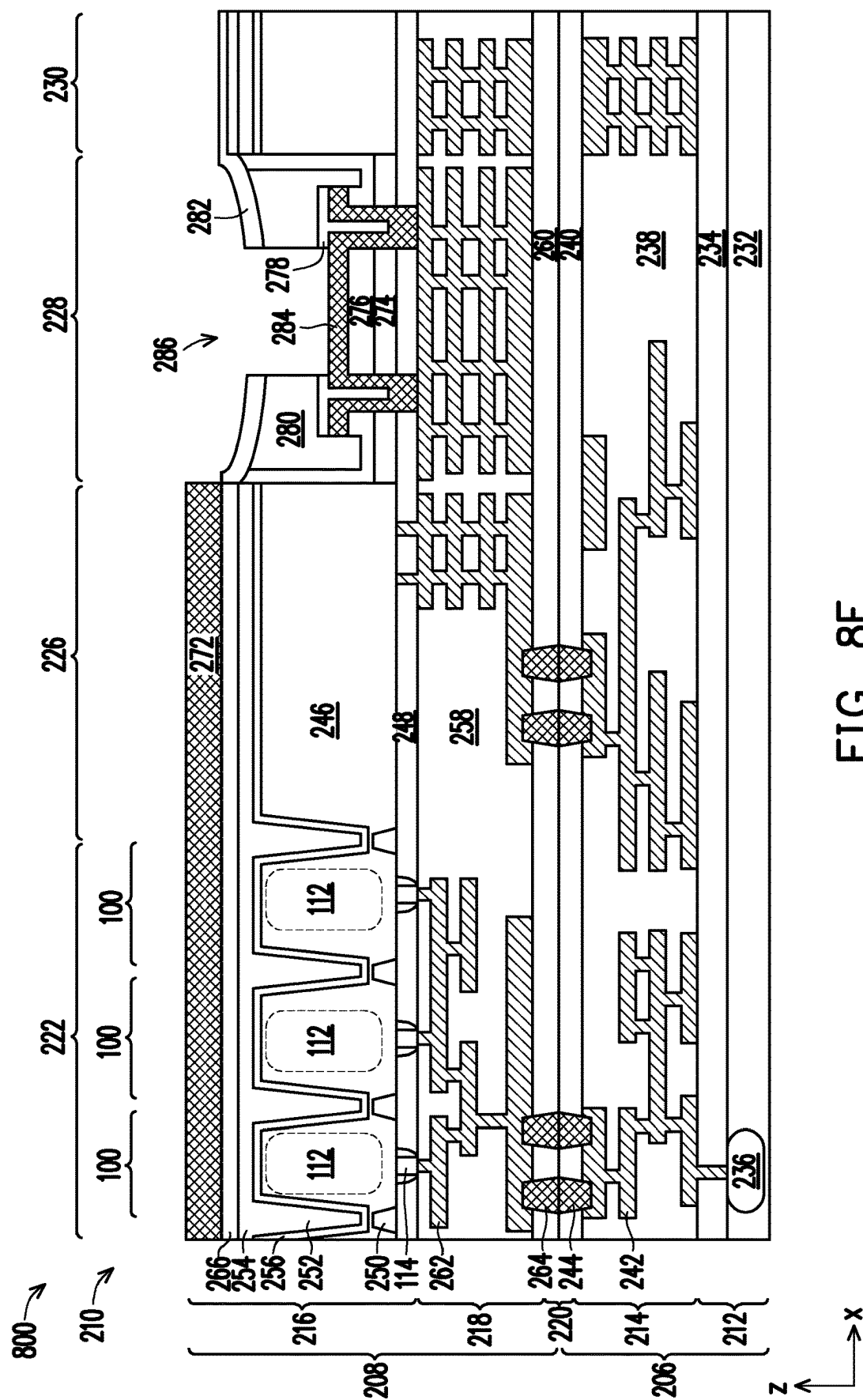


FIG. 8C





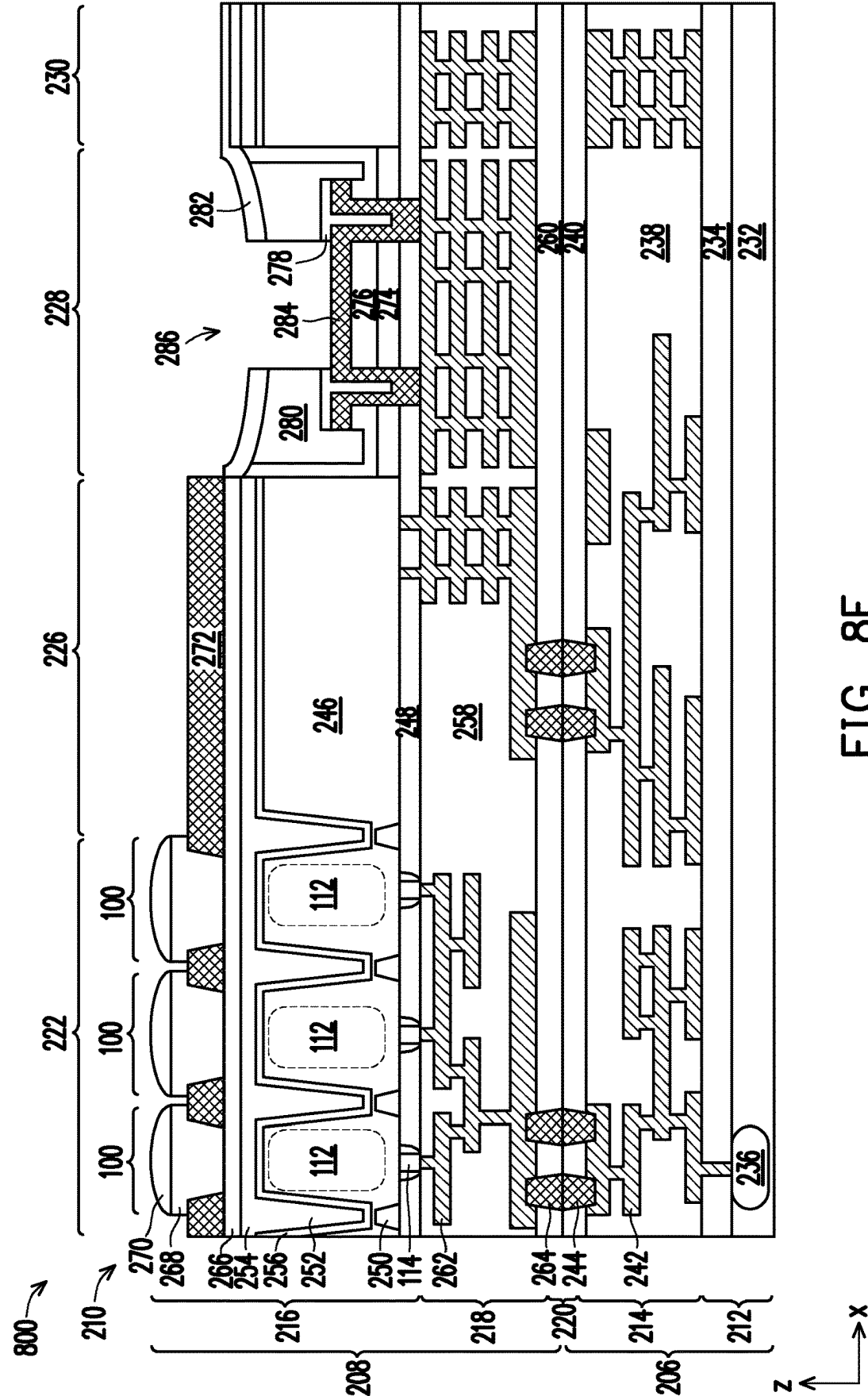
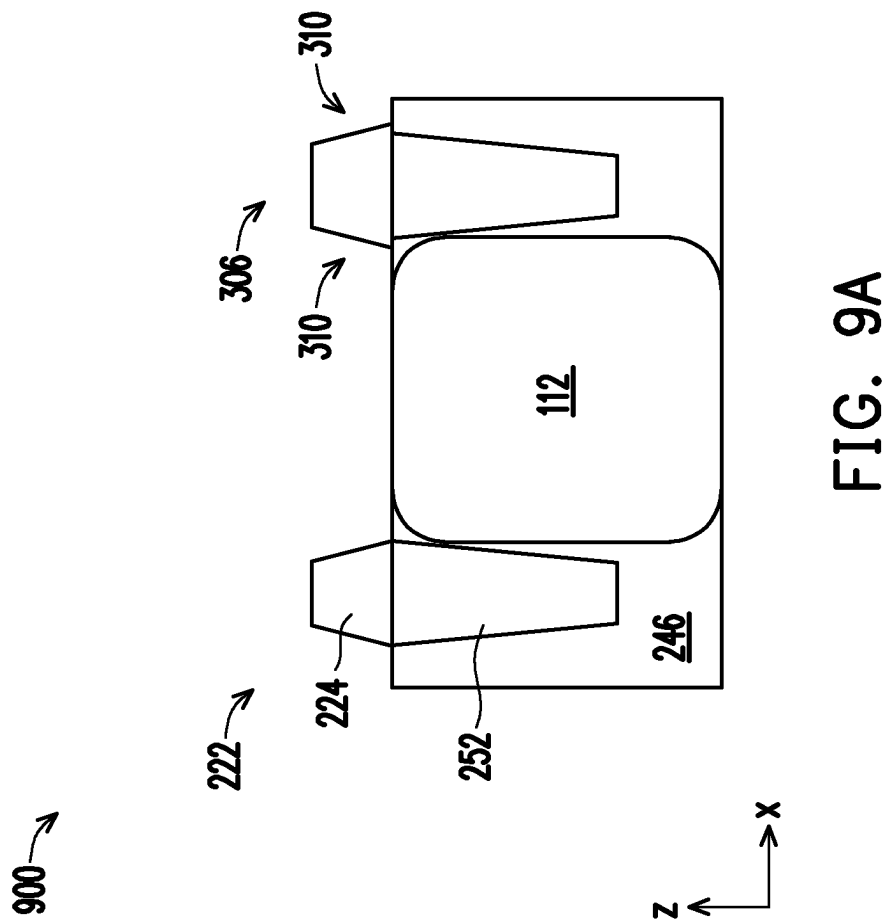


FIG. 8F



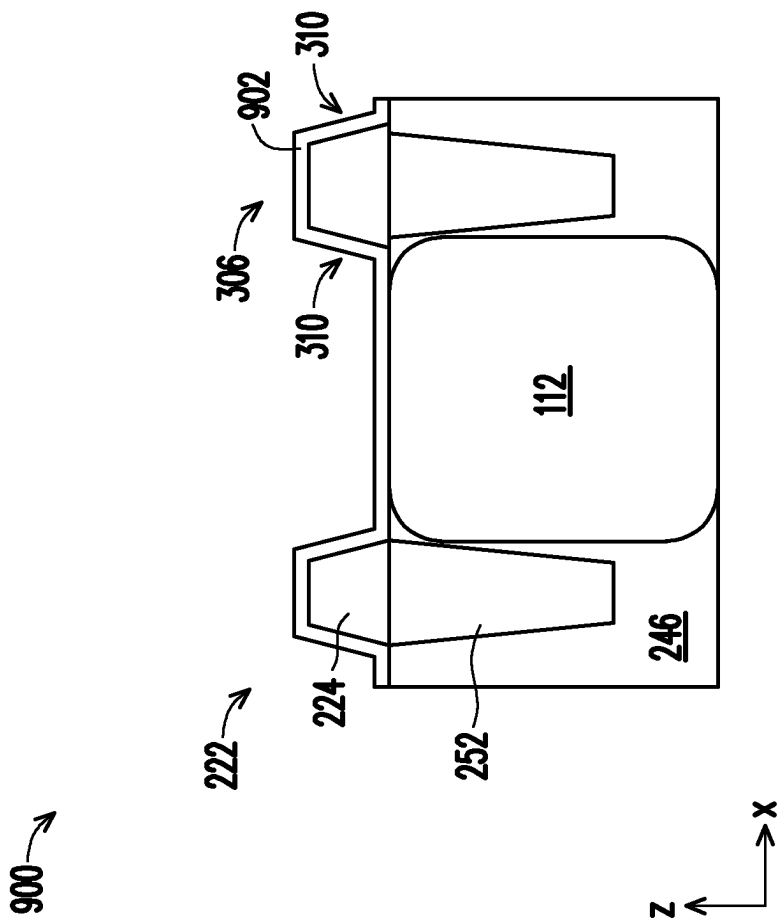


FIG. 9B

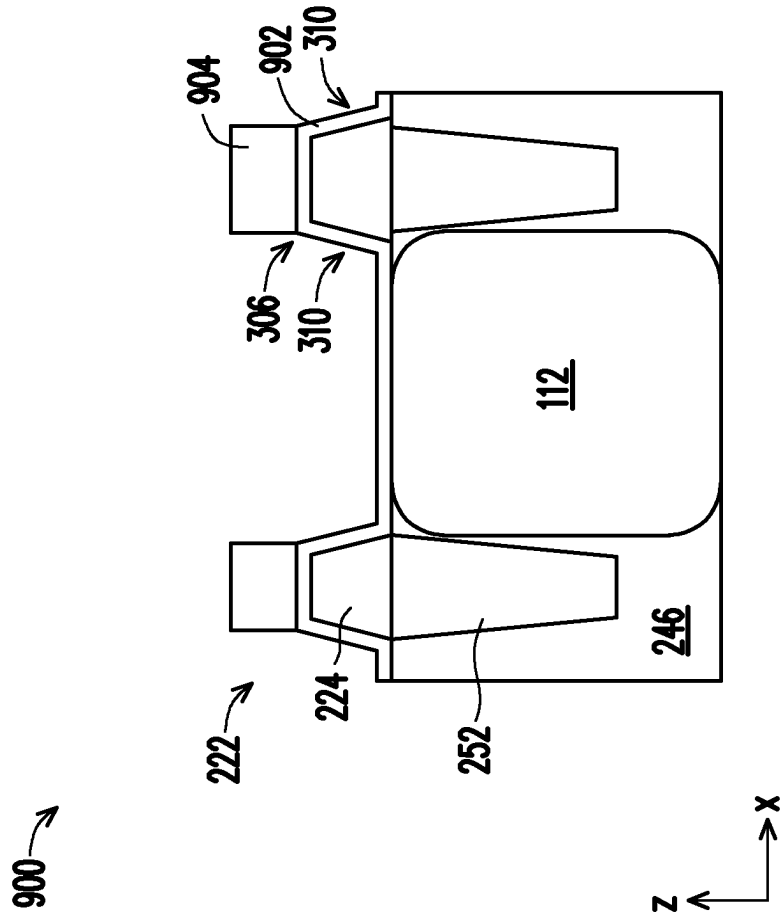


FIG. 9C

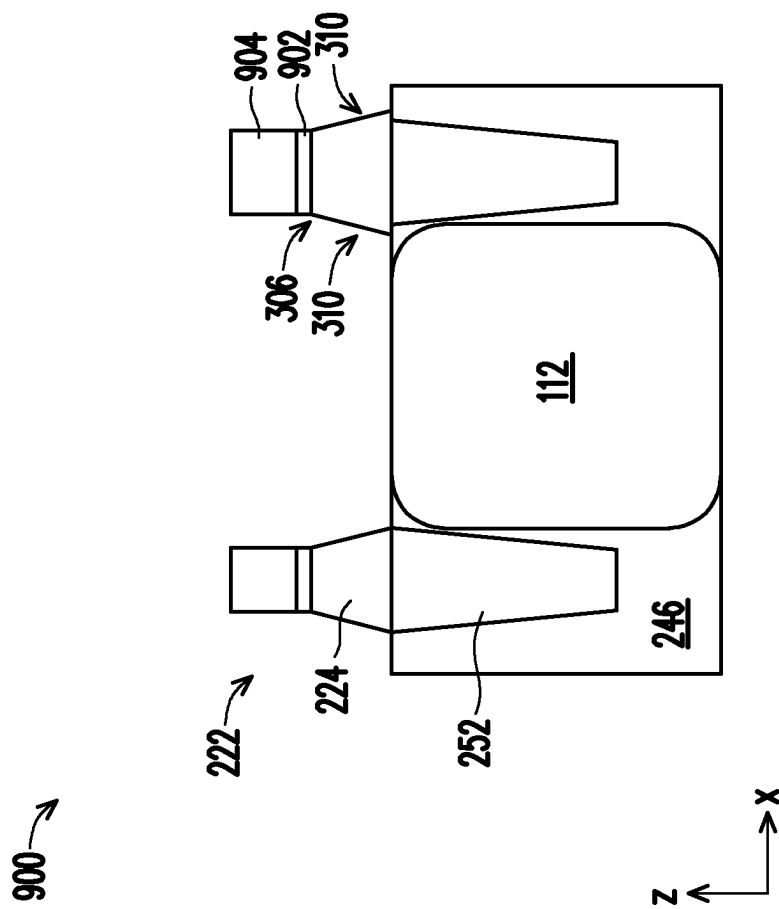
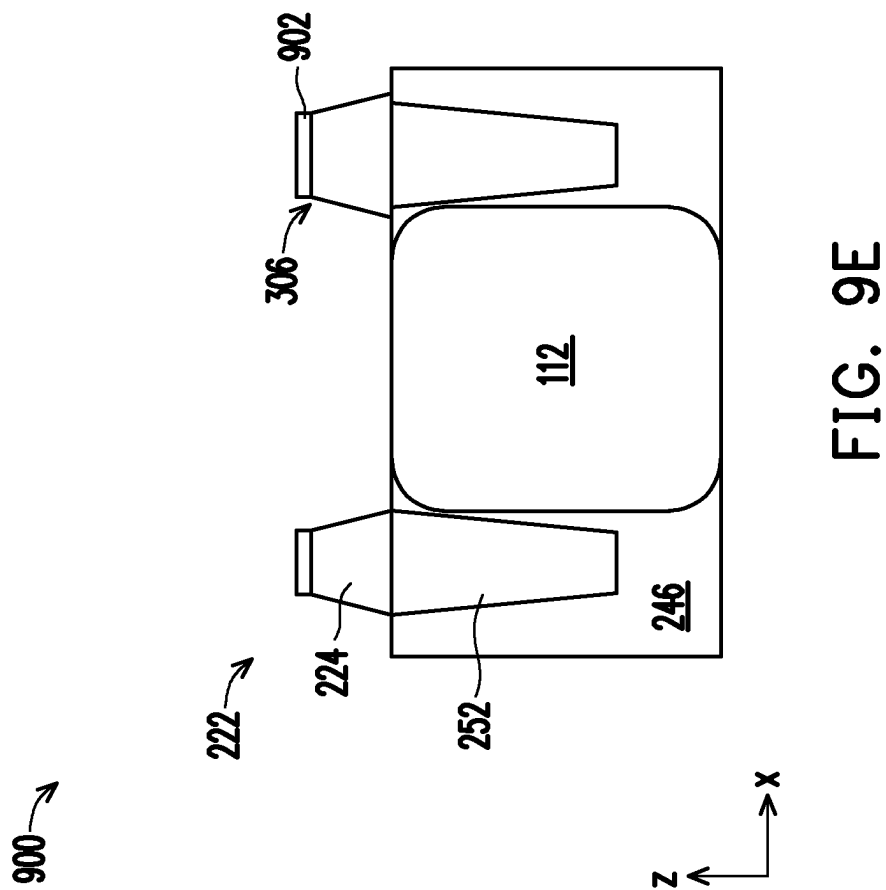


FIG. 9D



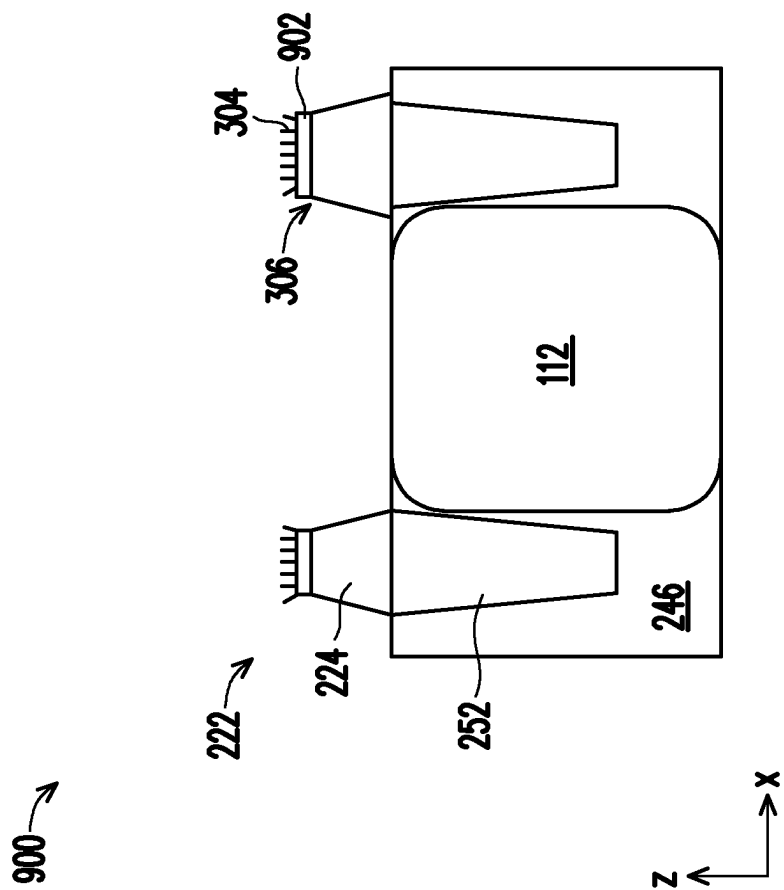


FIG. 9F

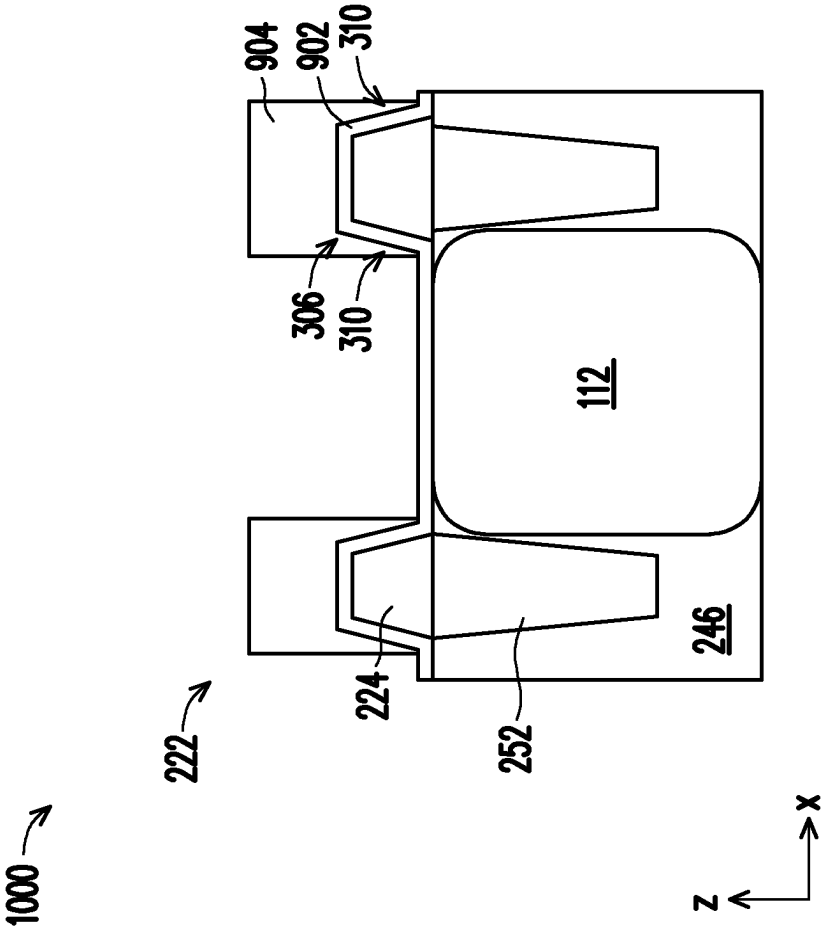


FIG. 10A

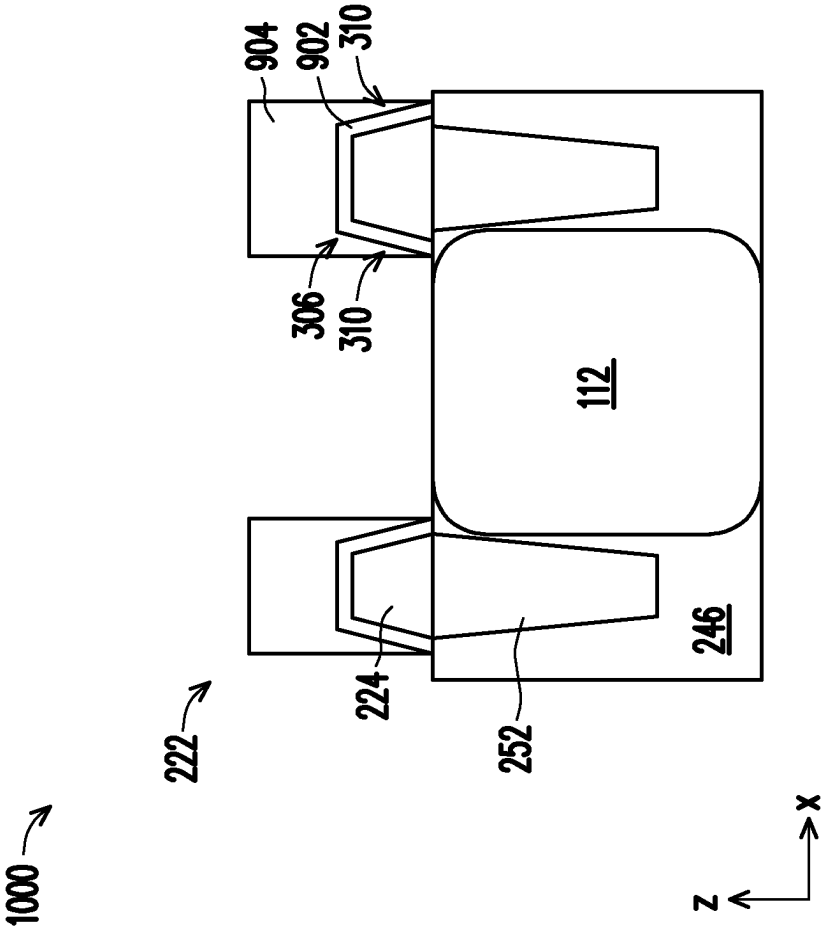


FIG. 10B

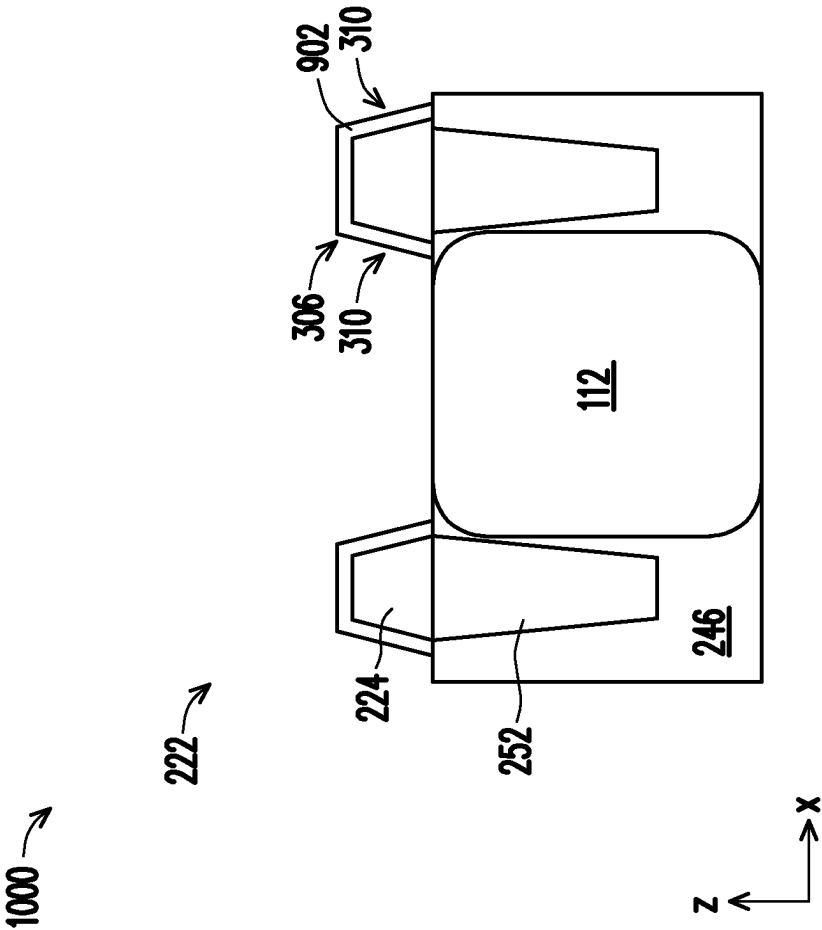


FIG. 10C

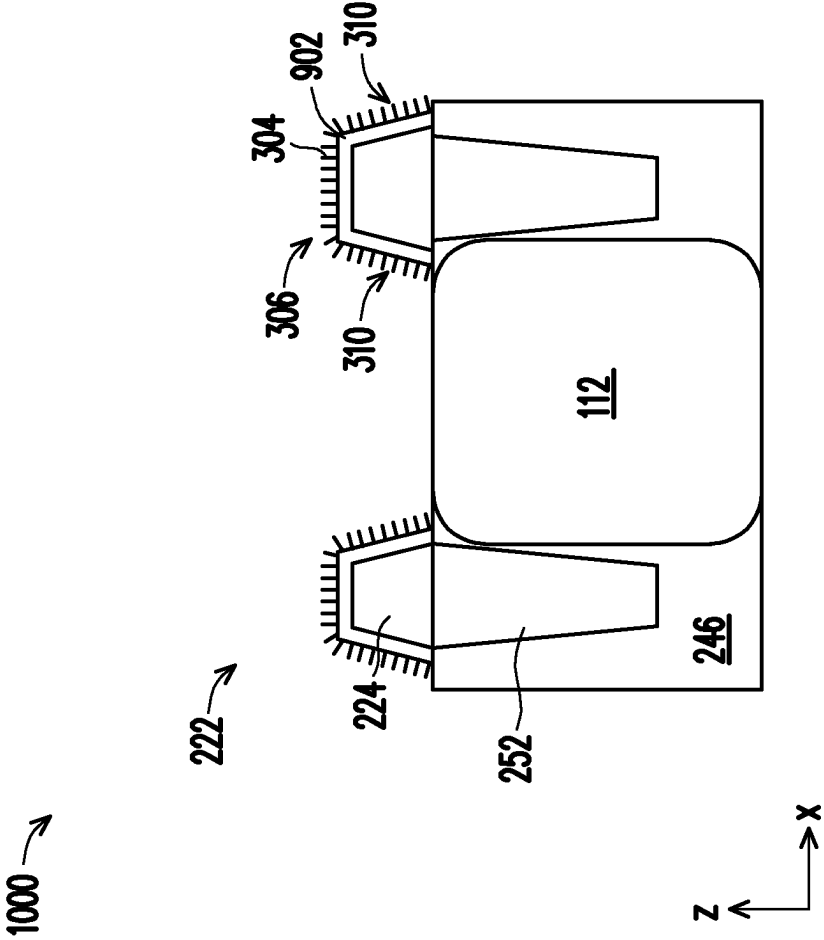
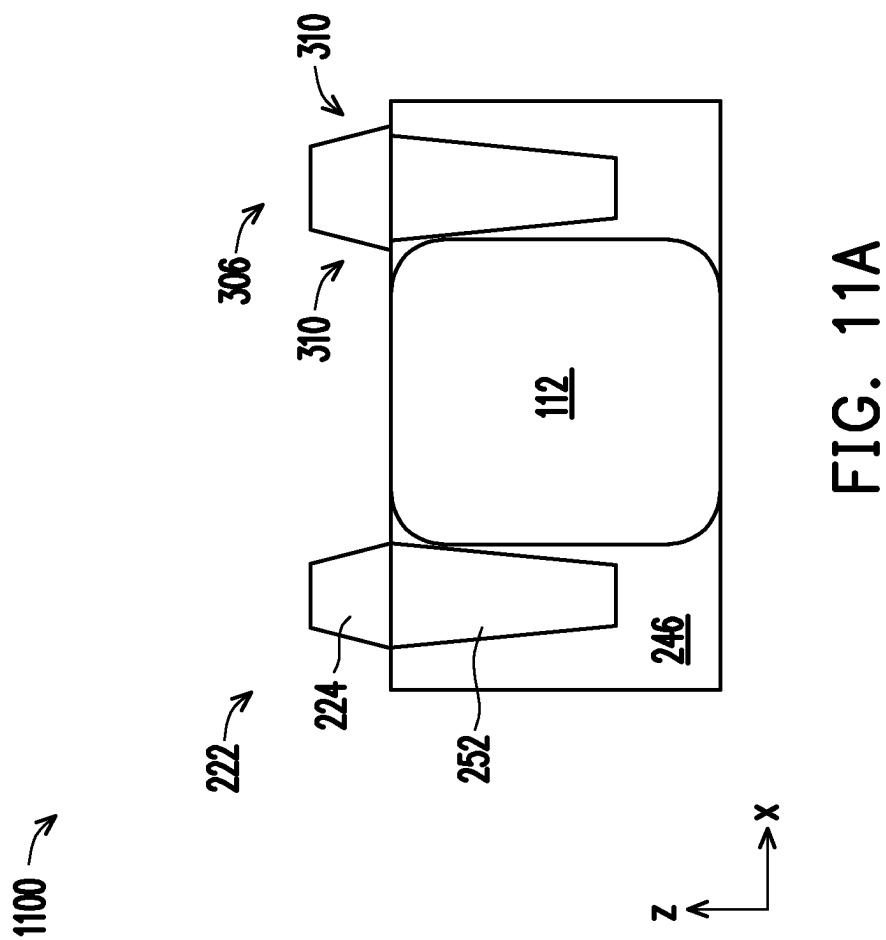


FIG. 10D



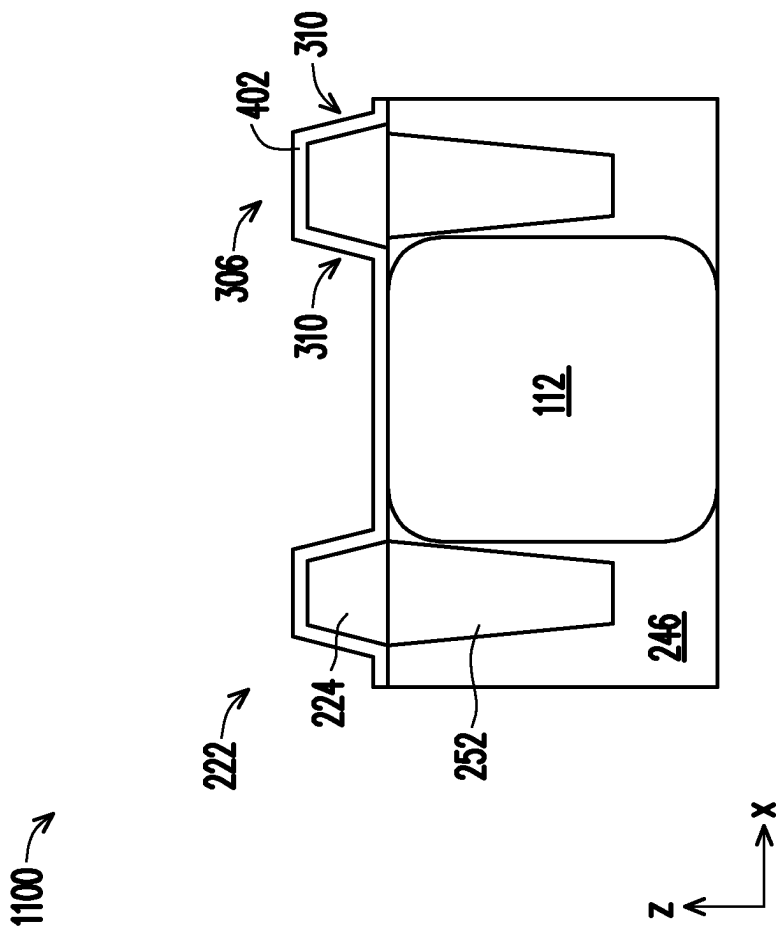


FIG. 11B

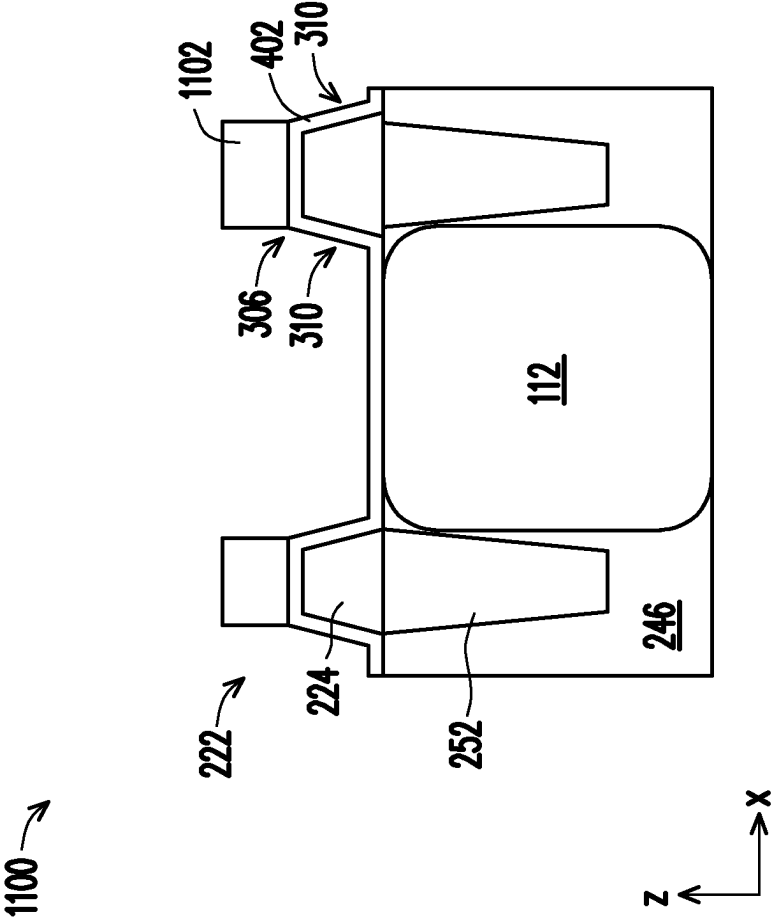


FIG. 11C

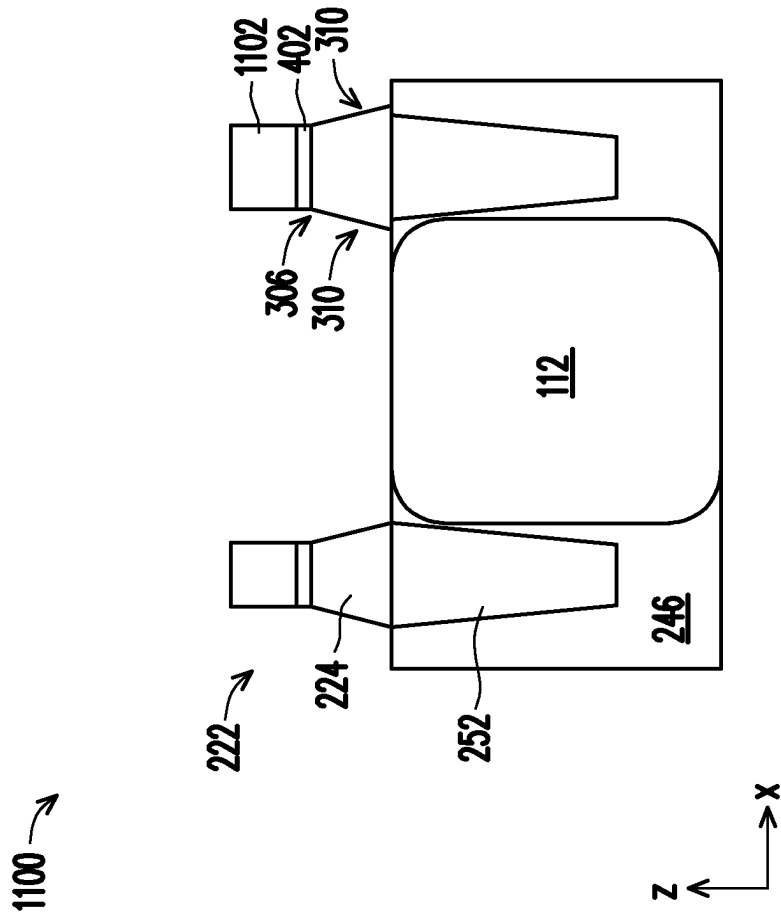


FIG. 11D

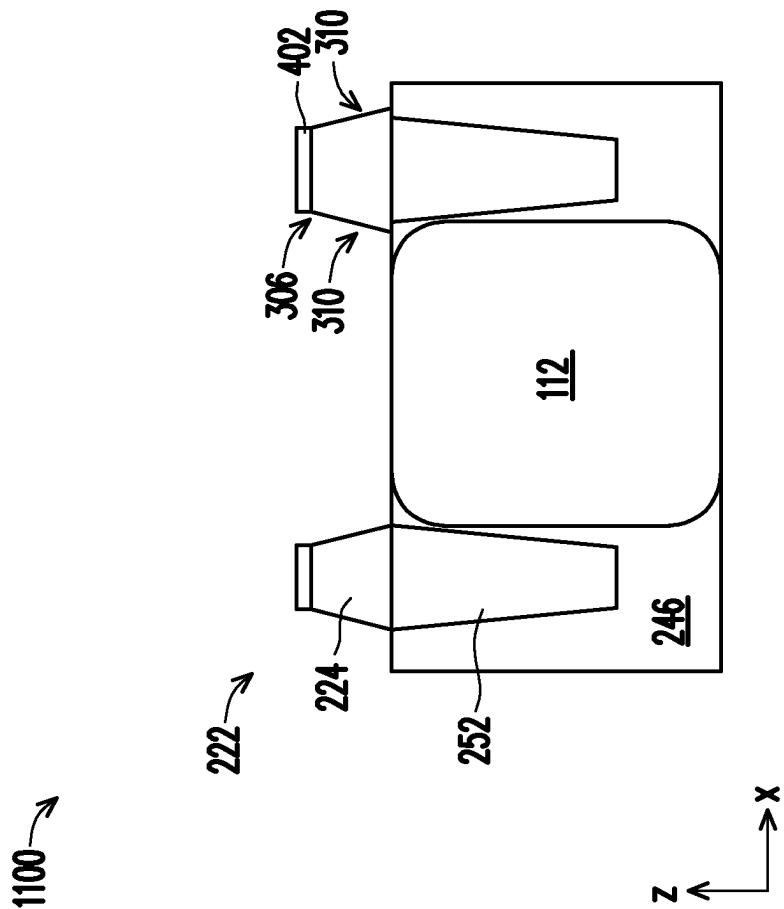


FIG. 11E

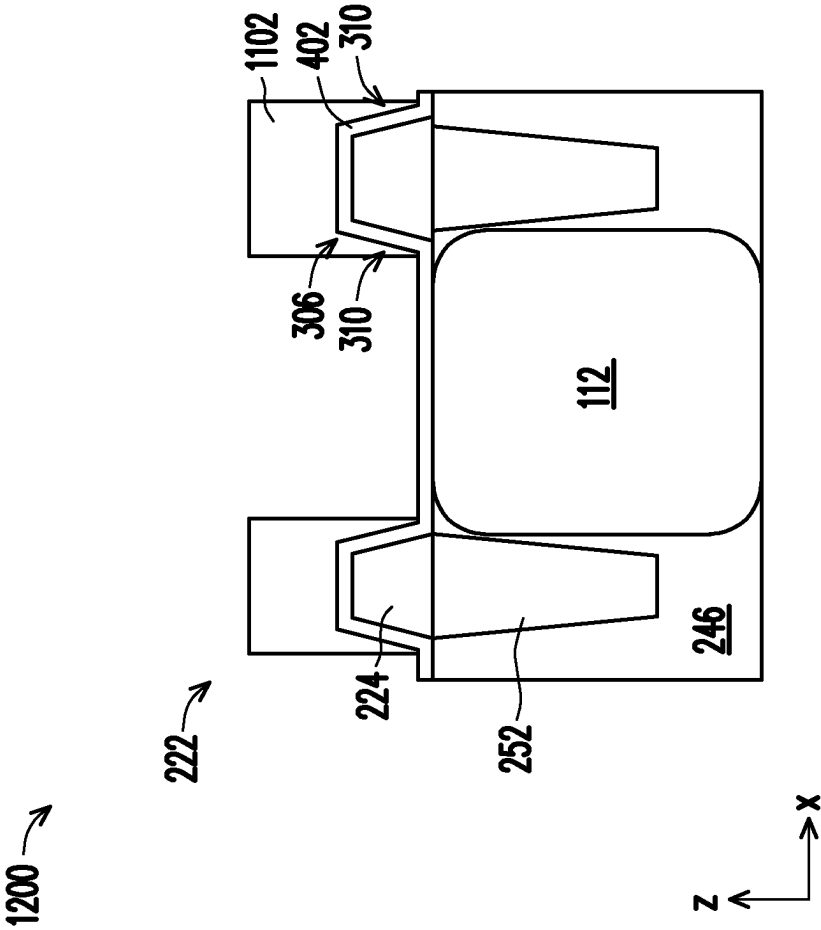


FIG. 12A

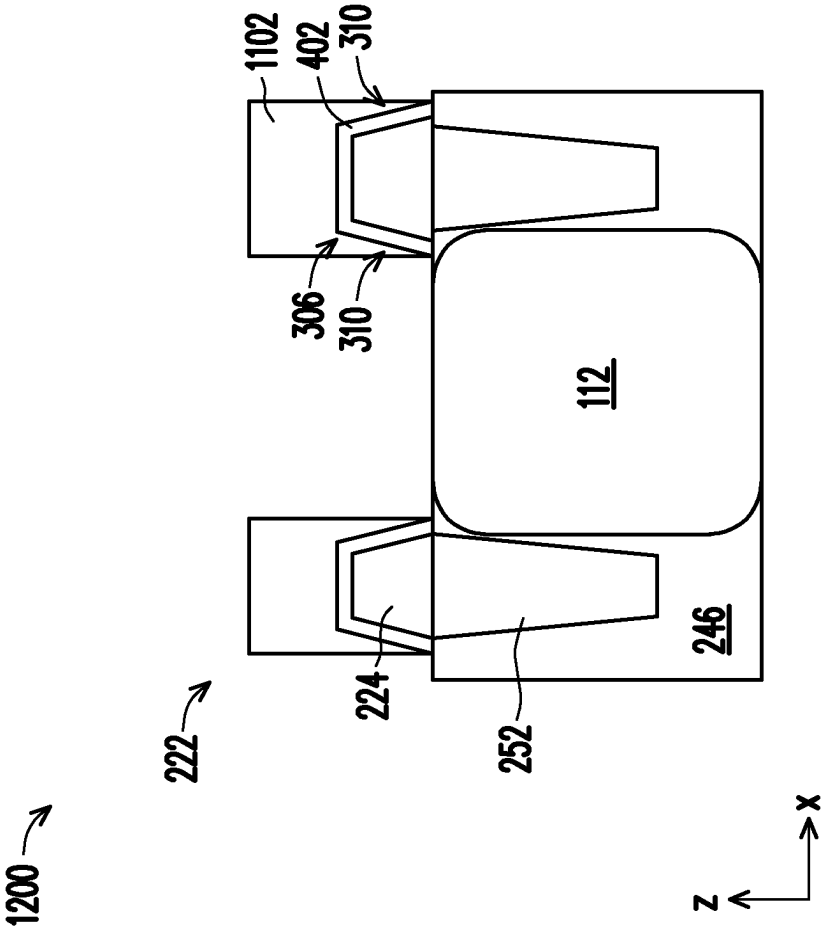


FIG. 12B

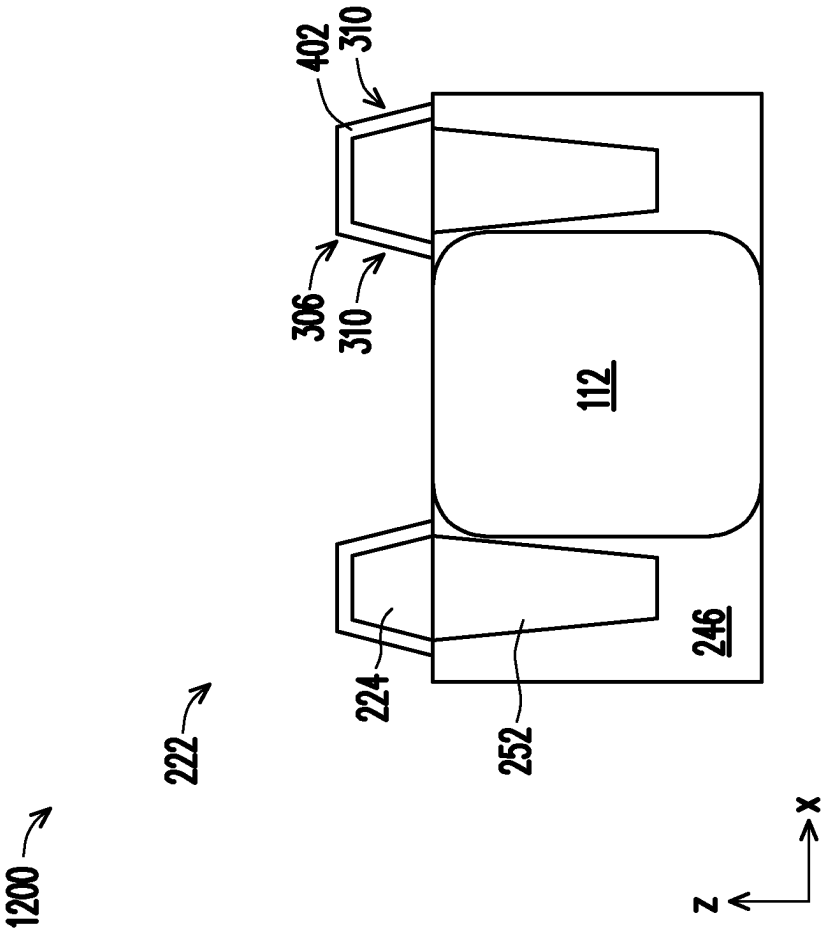
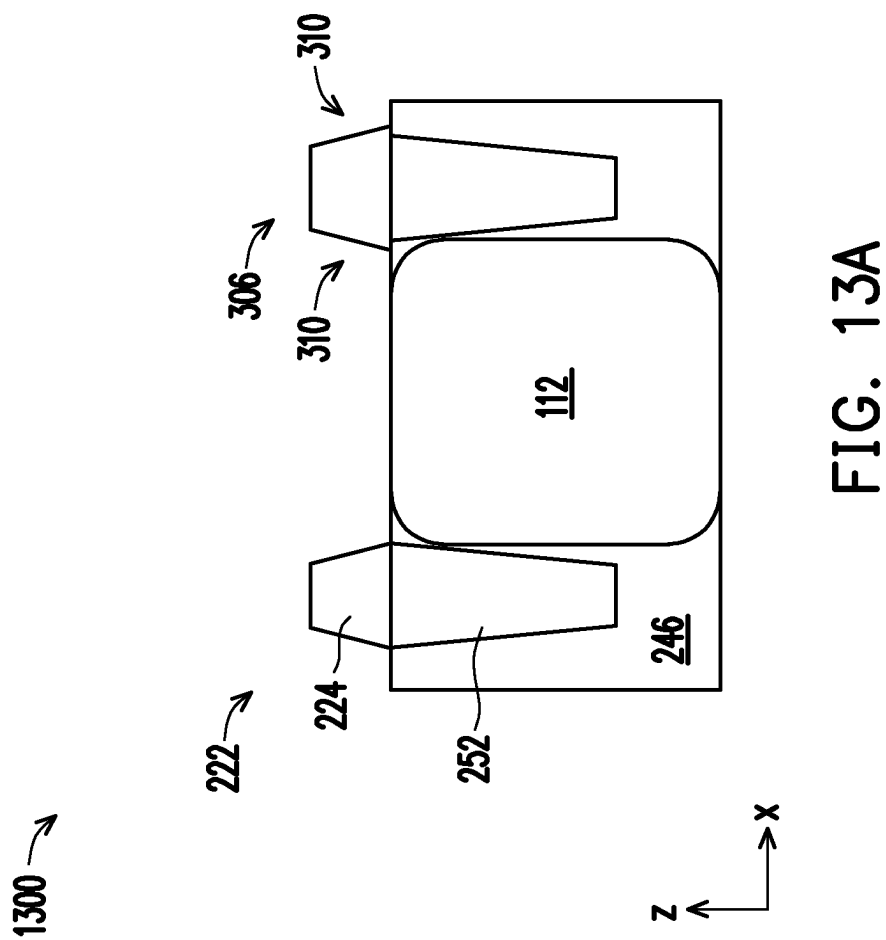


FIG. 12C



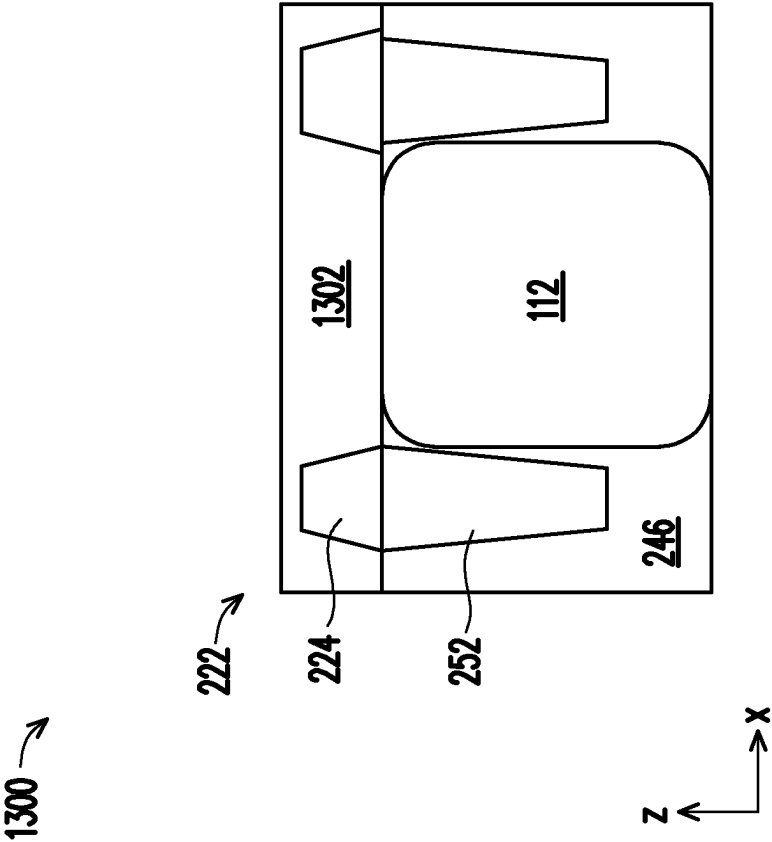


FIG. 13B

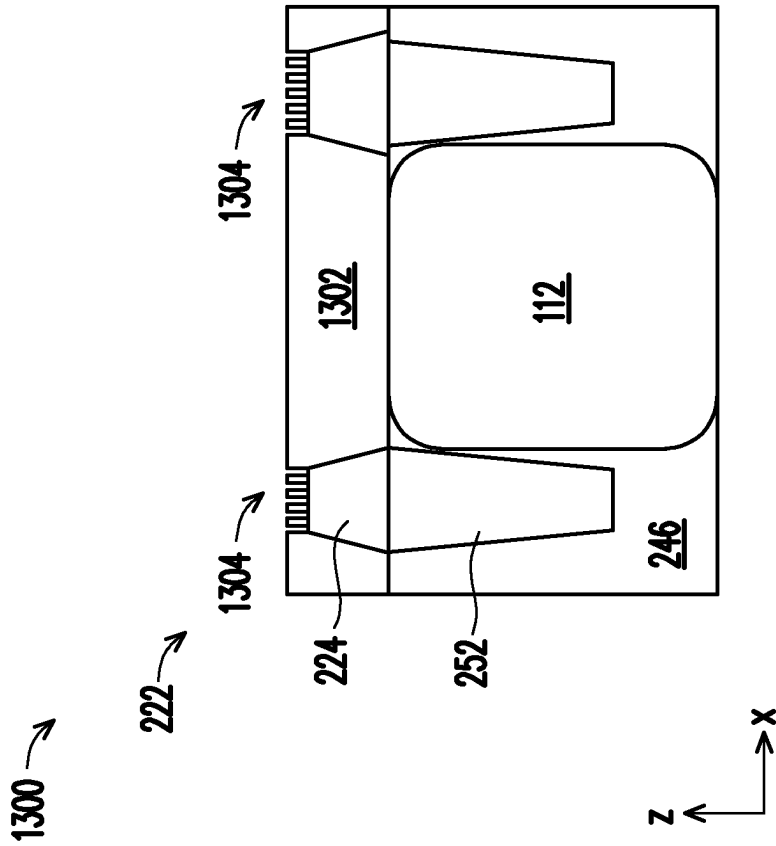


FIG. 13C

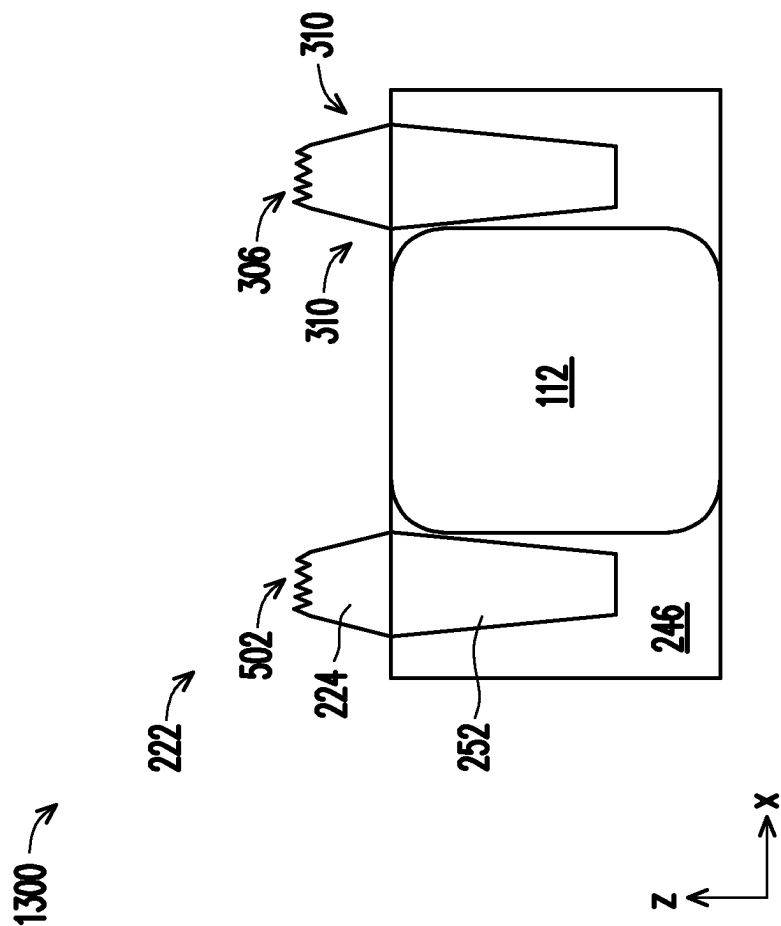


FIG. 13D

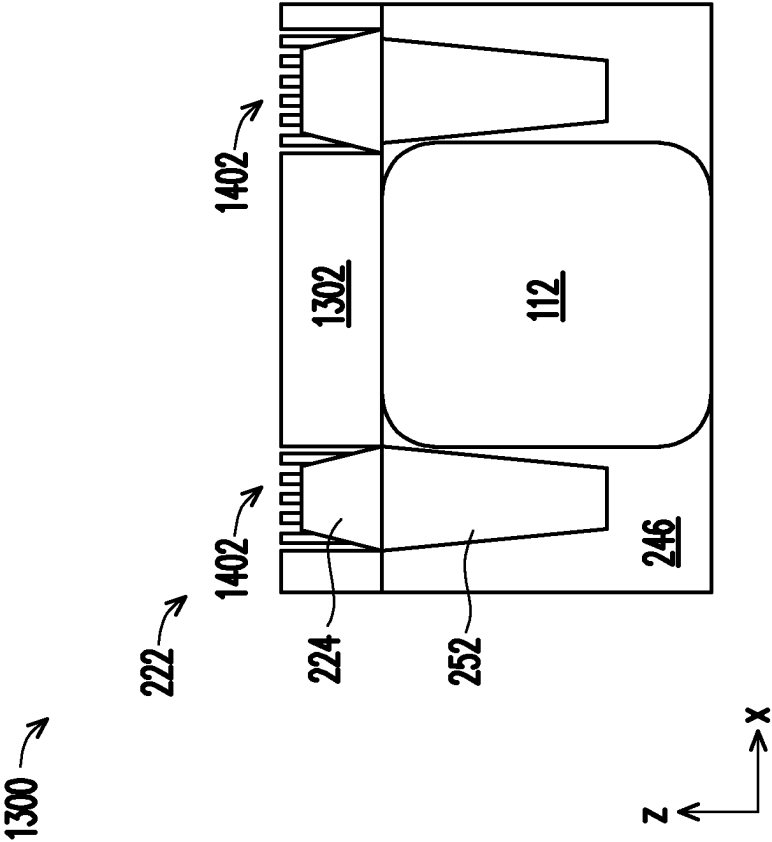


FIG. 14A

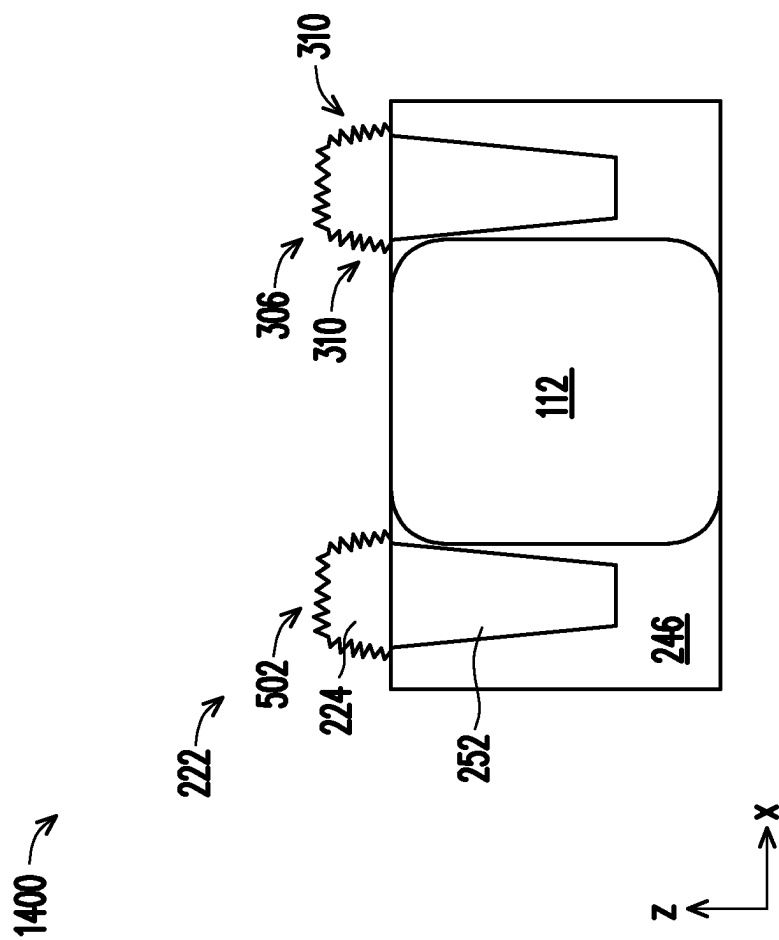


FIG. 14B

1500 →

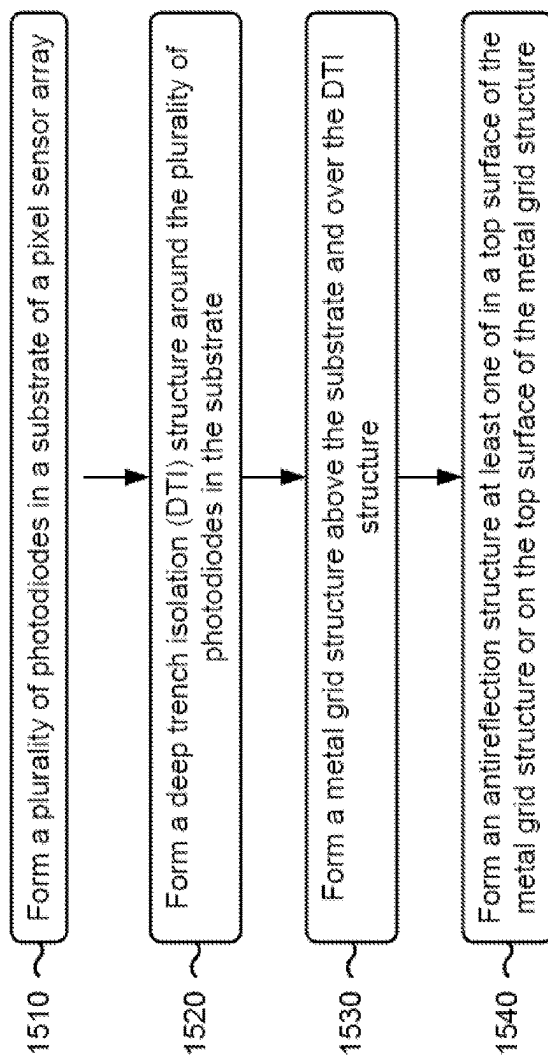


FIG. 15

PIXEL SENSOR ARRAYS AND METHODS OF FORMATION

BACKGROUND

[0001] A complementary metal oxide semiconductor (CMOS) image sensor may include a plurality of pixel sensors. A pixel sensor of the CMOS image sensor may include a transfer gate transistor, which may include a photodiode configured to convert photons of incident light into a photocurrent of electrons and a transfer gate configured to control the flow of the photocurrent between the photodiode and a drain region. The drain region may be configured to receive the photocurrent such that the photocurrent can be measured and/or transferred to other areas of the CMOS image sensor.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a diagram of an example of a pixel sensor described herein.

[0004] FIGS. 2A-2C are diagrams of examples of an image sensor device described herein.

[0005] FIGS. 3A-3D are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0006] FIGS. 4A-4D are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0007] FIGS. 5A-5J are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0008] FIGS. 6A-6E are diagrams of an example implementation of forming a circuitry die (or a portion thereof) described herein.

[0009] FIGS. 7A-7F are diagrams of an example implementation of forming a sensor die (or a portion thereof) described herein.

[0010] FIGS. 8A-8F are diagrams of an example implementation of forming an image sensor device (or a portion thereof) described herein.

[0011] FIGS. 9A-9F are diagrams of an example implementation of forming nanostructures on a metal grid structure of a pixel sensor array described herein.

[0012] FIGS. 10A-10D are diagrams of an example implementation of forming nanostructures on a metal grid structure of a pixel sensor array described herein.

[0013] FIGS. 11A-11E are diagrams of an example implementation of forming antireflective layers on a metal grid structure of a pixel sensor array described herein.

[0014] FIGS. 12A-12C are diagrams of an example implementation of forming antireflective layers on a metal grid structure of a pixel sensor array described herein.

[0015] FIGS. 13A-13D are diagrams of an example implementation of forming cavities in a metal grid structure of a pixel sensor array described herein.

[0016] FIGS. 14A and 14B are diagrams of an example implementation of forming cavities in a metal grid structure of a pixel sensor array described herein.

[0017] FIG. 15 is a flowchart of an example process associated with forming a metal grid structure of a pixel sensor array.

DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0019] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0020] An image sensor device (e.g., a complementary metal oxide semiconductor (CMOS) image sensor device or another type of image sensor device) is a type of electronic semiconductor device that uses pixel sensors to generate a photocurrent based on light received at the pixel sensors. The magnitude of the photocurrent may be based on the intensity of the light, based on the wavelength of the light, and/or based on another attribute of the light. The photocurrent is then processed to generate an electronic image, an electronic video, and/or another type of electronic signal.

[0021] Optical crosstalk can occur between adjacent pixel sensors in a pixel sensor array of an image sensor device. Optical crosstalk is a phenomenon whereby incident light passes through a pixel sensor at a non-orthogonal angle and is at least partially absorbed by a photodiode of an adjacent pixel sensor. Optical crosstalk can degrade the spatial resolution of the image sensor device, can reduce overall sensitivity of the image sensor device, can cause color mixing between pixel sensors of the image sensor device, and/or can lead to image noise after color correction.

[0022] In some cases, one or more structures are included in a pixel sensor array to reduce and/or minimize optical crosstalk between pixel sensors of the pixel sensor array. For example, a deep trench isolation (DTI) structure may be formed in a substrate such that the DTI structure surrounds the photodiodes of the pixel sensors in the pixel sensor array. As another example, a metal grid structure may be formed above the substrate, and may surround the pixel sensors in a similar manner as the DTI structure. The metal grid structure may be formed of one or more metal materials to facilitate reflection of incident light toward the photodiodes

of the pixel sensors, thereby improving the quantum efficiency (QE) of the pixel array. However, some incident light may be reflected away from a pixel sensor and toward another pixel sensor, resulting in the incident light passing through multiple different types of color filters. This may result in optical crosstalk and associated color mixing between pixel sensors.

[0023] As described herein, an image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

[0024] FIG. 1 is a diagram of an example of a pixel sensor 100 described herein. The pixel sensor 100 may include a front side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a front side of a sensor die), a back side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a back side of a sensor die), and/or another type of pixel sensor. The pixel sensor 100 may be electrically connected to a supply voltage (V_{dd}) 102 and an electrical ground 104.

[0025] The pixel sensor 100 includes a sensing region 106 that may be configured to sense and/or accumulate incident light (e.g., light directed toward the pixel sensor 100). The pixel sensor 100 also includes a control circuitry region 108. The control circuitry region 108 is electrically connected with the sensing region 106 and is configured to receive a photocurrent 110 that is generated by the sensing region 106. Moreover, the control circuitry region 108 is configured to transfer the photocurrent 110 from the sensing region 106 to downstream circuits such as amplifiers or analog-to-digital (AD) converters, among other examples.

[0026] The sensing region 106 includes a photodiode 112. The photodiode 112 may absorb and accumulate photons of the incident light, and may generate the photocurrent 110 based on absorbed photons. The magnitude of the photocurrent 110 is based on the amount of light collected in the photodiode 112. Thus, the accumulation of photons in the photodiode 112 generates a build-up of electrical charge that represents the intensity or brightness of the incident light (e.g., a greater amount of charge may correspond to a greater intensity or brightness, and a lower amount of charge may correspond to a lower intensity or brightness).

[0027] The photodiode 112 is electrically connected with a source of a transfer gate 114 in the control circuitry region 108. The transfer gate 114 is configured to control the transfer of the photocurrent 110 from the photodiode 112. The photocurrent 110 is provided from the source of the transfer gate 114 to a drain of the transfer gate 114 based on selectively switching a gate of the transfer gate 114. The gate of the transfer gate 114 may be selectively switched by applying a transfer voltage (V_{tx}) 116 to the transfer gate 114. In some implementations, the transfer voltage 116 being applied to the transfer gate 114 causes a conductive channel to form between the source and the drain of the transfer gate 114, which enables the photocurrent 110 to traverse along the conductive channel from the source to the drain. In some

implementations, the transfer voltage 116 being removed from the transfer gate 114 (or the absence of the transfer voltage 116) causes the conductive channel to be removed such that the photocurrent 110 cannot pass from the source to the drain.

[0028] The control circuitry region 108 further includes a reset gate 118. The reset gate 118 is electrically connected to the supply voltage 102. The reset gate 118 may be controlled by a reset voltage (V_{rst}) 120. The transfer gate 114 and the reset gate 118 may be electrically coupled with a floating diffusion node 122. The reset voltage 120 may be applied to the reset gate 118 to pull the drain of the transfer gate 114 to a high voltage (e.g., to the supply voltage 102) to “reset” the floating diffusion node 122 (e.g., by draining any residual charge in the floating diffusion node 122) prior to activation of the transfer gate 114 to transfer the photocurrent 110 from the photodiode 112 to the floating diffusion node 122.

[0029] The photocurrent 110 may be used to apply a floating diffusion voltage (V_{fd}) to a source follower gate 124 of the control circuitry region 108. This permits the photocurrent 110 to be observed without removing or discharging the photocurrent 110 from the floating diffusion node 122. The reset gate 118 may instead be used to remove or discharge the photocurrent 110 from the floating diffusion node 122.

[0030] The source follower gate 124 functions as a high impedance amplifier for the pixel sensor 100. The source follower gate 124 provides a voltage to current conversion of the floating diffusion voltage. The output of the source follower gate 124 is electrically connected with a row select gate 126, which is configured to control the flow of the photocurrent 110 to external circuitry. The row select gate 126 is controlled by selectively applying a select voltage (V_{di}) 128 to the gate of the row select gate 126. This permits the photocurrent 110 to flow to an output 130 of the pixel sensor 100.

[0031] As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.

[0032] FIGS. 2A-2C are diagrams of examples 200 of an image sensor device described herein. As shown in FIG. 2A, an image sensor device may be formed by bonding a circuitry wafer 202 and a sensor wafer 204. For example, a bonding tool may be used to perform a bonding operation to bond the circuitry wafer 202 and the sensor wafer 204 using a metal-to-metal bonding technique, a dielectric-to-dielectric bonding technique, and/or another bonding technique. In the bonding operation, circuitry dies 206 on the circuitry wafer 202 are bonded with associated sensor dies 208 on the sensor wafer 204 to image sensor devices 210. The image sensor devices 210 are then diced and packaged. Other processing steps may be performed to form the image sensor devices 210.

[0033] Each image sensor device 210 includes a circuitry die 206 and a sensor die 208. The circuitry die 206 and the sensor die 208 may be stacked or vertically arranged in the image sensor device 210. The sensor die 208 includes a pixel sensor array that includes a plurality of pixel sensors 100, or portions of a plurality of pixel sensors 100. In particular, the pixel sensor array includes at least the sensing regions 106 (and thus, the photodiodes 112) of the pixel sensors 100.

Accordingly, the sensor die **208** primarily is configured to sense photons of incident light and convert the photons to a photocurrent **110**.

[0034] The circuitry die **206** includes circuitry that is configured to measure, manipulate, and/or otherwise use the photocurrent **110**. Moreover, the circuitry die **206** includes at least a subset of the transistors of the control circuitry regions **108** of the pixel sensors **100**. For example, the circuitry die **206** may include the row select gates **126** of the pixel sensors **100**, the source follower gates **124** of the pixel sensor, and/or a combination thereof. This provides increased area on the sensor die **208** for the photodiodes **112**, which enables the size of the photodiodes **112** to be increased to increase the sensitivity and/or overall performance of the light sensing performance of the pixel sensor, and/or enables the size of the pixel sensors **100** to be decreased while maintaining the same size for the photodiodes **112**.

[0035] As further shown in FIG. 2A, the circuitry die **206** may include a device layer **212** and an interconnect layer **214**. The device layer **212** may include the devices (e.g., transistors) of the circuitry die **206**, and the interconnect layer **214** may include interconnects that enable signals and/or power to be provided to and/or from the devices in the device layer **212**. The sensor die **208** may also include a device layer **216** and an interconnect layer **218**. The device layer **216** may include portions of the pixel sensors **100**, including the photodiodes **112**, the transfer gates **114**, and the floating diffusion nodes **122**, among other examples. The interconnect layer **218** may include interconnects that enable signals and/or power to be provided to and/or from the device layer **216**.

[0036] The circuitry die **206** and the sensor die **208** may be bonded at a bonding interface **220**, which may be included between the interconnect layers **214** and **218**, and/or may be included in a portion of the interconnect layers **214** and/or **218**. The bonding interface **220** may include bonding pads, bonding vias, bonding dielectric layers, and/or other bonding structures.

[0037] FIG. 2B is a top-down view of an example pixel sensor array **222** included on a sensor die **208**. The pixel sensor array **222** may be included on a sensor die **208** of an image sensor device **210**. As shown in FIG. 2B, the pixel sensor array **222** may include a plurality of pixel sensors **100** (or portions of the plurality of plurality of pixel sensors **100**). For example, the pixel sensor array **222** may include the photodiodes **112** of the pixel sensors **100**. As further shown in FIG. 2B, the pixel sensors **100** may be arranged in a grid. In some implementations, the pixel sensors **100** are square-shaped (as shown in the example in FIG. 2B). In some implementations, the pixel sensors **100** include other shapes such as rectangle shapes, circle shapes, octagon shapes, diamond shapes, and/or other shapes.

[0038] In some implementations, the size of the pixel sensors **100** (e.g., the width or the diameter) of the pixel sensors **100** is approximately 1 micron. In some implementations, the size of the pixel sensors **100** (e.g., the width or the diameter) of the pixel sensors **100** is less than approximately 1 micron. For example, a width of one or more of the pixel sensors **100** may be included in a range of approximately 0.6 microns to approximately 0.7 microns. In these examples, the pixel sensors **100** may be referred to as sub-micron pixel sensors. Sub-micron pixel sensors may decrease the pixel sensor pitch (e.g., the distance between

adjacent pixel sensors) in the pixel sensor array **222**, which may enable increased pixel sensor density in the pixel sensor array **222** (which can increase the performance of the pixel sensor array **222**). However, other values for the range of the size of the pixel sensors **100** are within the scope of the present disclosure.

[0039] Each pixel sensor **100** may be configured to sense a particular wavelength range of incident light associated with a particular color component of the incident light. For example, a pixel sensor **100** may be configured to sense a wavelength range associated with a red component of incident light, and may therefore be referred to as a red pixel sensor. As another example, a pixel sensor **100** may be configured to sense a wavelength range associated with a blue component of incident light, and may therefore be referred to as a blue pixel sensor. As another example, a pixel sensor **100** may be configured to sense a wavelength range associated with a green component of incident light, and may therefore be referred to as a green pixel sensor. In some implementations, a plurality of pixel sensors **100** are configured to sense a wavelength range associated with a near infrared (NIR) component of incident light, and may therefore be referred to as NIR pixel sensors. The NIR pixel sensors may be included in the pixel sensor array **222** to improve low-light performance of the image sensor device **210** and/or to enable night-vision functionality to be realized for the image sensor device **210**.

[0040] As further shown in FIG. 2B, the photodiodes **112** of the pixel sensors **100** may be electrically and optically isolated by a metal grid structure **224** included in the pixel sensor array **222**. The photodiodes **112** may be formed in a substrate of the sensor die **208**, and the metal grid structure **224** may be included above the substrate. The metal grid structure **224** includes a plurality of intersecting metal lines around the perimeters of the pixel sensors **100**. The metal grid structure **224** may be formed of tungsten (W) and/or another suitable metal or metal alloy. The metal grid structure **224** may be included in the pixel sensor array **222** to reduce optical cross-talk between the pixel sensors **100**, which reduces color mixing between the pixel sensors **100**.

[0041] FIG. 2C illustrates a cross-section view of an image sensor device **210**. As shown in FIG. 2C, a circuitry die **206** and a sensor die **208** may be bonded at a bonding interface **220** such that the circuitry die **206** and the sensor die **208** are stacked or vertically arranged in a z-direction in the image sensor device **210**. As further shown in FIG. 2C, the image sensor device **210** includes the pixel sensor array **222** (e.g., including the pixel sensors **100**), a black level region (BLC) region **226** adjacent to (e.g., horizontally adjacent to) the pixel sensor array **222**, a bonding pad region **228** adjacent to (e.g., horizontally adjacent to) the BLC region **226**, and a seal ring region **230** adjacent to (e.g., horizontally adjacent to) the bonding pad region **228**, among other examples.

[0042] As further shown in FIG. 2C, the image sensor device **210** includes a plurality of layers, such as the device layer **212** and the interconnect layer **214** of the circuitry die **206**, and the device layer **216** and the interconnect layer **218** of the sensor die **208**. The device layer **212** of the circuitry die **206** includes a substrate **232** and a dielectric layer **234** above the substrate **232**. The substrate **232** may include silicon (Si) (e.g., a silicon substrate), a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), a silicon on insulator (SOI), or

another type of semiconductor material. The substrate **232** may include a semiconductor layer such as a silicon layer. The dielectric layer **234** may include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0043] Devices **236** may be included in and/or on the substrate **232** of the device layer **212**. The devices **236** may include one or more application-specific integrated circuit (ASIC) devices, one or more system-on-chip (SOC) devices, one or more transistors, and/or one or more other components configured to measure the magnitude of a photocurrent **110** generated by the pixel sensors **100** to determine light intensity of incident light and/or to generate images and/or video (e.g., digital images, digital video).

[0044] The interconnect layer **214** of the circuitry die **206** may include a dielectric layer **238**, a bonding layer **240**, a plurality of interconnect structures **242** in the dielectric layer **238**, and a plurality of bonding structures **244** in the bonding layer **240**. The dielectric layer **238** may include one or more interlayer dielectric (ILD) layers, one or more intermetal dielectric (IMD) layers, and/or one or more etch stop layers (ESLs), among other examples. The dielectric layer **238** and the bonding layer **240** may each include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0045] The interconnect structures **242** may each include conductive lines, trenches, vias, interconnects, metallization layers, and/or other types of electrically conductive structures that electrically connect the devices **236** to one or more other regions of the circuitry die **206** and/or to one or more regions of the sensor die **208**, among other examples. The bonding structures **244** may each include bonding pads, bonding vias, and/or other types of bonding structures. The interconnect structures **242** and the bonding structures **244** may each include one or more electrically conductive materials, such as, an electrically conductive metal, an electrically conductive metal alloy, an electrically conductive ceramic, tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), and/or gold (Au), among other examples of electrically conductive materials.

[0046] The device layer **216** of the sensor die **208** includes a substrate **246** and a dielectric layer **248** below the substrate **246**. The substrate **246** may include silicon (Si) (e.g., a silicon substrate), a silicon layer or another type of semiconductor layer, a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), an SOI, or another type of semiconductor material. The dielectric layer **248** may include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0047] The photodiodes **112** of the pixel sensors **100** are included in the substrate **246** of the sensor die **208**. The photodiodes **112** may each include one or more doped regions of substrate **246**. The substrate **246** may be doped with a plurality of types of ions to form a p-n junction or a

PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion) corresponding to a photodiode **112**. For example, the substrate **246** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **112** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **112**. A photodiode **112** may be configured to absorb photons of incident light. The absorption of photons causes the photodiode **112** to accumulate a charge (a photocurrent **110**) due to the photoelectric effect. Here, photons bombard the photodiode **112**, which causes emission of electrons of the photodiode **112**. The emission of electrons causes the formation of electron-hole pairs, where the electrons migrate toward the cathode of the photodiode **112** and the holes migrate toward the anode, which produces the photocurrent **110**.

[0048] The photodiodes **112** may be electrically isolated and/or optically isolated from one another by one or more isolation structures in the substrate **246**. Shallow trench isolation (STI) structures **250** extend into the substrate **246** from a bottom side of the substrate **246** (referred to as the front side of the substrate **246**), and a deep trench isolation (DTI) structure **252** extends into the substrate **246** from a top side of the substrate **246** (referred to as the back side of the substrate **246**) over the STI structures **250**. The combination of the STI structures **250** and the DTI structure **252** in the substrate **246** surround the pixel sensors **100** in the substrate **246** and provide the electrically isolation and/or optically isolation for the pixel sensors **100** in the substrate **246**.

[0049] The STI structures **250** may include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), and/or a silicon oxynitride (SiON), among other examples. The DTI structure **252** may include elongated structures of dielectric material **254** and a dielectric liner **256** between the dielectric material **254** and the substrate **246**. The DTI structure **252** extends along the sides of the photodiodes **112** and conforms to the top view shape of the metal grid structure **224** illustrated in FIG. 2B. The dielectric material **254** may also be included on the top side of the substrate **246** as a buffer layer. The dielectric liner **256** may be included on sidewalls and on a bottom surface of the DTI structure **252**, and may be included as an antireflective coating (ARC) and/or to further facilitate electrical and/or optical isolation of the pixel sensors **100**. In some implementations, the dielectric material **254** includes a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material. In some implementations, the dielectric liner **256** may include a high-k dielectric material such as a silicon nitride (Si_xN_y), a hafnium oxide (HfO_x), and/or another high-k dielectric material.

[0050] Transfer gates **114** of the pixel sensors **100** are included in the dielectric layer **248** and on the bottom side of the substrate **246**. The transfer gates **114** are electrically connected to the interconnect layer **218**, which enables inputs (e.g., gate voltages) to be provided to the transfer gates **114** to control the flow of photocurrents **110** from the photodiodes **112** to floating diffusion nodes **122** (not shown) of pixel sensors **100**.

[0051] The interconnect layer 218 may include a dielectric layer 258, a bonding layer 260, a plurality of interconnect structures 262 in the dielectric layer 258, and a plurality of bonding structures 264 in the bonding layer 260. The dielectric layer 258 may include one or more ILD layers, one or more IMD layers, and/or one or more ESLs, among other examples. The dielectric layer 258 and the bonding layer 260 may each include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0052] The interconnect structures 262 may each include conductive lines, trenches, vias, interconnects, metallization layers, and/or other types of electrically conductive structures that electrically connect the transfer gates 114 to one or more other regions of the sensor die 208 and/or to one or more regions of the circuitry die 206, among other examples. The bonding structures 264 may each include bonding pads, bonding vias, and/or other types of bonding structures. The interconnect structures 262 and the bonding structures 264 may each include one or more electrically conductive materials, such as, an electrically conductive metal, an electrically conductive metal alloy, an electrically conductive ceramic, tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), and/or gold (Au), among other examples of electrically conductive materials.

[0053] At the bonding interface 220, the bonding layers 240 and 260 may be bonded together (e.g., in a dielectric-to-dielectric bond), and the bonding structures 244 and 264 may be bonded together (e.g., in a metal-to-metal bond). Signals and/or power may be provided between the circuitry die 206 and the sensor die 208 through the bonding structures 244 and 264.

[0054] Above the top side of the substrate 246, a passivation layer 266 may be included on the buffer layer, and the metal grid structure 224 may be included above the passivation layer 266. The passivation layer 266 may include an oxide material such as a silicon oxide (SiOx). Additionally and/or alternatively, a silicon nitride (SiNx), a silicon carbide (SiC_x), or a mixture thereof, such as a silicon carbon nitride (SiCN), a silicon oxynitride (SiON), or another dielectric material is used for the passivation layer 266.

[0055] As shown in FIG. 2C, sections of the metal grid structure 224 may have an approximately trapezoidal cross-section shape or profile. Additionally and/or alternatively, sections of the metal grid structure 224 may have an approximately square-shaped cross-sectional profile. The sections of the metal grid structure 224 may be located over the DTI structure 252 and may be formed around the perimeter of the photodiodes 112 of the pixel sensors 100. Openings in the metal grid structure 224 are included above the photodiodes 112 to enable incident light to pass through the metal grid structure 224 and to the photodiodes 112. The metal grid structure 224 may be formed of a metal material, such as gold (Au), copper (Cu), silver (Ag), cobalt (Co), tungsten (W), titanium (Ti), ruthenium (Ru), a metal alloy (e.g., aluminum copper (AlCu)), and/or a combination thereof, among other examples.

[0056] Color filter regions 268 of the pixel sensors 100 be included in the openings in the metal grid structure 224. The color filter regions 268 may be included above the photo-

diodes 112 of the pixel sensors 100. The color filter regions 268 may be included above the photodiodes 112. Each color filter region 268 may be configured to filter incident light to allow a particular wavelength of the incident light to pass to a photodiode 112. For example, a color filter region 268 may filter incident light to allow red light to pass through the color filter region 268 to an associated photodiode 112. As another example, a color filter region 268 may filter incident light to allow green light to pass through the color filter region 268 to an associated photodiode 112. As another example, a color filter region 268 may filter incident light to allow blue light to pass through the color filter region 268 to an associated photodiode 112.

[0057] A blue color filter region 268 may permit the component of incident light near a 450 nanometer wavelength to pass through and may block other wavelengths from passing. A green color filter region 268 may permit the component of incident light near a 550 nanometer wavelength to pass and may block other wavelengths from passing. A red color filter region 268 may permit the component of incident light near a 650 nanometer wavelength to pass and may block other wavelengths from passing. A yellow color filter region 268 may permit the component of incident light near a 580 nanometer wavelength to pass and may block other wavelengths from passing.

[0058] In some implementations, a color filter region 268 may be non-discriminating or non-filtering, which may define a white pixel sensor. A non-discriminating or non-filtering color filter region 268 may include a material that permits all wavelengths of light to pass into the associated photodiode 112 (e.g., for purposes of determining overall brightness to increase light sensitivity for the image sensor). In some implementations, a color filter region 268 may be a NIR bandpass color filter region 268, which may define an NIR pixel sensor. An NIR bandpass color filter region 268 may include a material that permits the portion of incident light in an NIR wavelength range to pass to an associated photodiode 112 while blocking visible light from passing.

[0059] Micro-lenses 270 may be included over and/or on the color filter regions 268. The micro-lenses 270 may include a respective micro-lens for each of the pixel sensors 100. A micro-lens may be formed to focus incident light toward a photodiode 112 of an associated pixel sensor 100.

[0060] As further shown in FIG. 2C, a metal layer 272 may be included above the substrate 246 in the BLC region 226 of the substrate 246. The metal layer 272 may be included as a light-blocking layer to prevent incident light from entering the portion of substrate 246 in the BLC region 226. The portion of substrate 246 in the BLC region 226 is thus a sensing region that is kept “dark” so that dark current measurements may be performed in the BLC region 226. A dark current measurement may be performed to measure the amount of charge (dark current) in the substrate 246 that is generated from sources other than incident light (e.g., from thermal energy in the substrate 246) so that the dark current measurement may be used for black level correction (or black level calibration) for the pixel sensor array 222.

[0061] As further shown in FIG. 2C, the bonding pad region 228 may include a plurality of dielectric layers 274, 276, 278, 280, and 282 that electrically isolate a bonding pad structure 284. The bonding pad structure 284 is electrically coupled and/or physically coupled with one or more of the interconnect structures 262 in the interconnect layer 218 of

the sensor die 208. A bonding pad opening 286 is included above the bonding pad structure 284 to enable an external electrical connection to be formed to the bonding pad structure 284.

[0062] The plurality of dielectric layers 274, 276, 278, 280, and 282 may each include one or more dielectric materials, such as a silicon oxide (SiO_x), a silicon nitride (Si₃N₄), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples. The bonding pad structure 284 may include a metal material, such as gold (Au), copper (Cu), silver (Ag), cobalt (Co), tungsten (W), titanium (Ti), ruthenium (Ru), a metal alloy (e.g., aluminum copper (AlCu)), and/or a combination thereof, among other examples.

[0063] The seal ring region 230 includes a plurality of stacked interconnect structures 242 in the interconnect layer 214 and a plurality of stacked interconnect structures 262 in the interconnect layer 218 to seal the structures and layers of the image sensor device 210 to prevent ingress of humidity and other contaminants, as well as to provide structural rigidity to the image sensor device 210.

[0064] As indicated above, FIGS. 2A-2C are provided as examples. Other examples may differ from what is described with regard to FIGS. 2A-2C.

[0065] FIGS. 3A-3D are diagrams of examples of pixel sensor arrays 222 of a sensor die 208 described herein. FIGS. 3A-3D illustrate cross-section views of the examples of the pixel sensor arrays 222. The examples illustrated in FIGS. 3A-3D each include an example implementation of an antireflection structure for the metal grid structure 224 of the pixel sensor arrays 222. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array 222 to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor 100 toward another pixel sensor 100. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors 100 of the associated pixel sensor array 222. One or more of the examples of the pixel sensor arrays 222 of FIGS. 3A-3D may be included in a sensor die 208 of an image sensor device 210 illustrated and described in connection with FIGS. 2A-2C.

[0066] As shown in FIG. 3A, an example 300 of a pixel sensor array 222 includes a plurality of pixel sensors 100. The pixel sensors 100 each include a photodiode 112 in the substrate 246 of the sensor die 208. A DTI structure 252 is included around the photodiodes 112 in the substrate 246, and a metal grid structure 224 is included on the substrate 246 above the DTI structure 252. The metal grid structure 224 extends above the substrate 246 and surrounds the photodiodes 112. Color filter regions 268 are included between the openings of the metal grid structure 224, and micro-lenses 270 are included on the color filter regions. A passivation layer 302 may additionally be included on the metal grid structure 224, or may be omitted.

[0067] As further shown in the example 300, nanostructures 304 are included on top surfaces 306 of the metal grid structure 224. The nanostructures 304 may include nanotubes, nanowires, and/or another type of nanostructures that extend from the top surfaces 306 of the metal grid structure 224. The nanostructures 304 may be formed of carbon (C)

and/or another suitable material. The nanostructures 304 may reduce optical crosstalk in the pixel sensor array 222 in that the nanostructures 304 may absorb photons of incident light that might otherwise reflect off of the top surfaces 306 of the metal grid structure 224.

[0068] A length of the nanostructures 304 (indicated in FIG. 3A as dimension D1) may be included in a range of approximately 80 nanometers to approximately 1000 nanometers. If the length of the nanostructures 304 is less than approximately 80 nanometers, the nanostructures 304 may not provide sufficient optical absorption to reduce or prevent optical crosstalk, whereas the nanostructures 304 may reduce absorption of photons in the photodiodes 112 (which may reduce the quantum efficiency of the pixel sensors 100) if the length of the nanostructures 304 is greater than approximately 1000 nanometers. If the length of the nanostructures 304 is included in the range of approximately 80 nanometers to approximately 1000 nanometers, the nanostructures 304 may provide sufficient optical absorption to reduce or prevent optical crosstalk while enabling a high quantum efficiency to be achieved for the pixel sensors 100. However, other values for the length of the nanostructures 304, and ranges other than approximately 80 nanometers to approximately 1000 nanometers, are within the scope of the present disclosure.

[0069] The nanostructures 304 may extend in a direction that is approximately perpendicular to the top surfaces 306 of the metal grid structure 224. Additionally and/or alternatively, at least a subset of the nanostructures 304 may extend in a direction that is angled (e.g., angled at approximately 60 degrees, angled at approximately 75 degrees) relative to the top surfaces 306 of the metal grid structure 224.

[0070] As further shown in FIG. 3A, the metal grid structures 224 in the example 300 of the pixel sensor array 222 may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface 306 of the metal grid structure 224 may be less than a cross-sectional width of a bottom surface of the metal grid structure 224. Thus, the cross-sectional width increases from the top surface 306 to the bottom surface for the metal grid structure 224.

[0071] FIG. 3B illustrates an example 308 of a pixel sensor array 222, which is similar to the example 300 of the pixel sensor array 222 except that the example 308 of the pixel sensor array 222 further includes nanostructures 304 on sidewalls 310 of the metal grid structure 224. Thus, the nanostructures 304 are included on the top surfaces 306 and on the sidewalls 310 of the metal grid structure 224. The nanostructures 304 may provide further optical crosstalk reduction by absorbing photons of incident light that are directed toward the sidewalls 310 of the metal grid structure 224.

[0072] FIG. 3C illustrates an example 312 of a pixel sensor array 222, which is similar to the example 300 of the pixel sensor array 222 except that the metal grid structure 224 in the example 312 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224.

[0073] FIG. 3D illustrates an example 314 of a pixel sensor array 222, which is similar to the example 312 of the pixel sensor array 222 except that the example 314 of the pixel sensor array 222 further includes nanostructures 304

on the sidewalls 310 of the metal grid structure 224. Thus, the nanostructures 304 are included on the top surfaces 306 and on the sidewalls 310 of the metal grid structure 224, where the sidewalls 310 are approximately perpendicular to the top surfaces 306. The nanostructures 304 may provide further optical crosstalk reduction by absorbing photons of incident light that are directed toward the sidewalls 310 of the metal grid structure 224.

[0074] As indicated above, FIGS. 3A-3D is provided as examples. Other examples may differ from what is described with regard to FIGS. 3A-3D.

[0075] FIGS. 4A-4D are diagrams of examples of pixel sensor arrays 222 of a sensor die 208 described herein. FIGS. 4A-4D illustrate cross-section views of the examples of the pixel sensor arrays 222. The examples illustrated in FIGS. 4A-4D each include an example implementation of an antireflection structure for the metal grid structure 224 of the pixel sensor arrays 222. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array 222 to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor 100 toward another pixel sensor 100. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors 100 of the associated pixel sensor array 222. One or more of the examples of the pixel sensor arrays 222 of FIGS. 4A-4D may be included in a sensor die 208 of an image sensor device 210 illustrated and described in connection with FIGS. 2A-2C.

[0076] As shown in FIG. 4A, an example 400 of a pixel sensor array 222 includes a plurality of pixel sensors 100. The pixel sensors 100 each include a photodiode 112 in the substrate 246 of the sensor die 208. A DTI structure 252 is included around the photodiodes 112 in the substrate 246, and a metal grid structure 224 is included on the substrate 246 above the DTI structure 252. The metal grid structure 224 extends above the substrate 246 and surrounds the photodiodes 112. Color filter regions 268 are included between the openings of the metal grid structure 224, and micro-lenses 270 are included on the color filter regions. A passivation layer 302 may additionally be included on the metal grid structure 224, or may be omitted.

[0077] As further shown in the example 400, antireflective layers 402 are included on the top surfaces 306 of the metal grid structure 224. The antireflective layers 402 may each include a layer of antireflective material that absorbs photons of incident light. The layer of antireflective material may have one or more properties that, when included on the metal grid structure 224, result in cancelation of a broad range of wavelengths of incident light. Examples of such properties include refractive index, optical band gap, and/or thickness, among other examples. The properties of the layer of antireflective material may result in formation of destructive waves that are opposing and out of phase of the incident light, and these destructive waves cancel the incident light, which prevents or reduces the incident light from being reflected off of the metal grid structure 224. Thus, the antireflective material may be selected based on the material used for the metal grid structure 224. For example, the antireflective material may be selected such that the refractive index; optical band gap; and/or another property of the antireflective material, in combination with the refractive index; optical band gap; and/or another property of the metal

material of the metal grid structure, results in formation of destructive waves that cancel the incident light. Examples of such antireflective materials include zinc oxide (ZnO), aluminum-doped zinc oxide (Al: ZnO or AZO), and/or aluminum oxide (Al_xO_y , such as Al_2O_3), among other examples.

[0078] A thickness of the antireflective layers 402 (indicated in FIG. 4A as dimension D2) may be included in a range of approximately 50 nanometers to approximately 200 nanometers. If the thickness of the antireflective layers 402 is less than approximately 50 nanometers or greater than approximately 200 nanometers, the properties of the antireflective layers 402 may not combine with the properties of the metal material of the metal grid structure 224 to cancel out incident light to reduce or prevent optical crosstalk. If the thickness of the antireflective layers 402 is included in the range of approximately 50 nanometers to approximately 200 nanometers, the properties of the antireflective layers 402 may combine with the properties of the metal material of the metal grid structure 224 to cancel out incident light to reduce or prevent optical crosstalk. However, other values for the thickness of the antireflective layers 402, and ranges other than approximately 50 nanometers to approximately 200 nanometers, are within the scope of the present disclosure.

[0079] As further shown in FIG. 4A, the metal grid structures 224 in the example 400 of the pixel sensor array 222 may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface 306 of the metal grid structure 224 may be less than a cross-sectional width of a bottom surface of the metal grid structure 224. Thus, the cross-sectional width increases from the top surface 306 to the bottom surface for the metal grid structure 224.

[0080] FIG. 4B illustrates an example 404 of a pixel sensor array 222, which is similar to the example 400 of the pixel sensor array 222 except that the antireflective layers 402 are included on the sidewalls 310 of the metal grid structure 224 in the example 404 of the pixel sensor array 222. Thus, the antireflective layers 402 are included on the top surfaces 306 and on the sidewalls 310 of the metal grid structure 224. The antireflective layers 402 may provide further optical crosstalk reduction by canceling out reflections of incident light that might otherwise reflect off of the sidewalls 310 of the metal grid structure 224.

[0081] FIG. 4C illustrates an example 406 of a pixel sensor array 222, which is similar to the example 400 of the pixel sensor array 222 except that the metal grid structure 224 in the example 406 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224.

[0082] FIG. 4D illustrates an example 408 of a pixel sensor array 222, which is similar to the example 406 of the pixel sensor array 222 except that the antireflective layers 402 are included on the sidewalls 310 of the metal grid structure 224 in the example 404 of the pixel sensor array 222. Thus, the antireflective layers 402 are included on the top surfaces 306 and on the sidewalls 310 of the metal grid structure 224, where the sidewalls 310 are approximately perpendicular to the top surfaces 306. The antireflective layers 402 may provide further optical crosstalk reduction

by canceling out reflections of incident light that might otherwise reflect off of the sidewalls 310 of the metal grid structure 224.

[0083] As indicated above, FIGS. 4A-4D is provided as examples. Other examples may differ from what is described with regard to FIGS. 4A-4D.

[0084] FIGS. 5A-5J are diagrams of examples of pixel sensor arrays 222 of a sensor die 208 described herein. FIGS. 5A-5J illustrate cross-section views of the examples of the pixel sensor arrays 222. The examples illustrated in FIGS. 5A-5J each include an example implementation of an antireflection structure for the metal grid structure 224 of the pixel sensor arrays 222. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array 222 to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor 100 toward another pixel sensor 100. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors 100 of the associated pixel sensor array 222. One or more of the examples of the pixel sensor arrays 222 of FIGS. 5A-5J may be included in a sensor die 208 of an image sensor device 210 illustrated and described in connection with FIGS. 2A-2C.

[0085] As shown in FIG. 5A, an example 500 of a pixel sensor array 222 includes a plurality of pixel sensors 100. The pixel sensors 100 each include a photodiode 112 in the substrate 246 of the sensor die 208. A DTI structure 252 is included around the photodiodes 112 in the substrate 246, and a metal grid structure 224 is included on the substrate 246 above the DTI structure 252. The metal grid structure 224 extends above the substrate 246 and surrounds the photodiodes 112. Color filter regions 268 are included between the openings of the metal grid structure 224, and micro-lenses 270 are included on the color filter regions. A passivation layer 302 may additionally be included on the metal grid structure 224, or may be omitted.

[0086] As further shown in the example 500, cavities 502 are included in the top surfaces 306 of the metal grid structure 224 as the antireflection structures. The cavities 502 result in the top surfaces 306 of the metal grid structure 224 having a plurality of concave surfaces (or a plurality of recessed surfaces). The cavities 502 may be spaced apart from each other in the top surfaces 306 and may each have an approximately curved, rounded, and/or semi-circular cross-sectional profile. The curved or rounded surfaces of the cavities 502 scatter photons of incident light. Scattering the photons of incident light around a photodiode 112 causes the incident light to be directed in multiple directions off of the top surfaces 306 of the metal grid structure 224, thereby reducing the concentration of photons reflected toward other photodiodes 112, and thereby reducing optical crosstalk in the pixel sensor array 222.

[0087] The cavities 502 may have one or more dimensions, such as a depth (illustrated in FIG. 5A as dimension D3), a spacing between adjacent cavities 502 (illustrated in FIG. 5A as dimension D4), and/or a width (illustrated in FIG. 5A as dimension D5), among other examples. One or more dimensions of the cavities 502, the quantity of cavities 502, and/or another parameter of the cavities 502 may be selected to reduce, minimize, and/or prevent optical crosstalk from resulting from reflections of incident light off of the metal grid structure 224. The one or more dimensions of

the cavities 502, the quantity of cavities 502, and/or another parameter of the cavities 502 may be selected based on a size of the pixel sensors 100 (e.g., a width of the pixel sensors 100, based on a wavelength or wavelengths of incident light that is to be sensed by the pixel sensor array 222, spacing between adjacent pixel sensors 100, and/or another parameter).

[0088] In some implementations, the depth of a cavity 502 (dimension D3) is included in a range of approximately 1 nanometer to approximately 10 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the spacing between adjacent cavities 502 (dimension D4) is included in a range of approximately 1 nanometer to approximately 5 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the width of a cavity 502 (dimension D5) is included in a range of approximately 1 nanometer to approximately 50 nanometers. However, other values for the range are within the scope of the present disclosure.

[0089] As further shown in FIG. 5A, the metal grid structures 224 in the example 500 of the pixel sensor array 222 may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface 306 of the metal grid structure 224 may be less than a cross-sectional width of a bottom surface of the metal grid structure 224. Thus, the cross-sectional width increases from the top surface 306 to the bottom surface for the metal grid structure 224.

[0090] FIG. 5B illustrates an example 504 of a pixel sensor array 222, which is similar to the example 500 of the pixel sensor array 222 except that the cavities 502 have an approximately V-shaped (or triangular) cross-sectional profile in the example 504 of the pixel sensor array 222. The cavities 502 in the example 504 may have one or more dimensions, such as a depth (illustrated in FIG. 5B as dimension D6), a spacing between adjacent cavities 502 (illustrated in FIG. 5B as dimension D7), a sidewall angle relative to a bottom surface of the metal grid structure 224 (illustrated in FIG. 5B as dimension D8), and/or a width (illustrated in FIG. 5B as dimension D9), among other examples. One or more dimensions of the cavities 502, the quantity of cavities 502, and/or another parameter of the cavities 502 may be selected to reduce, minimize, and/or prevent optical crosstalk from resulting from reflections of incident light off of the metal grid structure 224. The one or more dimensions of the cavities 502, the quantity of cavities 502, and/or another parameter of the cavities 502 may be selected based on a size of the pixel sensors 100 (e.g., a width of the pixel sensors 100, based on a wavelength or wavelengths of incident light that is to be sensed by the pixel sensor array 222, spacing between adjacent pixel sensors 100, and/or another parameter).

[0091] In some implementations, the depth of a cavity 502 (dimension D6) is included in a range of approximately 1 nanometer to approximately 10 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the spacing between adjacent cavities 502 (dimension D7) is included in a range of approximately 1 nanometer to approximately 5 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the angle of the sidewalls of a cavity 502 (dimension D8) is included in a range of approximately 30 degrees to approxi-

mately 75 degrees. However, other values for the range are within the scope of the present disclosure. In some implementations, the width of a cavity 502 (dimension D9) is included in a range of approximately 1 nanometer to approximately 50 nanometers. However, other values for the range are within the scope of the present disclosure.

[0092] FIG. 5C illustrates an example 506 of a pixel sensor array 222, which is similar to the example 500 of the pixel sensor array 222 except that the cavities 502 having approximately curved, rounded, and/or semi-circular cross-sectional profiles are also included in the sidewalls 310 of the metal grid structure 224 in the example 504 of the pixel sensor array 222. Thus, the cavities 502 having approximately curved, rounded, and/or semi-circular cross-sectional profiles are included in the top surfaces 306 and the sidewalls 310 of the metal grid structure 224. The cavities 502 in the sidewalls 310 may provide further optical crosstalk reduction by scattering incident light directed to the sidewalls 310 of the metal grid structure 224.

[0093] FIG. 5D illustrates an example 508 of a pixel sensor array 222, which is similar to the example 504 of the pixel sensor array 222 except that the cavities 502 having approximately V-shaped (or triangular) cross-sectional profiles are also included in the sidewalls 310 of the metal grid structure 224 in the example 504 of the pixel sensor array 222. Thus, the cavities 502 having approximately V-shaped (or triangular) cross-sectional profiles are included in the top surfaces 306 and the sidewalls 310 of the metal grid structure 224. The cavities 502 in the sidewalls 310 may provide further optical crosstalk reduction by scattering incident light directed to the sidewalls 310 of the metal grid structure 224.

[0094] FIG. 5E illustrates an example 510 of a pixel sensor array 222, which is similar to the examples 506 and 508 of the pixel sensor arrays 222 except that the example 510 of the pixel sensor array 222 in FIG. 5E includes cavities 502 in the top surfaces 306 of the metal grid structure 224 and cavities 502 in the sidewalls 310 of the metal grid structure 224 that have different cross-sectional profiles. For example, the cavities 502 in the top surfaces 306 of the metal grid structure 224 may have approximately V-shaped (or triangular) cross-sectional profiles, and the cavities 502 in the sidewalls 310 of the metal grid structure 224 may have approximately curved, rounded, and/or semi-circular cross-sectional profiles (as illustrated in the example in FIG. 5E). As another example, the cavities 502 in the top surfaces 306 of the metal grid structure 224 may have approximately curved, rounded, and/or semi-circular cross-sectional profiles, and the cavities 502 in the sidewalls 310 of the metal grid structure 224 may have approximately V-shaped (or triangular) cross-sectional profiles. This enables greater flexibility in configuring the metal grid structure 224 for further optical crosstalk reduction for the pixel sensor array 222.

[0095] FIG. 5F illustrates an example 512 of a pixel sensor array 222, which is similar to the example 500 of the pixel sensor array 222 except that the metal grid structure 224 in the example 512 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224.

[0096] FIG. 5G illustrates an example 514 of a pixel sensor array 222, which is similar to the example 504 of the

pixel sensor array 222 except that the metal grid structure 224 in the example 514 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224.

[0097] FIG. 5H illustrates an example 516 of a pixel sensor array 222, which is similar to the example 506 of the pixel sensor array 222 except that the metal grid structure 224 in the example 516 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224, and the cavities 502 in the sidewalls 310 of the metal grid structure 224 may have approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0098] FIG. 5I illustrates an example 518 of a pixel sensor array 222, which is similar to the example 508 of the pixel sensor array 222 except that the metal grid structure 224 in the example 518 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224, and the cavities 502 in the sidewalls 310 of the metal grid structure 224 may have approximately V-shaped (or triangular) cross-sectional profiles.

[0099] FIG. 5J illustrates an example 520 of a pixel sensor array 222, which is similar to the example 510 of the pixel sensor array 222 except that the metal grid structure 224 in the example 520 of the pixel sensor array 222 has a square-shaped cross-section shape or profile. Thus, the sidewalls 310 of the metal grid structure 224 are approximately perpendicular to the top surfaces 306 of the metal grid structure 224. The cavities 502 in the sidewalls 310 may have approximately V-shaped (or triangular) cross-sectional profiles, or may have approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0100] As indicated above, FIGS. 5A-5J is provided as examples. Other examples may differ from what is described with regard to FIGS. 5A-5J.

[0101] FIGS. 6A-6E are diagrams of an example implementation 600 of forming a circuitry die 206 (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 6A-6E may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0102] Turning to FIG. 6A, the substrate 232 of the device layer 212 of the circuitry die 206 is provided. The substrate 232 may be provided in the form of a semiconductor wafer such as a silicon (Si) wafer may be provided as an SOI wafer, and/or another type of semiconductor work piece.

[0103] As shown in FIG. 6B, one or more devices 236 may be formed in and/or on the substrate 232. One or more semiconductor processing tools may be used to form one or more portions of the devices 236. For example, a deposition tool may be used to perform various deposition operations to deposit layers of the devices 236, and/or to deposit photoresist layers for etching the substrate 232 and/or portions of the deposited layers. As another example, an exposure tool may be used to expose the photoresist layers to form patterns

in the photoresist layers. As another example, a developer tool may develop the patterns in the photoresist layers. As another example, an etch tool may be used to etch the substrate **232** and/or portions of the deposited layers to form the devices **236**. As another example, a planarization tool may be used to planarize portions of the devices **236**. As another example, a plating tool may be used to deposit metal structures and/or layers of the devices **236**.

[0104] As further shown in FIG. 6B, a dielectric layer **234** may be deposited over and/or on the substrate **232** and over and/or on the devices **236**. A deposition tool may be used to deposit the dielectric layer **234** using a physical vapor deposition (PVD) technique, an atomic layer deposition (ALD) technique, a chemical vapor deposition (CVD) technique, an oxidation technique, another type of deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer **234** after the dielectric layer **234** is deposited.

[0105] As shown in FIG. 6C, a first portion of an interconnect layer **214** of the circuitry die **206** is formed above the device layer **212**. To form the first portion of the interconnect layer **214**, a deposition tool may be used to deposit a dielectric layer **238** (which may include one or more ILD layers, one or more IMD layers, one or more ESLs, and/or one or more of another type of dielectric layer) using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer **238** after the dielectric layer **238** is deposited.

[0106] A deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, and/or another semiconductor processing tool may be used to perform various operations to form interconnect structures **242** in the first portion of the interconnect layer **214**. A deposition tool and/or a plating tool may be used to deposit the interconnect structures **242** using a PVD technique, an ALD technique, a CVD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may be used to planarize the interconnect structures **242** after the interconnect structures **242** are deposited.

[0107] In some implementations, first portion of the interconnect layer **214** is built up in the z-direction in a plurality of via layers (V-layers) and metallization layers (M-layers). For example, a first portion of the dielectric layer **238** may be formed, recesses may be formed in the first portion of the dielectric layer **238**, and first interconnect structures **242** (e.g., a V0 via layer, an M0 metallization layer) may be formed in the recesses. A second portion of the dielectric layer **238** may be formed, recesses may be formed in the second portion of the dielectric layer **238**, and second interconnect structures **242** (e.g., a V1 via layer, an M1 metallization layer) may be formed in the recesses. The remaining via layers and/or metallization layers of the first portion of the interconnect layer **214** may be formed in a similar manner.

[0108] As shown in FIGS. 6D and 6E, a second portion of the interconnect layer **214** may be formed, and the second portion of the interconnect layer **214** may include a bonding layer **240** and bonding structures **244**. As shown in FIG. 6D, the bonding layer **240** may be formed over and/or on the dielectric layer **238**, and over and/or on the top-most interconnect structures **242**. A deposition tool may be used to

deposit the bonding layer **240** using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the bonding layer **240** after the bonding layer **240** is deposited.

[0109] As shown in FIG. 6E, the bonding structures **244** may be formed in the bonding layer **240**. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the bonding layer **240**. An etch tool may be used to etch the bonding layer **240** (e.g., using a wet etch technique, a dry etch technique) to form recesses in the bonding layer **240**. A deposition tool and/or a plating tool may be used to deposit the bonding structures **244** in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the bonding structures **244** after the bonding structures **244** are deposited.

[0110] As indicated above, FIGS. 6A-6E are provided as an example. Other examples may differ from what is described with regard to FIGS. 6A-6E.

[0111] FIGS. 7A-7F are diagrams of an example implementation **700** of forming a sensor die **208** (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 7A-7F may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0112] Turning to FIG. 7A, the substrate **246** of the device layer **216** of the sensor die **208** is provided. The substrate **246** may be provided in the form of a semiconductor wafer such as a silicon (Si) wafer may be provided as an SOI wafer, and/or another type of semiconductor work piece.

[0113] As shown in FIG. 7B, photodiodes **112** of pixel sensors **100** of a pixel sensor array **222** of the sensor die **208** may be formed in the substrate **246** in the device layer **216** of the sensor die **208**. In some implementations, an ion implantation tool may be used to implant ions into the substrate **246** to form a P-N junction between a p-doped region of the substrate **246** and an n-doped region of the substrate **246**, or to form a P-I-N junction between p-doped region of the substrate **246**, an n-doped region of the substrate **246**, and an intrinsic (e.g., undoped) semiconductor region for a photodiode **112**.

[0114] As further shown in FIG. 7B, STI structures **250** may be formed in the substrate **246** (e.g., from the front side of the substrate **246**) such that the STI structures **250** are located between the photodiodes **112**. In some implementations, the STI structures **250** are formed after the photodiodes **112** are formed. In some implementations, the STI structures **250** are formed prior to formation of the photodiodes **112**. A deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the substrate **246**. An etch tool may be used to etch into the substrate **246** from the front side of the substrate **246** (e.g., using a wet etch technique, a dry etch technique) to form the recesses in the front side of the substrate **246**. A deposition tool may be used to deposit the STI structures **250** in the recesses using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another

deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the STI structures 250 after the STI structures 250 are deposited.

[0115] As shown in FIG. 7C, transfer gates 114 of the pixel sensors 100 may be formed over and/or on the front side surface of the substrate 246. Forming a transfer gate 114 may include depositing a gate dielectric on the front side surface of the substrate 246, depositing a gate electrode on the gate dielectric layer, and/or forming sidewall spacers on sidewalls of the gate electrode, among other examples.

[0116] As further shown in FIG. 7C, a dielectric layer 248 may be formed over and/or on the front side of the substrate 246, and over and/or on the transfer gates 114. A deposition tool may be used to deposit the dielectric layer 248 using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the dielectric layer 248 after the dielectric layer 248 is deposited.

[0117] As shown in FIG. 7D, a first portion of an interconnect layer 218 of the sensor die 208 is formed above the device layer 216. To form the first portion of the interconnect layer 218, a deposition tool may be used to deposit a dielectric layer 258 (which may include one or more ILD layers, one or more IMD layers, one or more ESLs, and/or one or more of another type of dielectric layer) using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer 258 after the dielectric layer 258 is deposited.

[0118] A deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, and/or another semiconductor processing tool may be used to perform various operations to form interconnect structures 262 in the first portion of the interconnect layer 218. A deposition tool and/or a plating tool may be used to deposit the interconnect structures 262 using a PVD technique, an ALD technique, a CVD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may be used to planarize the interconnect structures 262 after the interconnect structures 262 are deposited.

[0119] In some implementations, first portion of the interconnect layer 218 is built up in the z-direction in a plurality of via layers (V-layers) and metallization layers (M-layers). For example, a first portion of the dielectric layer 258 may be formed, recesses may be formed in the first portion of the dielectric layer 258, and first interconnect structures 262 (e.g., a V0 via layer, an M0 metallization layer) may be formed in the recesses. A second portion of the dielectric layer 258 may be formed, recesses may be formed in the second portion of the dielectric layer 258, and second interconnect structures 262 (e.g., a V1 via layer, an M1 metallization layer) may be formed in the recesses. The remaining via layers and/or metallization layers of the first portion of the interconnect layer 218 may be formed in a similar manner.

[0120] As shown in FIGS. 7E and 7F, a second portion of the interconnect layer 218 may be formed, and the second portion of the interconnect layer 218 may include a bonding layer 260 and bonding structures 264. As shown in FIG. 7E, the bonding layer 260 may be formed over and/or on the

dielectric layer 260, and over and/or on the top-most interconnect structures 262. A deposition tool may be used to deposit the bonding layer 260 using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the bonding layer 260 after the bonding layer 260 is deposited.

[0121] As shown in FIG. 7F, the bonding structures 264 may be formed in the bonding layer 260. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the bonding layer 260. An etch tool may be used to etch the bonding layer 260 (e.g., using a wet etch technique, a dry etch technique) to form recesses in the bonding layer 260. A deposition tool and/or a plating tool may be used to deposit the bonding structures 264 in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the bonding structures 264 after the bonding structures 264 are deposited.

[0122] As indicated above, FIGS. 7A-7F are provided as an example. Other examples may differ from what is described with regard to FIGS. 7A-7F.

[0123] FIGS. 8A-8F are diagrams of an example implementation 800 of forming an image sensor device 210 (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 8A-8F may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0124] As shown in FIGS. 8A and 8B, a bonding operation is performed to bond a circuitry die 206 and a sensor die 208 to form the image sensor device 210. The circuitry die 206 and the sensor die 208 may be bonded at a bonding interface 220, which may include the bonding layers 240 and 260 (respectively of the circuitry die 206 and the sensor die 208), and the bonding structures 244 and 264 (respectively of the circuitry die 206 and the sensor die 208). A bonding tool may be used to form a dielectric-to-dielectric bond between the bonding layers 240 and 260 at the bonding interface 220, and to form a metal-to-metal bond between the bonding structures 244 and 264 at the bonding interface 220.

[0125] As shown in FIG. 8B, after bonding, the circuitry die 206 and the sensor die 208 are stacked or vertically arranged in the z-direction in the image sensor device 210. The interconnect layer 214 of the circuitry die 206 and the interconnect layer 218 of the sensor die 208 are facing toward each other in the image sensor die 210, and the device layer 212 of the circuitry die 206 and the device layer 216 of the sensor die 208 are facing away from each other.

[0126] As shown in FIG. 8C, the DTI structure 252 may be formed in the substrate 246 (e.g., in the back side of the substrate 246) and around the photodiodes 112 in the substrate 246. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the substrate 246. An etch tool may be used to etch the substrate 246 (e.g., using a wet etch technique, a dry etch technique) from the back side of the substrate 246

to form trenches in the back side of the substrate **246**. The trenches are above the STI structures **250** and alongside the photodiodes **112**.

[0127] A deposition tool may be used to conformally deposit a dielectric liner **256** of the DTI structure **252** in the trenches and on the back side surface of the substrate **246** using a CVD technique, an ALD technique, and/or another conformal deposition technique. A deposition tool may be used to deposit the dielectric material **254** of the DTI structure **252** on the dielectric liner **256** in the trenches and above the substrate **246** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the dielectric material **254** above the substrate **246**, which may remain as a buffer layer.

[0128] As shown in FIG. 8D, a passivation layer **266** may be formed over and/or on the buffer layer, and a metal layer **272** may be formed over and/or on the passivation layer over the back side of the substrate **246**. A deposition tool and/or a plating tool may be used to deposit the passivation layer **266** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the passivation layer **266** after the passivation layer **266** is deposited. A deposition tool and/or a plating tool may be used to deposit the metal layer **272** using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the metal layer **272** after the metal layer **272** is deposited.

[0129] As shown in FIG. 8E, various layers and/or structures may be formed in the bonding pad region **228** of the image sensor device **210**. For example, a recess may be formed through the metal layer **272**, through the passivation layer **266**, through the buffer layer (corresponding to the dielectric material **254** above the substrate **246**), through the dielectric liner **256**, and/or through the substrate **246** to the dielectric layer **248**. In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the metal layer **272**. An etch tool may be used to etch the through the metal layer **272**, through the passivation layer **266**, through the buffer layer, through the dielectric liner **256**, through the substrate **246** (e.g., using a wet etch technique, a dry etch technique) from the back side of the substrate **246** to form the recess.

[0130] A dielectric layer **274** may be formed in the recess on the dielectric layer **248**. A dielectric layer **276** may be formed on the dielectric layer **274**. A deposition tool may be used to deposit the dielectric layers **274** and **276** in the recess using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique.

[0131] Openings may be formed through the dielectric layers **248**, **274**, and **276** such that an interconnect structure **262** in the interconnect layer **218** is exposed through the recesses. A bonding pad structure **284** may be formed in the openings such that the bonding pad structure **284** lands on the interconnect structure **262**. The bonding pad structure **284** is also formed on the dielectric layer **276**.

[0132] In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a

patterned masking layer on the dielectric layer **276**. An etch tool may be used to etch the through the dielectric layer **276**, through the dielectric layer **274**, and through the dielectric layer **248** (e.g., using a wet etch technique, a dry etch technique) to form the recesses. A deposition tool and/or a plating tool may be used to deposit the bonding pad structure **284** in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique.

[0133] A dielectric layer **278** may be formed on the bonding pad structure **284**, and dielectric layers **280** and **282** may be deposited to fill in the recess in the bonding pad region **228**. A bonding pad opening **286** may be formed through the dielectric layers **278**, **280**, and **282** to expose the bonding pad structure **284**.

[0134] A deposition tool may be used to deposit the dielectric layers **278**, **280**, and **282** in the recess using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the dielectric layer **282**. An etch tool may be used to etch the through the dielectric layers **278**, **280**, and **282** (e.g., using a wet etch technique, a dry etch technique) to form the bonding pad opening **286**.

[0135] As shown in FIG. 8F, the metal layer **272** in the pixel sensor array **222** is etched to form the metal grid structure **224**. Techniques described in connection with FIGS. 9A-9F, 10A-10D, 11A-11E, 12A-12C, 13A-13D, 14A, and/or 14B may be used to form antireflection structures in and/or on the metal grid structure **224**.

[0136] In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the metal layer **272**. An etch tool may be used to etch the through the metal layer **272** to the passivation layer **266** (e.g., using a wet etch technique, a dry etch technique) remove portions of the metal layer **272**. Remaining portions of the metal layer **272** in the pixel sensor array **222** correspond to the metal grid structure **224** above the DTI structure **252**.

[0137] As further shown in FIG. 8F, color filter regions **268** are formed in openings in the metal grid structure **224** such that the color filter regions **268** are located above and/or over the photodiodes **112** of the pixel sensors **100**. Microlenses **270** are formed on the color filter regions **268**.

[0138] As indicated above, FIGS. 8A-8F are provided as an example. Other examples may differ from what is described with regard to FIGS. 8A-8F.

[0139] FIGS. 9A-9F are diagrams of an example implementation **900** of forming nanostructures **304** on a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. 9A-9F may be used to form nanostructures **304** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. 3A and 3C (e.g., the examples in which the nanostructures **304** are included on the top surfaces **306** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 9A-9F may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0140] As shown in FIG. 9A, the metal grid structure 224 may be formed above the backside surface of the substrate 246 such that the metal grid structure 224 is included above the DTI structure 252. An etch tool may be used to perform an etch operation to form the metal grid structure 224 from a metal layer (e.g., the metal layer 272). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure 224 such that the sidewalls 310 of the metal grid structure 224 are angled relative to a top surface 306 of the metal grid structure 224. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure 224. One or more parameters of the etch operation may be selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure 224. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure 224. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0141] As shown in FIG. 9B, an antireflective layer 902 may be formed over and/or on the metal grid structure 224, as well as on the backside surface of the substrate 246. A deposition tool may be used to deposit the antireflective layer 902 using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In particular, a conformal deposition technique may be used to conformally deposit the antireflective layer 902 such that the antireflective layer 902 conforms to the profile of the metal grid structure 224. The antireflective layer 902 may include one or more antireflective materials, such as zinc oxide (ZnO), aluminum-doped zinc oxide (Al: ZnO or AZO), and/or aluminum oxide (Al_2O_3 , such as Al_3O_4), among other examples. The antireflective layer 902 may be formed as a substrate on which the nanostructures 304 are formed or grown.

[0142] As shown in FIG. 9C, a patterned masking layer 904 may be formed on the top surfaces 306 of the metal grid structure 224. A deposition tool may be used to form a masking layer on the backside of the substrate 246 and on the metal grid structure 224 (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer to form a pattern in the masking layer. A developer tool may be used to develop the pattern to form the patterned masking layer 904.

[0143] As shown in FIG. 9D, an etch tool may be used to etch the antireflective layer 902 to remove portions of the antireflective layer 902 from the backside of the substrate 246 and from the sidewalls 310 of the metal grid structure 224. Thus, the antireflective layer 902 remains only on the top surfaces 306 of the metal grid structure 224. The portions of the antireflective layer 902 may be removed using a wet etch technique, a dry etch technique, and/or another suitable etch technique.

[0144] As shown in FIG. 9E, the patterned masking layer 904 may be removed from the top surfaces 306 of the metal grid structure 224. The patterned masking layer 904 may be removed by etching the patterned masking layer 904, by

ashing the patterned masking layer 904, and/or using another suitable masking layer removal technique.

[0145] As shown in FIG. 9F, the nanostructures 304 are formed on the antireflective layer 902 that is on the top surfaces 306 of the metal grid structure 224. A thermal anneal operation may be performed to form the nanostructures 304. For example, carbon (or graphite) powder may be deposited onto the antireflective layer 902 and annealed to cause the nanostructures 304 to grow from the carbon powder in a highly crystalline manner.

[0146] As indicated above, FIGS. 9A-9F are provided as an example. Other examples may differ from what is described with regard to FIGS. 9A-9F.

[0147] FIGS. 10A-10D are diagrams of an example implementation 1000 of forming nanostructures 304 on a metal grid structure 224 of a pixel sensor array 222 described herein. The techniques described in connection with FIGS. 10A-10D may be used to form nanostructures 304 in the examples of the pixel sensor arrays 222 illustrated and described in connection with FIGS. 3B and 3D (e.g., the examples in which the nanostructures 304 are included on the top surfaces 306 and sidewalls 310 of the metal grid structures 224). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 10A-10D may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0148] The example implementation 1000 is similar to the example implementation 900 of forming nanostructures 304 in that the metal grid structure 224 is formed, the antireflective layer 902 is formed on the top surfaces 306 and the sidewalls 310 of the metal grid structure 224 as well as on the backside of the substrate 246, and a patterned masking layer 904 is used to remove portions of the antireflective layer 902.

[0149] However, and as shown in FIG. 10A, the patterned masking layer 904 is also formed on the sidewalls 310 of the metal grid structure 224 in addition to the top surfaces 306 of the metal grid structure 224. As shown in FIG. 10B, this enables the portions of the antireflective layer 902 to be removed from the backside of the substrate 246 without removing the antireflective layer 902 from the top surfaces 306 and the sidewalls 310 of the metal grid structure 224. Thus, in the example implementation 1000 of forming nanostructures 304, the antireflective layer 902 is included on the top surfaces 306 and the sidewalls 310 of the metal grid structure 224. As shown in FIG. 10C, the patterned masking layer 904 is removed after the antireflective layer 902 is etched.

[0150] As shown in FIG. 10D, retaining the antireflective layer 902 on the sidewalls 310 of the metal grid structure 224 in the example implementation 1000 of forming nanostructures 304 enables the nanostructures 304 to be grown on the sidewalls 310 of the metal grid structure 224 in addition to the top surfaces 306 of the metal grid structure 224.

[0151] As indicated above, FIGS. 10A-10D are provided as an example. Other examples may differ from what is described with regard to FIGS. 10A-10D.

[0152] FIGS. 11A-11E are diagrams of an example implementation 1100 of forming antireflective layers 402 on a metal grid structure 224 of a pixel sensor array 222

described herein. The techniques described in connection with FIGS. 11A-11E may be used to form antireflective layers 402 in the examples of the pixel sensor arrays 222 illustrated and described in connection with FIGS. 4A and 4C (e.g., the examples in which the antireflective layers 402 are included on the top surfaces 306 of the metal grid structures 224). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 11A-11E may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0153] As shown in FIG. 11A, the metal grid structure 224 may be formed above the backside surface of the substrate 246 such that the metal grid structure 224 is included above the DTI structure 252. An etch tool may be used to perform an etch operation to form the metal grid structure 224 from a metal layer (e.g., the metal layer 272). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure 224 such that the sidewalls 310 of the metal grid structure 224 are angled relative to a top surface 306 of the metal grid structure 224. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure 224. One or more parameters of the etch operation may be selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure 224. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure 224. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0154] As shown in FIG. 11B, the antireflective layer 402 may be formed over and/or on the metal grid structure 224, as well as on the backside surface of the substrate 246. A deposition tool may be used to deposit the antireflective layer 902 using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In particular, a conformal deposition technique may be used to conformally deposit the antireflective layer 402 such that the antireflective layer 402 conforms to the profile of the metal grid structure 224.

[0155] As shown in FIG. 11C, a patterned masking layer 1102 may be formed on the top surfaces 306 of the metal grid structure 224. A deposition tool may be used to form a masking layer on the backside of the substrate 246 and on the metal grid structure 224 (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer to form a pattern in the masking layer. A developer tool may be used to develop the pattern to form the patterned masking layer 1102.

[0156] As shown in FIG. 11D, an etch tool may be used to etch the antireflective layer 402 remove portions of the antireflective layer 402 from the backside of the substrate 246 and from the sidewalls 310 of the metal grid structure 224. Thus, the antireflective layer 402 remains only on the top surfaces 306 of the metal grid structure 224. The portions

of the antireflective layer 402 may be removed using a wet etch technique, a dry etch technique, and/or another suitable etch technique.

[0157] As shown in FIG. 11E, the patterned masking layer 1102 may be removed from the top surfaces 306 of the metal grid structure 224. The patterned masking layer 1102 may be removed by etching the patterned masking layer 1102, by ashing the patterned masking layer 1102, and/or using another suitable masking layer removal technique.

[0158] As indicated above, FIGS. 11A-11E are provided as an example. Other examples may differ from what is described with regard to FIGS. 11A-11E.

[0159] FIGS. 12A-12C are diagrams of an example implementation 1200 of forming antireflective layers 402 on a metal grid structure 224 of a pixel sensor array 222 described herein. The techniques described in connection with FIGS. 12A-12C may be used to form antireflective layers 402 in the examples of the pixel sensor arrays 222 illustrated and described in connection with FIGS. 4B and 4D (e.g., the examples in which the antireflective layers 402 are included on the top surfaces 306 and sidewalls 310 of the metal grid structures 224). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 12A-12C may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0160] The example implementation 1200 is similar to the example implementation 1100 of forming antireflective layers 402 in that the metal grid structure 224 is formed, the antireflective layers 402 is formed on the top surfaces 306 and the sidewalls 310 of the metal grid structure 224 as well as on the backside of the substrate 246, and a patterned masking layer 1102 is used to remove portions of the antireflective layer 902.

[0161] However, and as shown in FIG. 12A, the patterned masking layer 1102 is also formed on the sidewalls 310 of the metal grid structure 224 in addition to the top surfaces 306 of the metal grid structure 224. As shown in FIG. 12B, this enables the portions of the antireflective layer 402 to be removed from the backside of the substrate 246 without removing the antireflective layer 402 from the top surfaces 306 and the sidewalls 310 of the metal grid structure 224. Thus, in the example implementation 1200 of forming antireflective layers 402, the antireflective layers 402 are included on the top surfaces 306 and on the sidewalls 310 of the metal grid structure 224. As shown in FIG. 12C, the patterned masking layer 904 is removed after the antireflective layer 402 is etched.

[0162] As indicated above, FIGS. 12A-12C are provided as an example. Other examples may differ from what is described with regard to FIGS. 12A-12C.

[0163] FIGS. 13A-13D are diagrams of an example implementation 1300 of forming cavities 502 in a metal grid structure 224 of a pixel sensor array 222 described herein. The techniques described in connection with FIGS. 13A-13D may be used to form cavities 502 in the examples of the pixel sensor arrays 222 illustrated and described in connection with FIGS. 5A, 5B, 5F, and 5G (e.g., the examples in which the cavities 502 are included in the top surfaces 306 of the metal grid structures 224). In some implementations, one or more of the semiconductor processing operations

described in connection with FIGS. 13A-13D may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0164] As shown in FIG. 13A, the metal grid structure 224 may be formed above the backside surface of the substrate 246 such that the metal grid structure 224 is included above the DTI structure 252. An etch tool may be used to perform an etch operation to form the metal grid structure 224 from a metal layer (e.g., the metal layer 272). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure 224 such that the sidewalls 310 of the metal grid structure 224 are angled relative to a top surface 306 of the metal grid structure 224. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure 224. One or more parameters of the etch operation may be selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure 224. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure 224. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0165] As shown in FIGS. 13B and 13C, a masking layer 1302 may be formed on the metal grid structure 224 and on the backside of the substrate 246, and a pattern 1304 may be formed in the masking layer 1302. A deposition tool may be used to form a masking layer 1302 on the backside of the substrate 246 and on the metal grid structure 224 (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer 1302 to form the pattern 1304 in the masking layer. A developer tool may be used to develop the pattern 1304. The pattern 1304 may include openings in the masking layer 1302 above the top surfaces 306 of the metal grid structure 224.

[0166] An etch tool may be used to etch the top surfaces 306 of the metal grid structure 224 based on the pattern 1304 in the masking layer 1302 to form the cavities 502 in the top surfaces 306 of the metal grid structure 224, as shown in FIG. 13D.

[0167] In some implementations, the cavities 502 may be formed using a wet etch technique, in which a wet etchant is provided to the top surfaces 306 of the metal grid structure 224 through the pattern 1304 in the masking layer 1302, and the wet etchant removes material from the top surfaces 306 of the metal grid structure 224. This results in formation of the cavities in the top surfaces 306 of the metal grid structure 224. In some implementations, using a wet etch technique results in the cavities 502 having approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0168] In some implementations, the cavities 502 may be formed using a dry etch technique, in which a plasma and/or another type of dry etchant is provided to the top surfaces 306 of the metal grid structure 224 through the pattern 1304 in the masking layer 1302, and the dry etchant removes

material from the top surfaces 306 of the metal grid structure 224. This results in formation of the cavities in the top surfaces 306 of the metal grid structure 224. In some implementations, using a dry etch technique results in the cavities 502 having approximately V-shaped (or triangular) cross-sectional profiles.

[0169] As indicated above, FIGS. 13A-13D are provided as an example. Other examples may differ from what is described with regard to FIGS. 13A-13D.

[0170] FIGS. 14A and 14B are diagrams of an example implementation 1400 of forming cavities 502 in a metal grid structure 224 of a pixel sensor array 222 described herein. The techniques described in connection with FIGS. 14A and 14B may be used to form cavities 502 in the examples of the pixel sensor arrays 222 illustrated and described in connection with FIGS. 5C, 5D, 5E, 5H, 5I, and 5J (e.g., the examples in which the cavities 502 are included in the top surfaces 306 and in the sidewalls 310 of the metal grid structures 224). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 14A and 14B may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0171] The example implementation 1400 is similar to the example implementation 1300 of forming cavities 502 in the metal grid structure 224. However, as shown in FIG. 14A, a pattern 1402 is formed in the masking layer 1302, and the pattern 1402 includes openings over the sidewalls 310 of the metal grid structure 224 as well as over the top surfaces 306 of the metal grid structure 224. An etch tool may be used to etch the top surfaces 306 of the metal grid structure 224 based on the pattern 1402 in the masking layer 1302 to form the cavities 502 in the top surfaces 306 of the metal grid structure 224 as well as to form cavities 502 in the sidewalls 310 of the metal grid structure 224, as shown in FIG. 14B.

[0172] As indicated above, FIGS. 14A and 14B are provided as an example. Other examples may differ from what is described with regard to FIGS. 14A and 14B.

[0173] FIG. 15 is a flowchart of an example process 1500 associated with forming a metal grid structure of a pixel sensor array. In some implementations, one or more process blocks of FIG. 15 are performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0174] As shown in FIG. 15, process 1500 may include forming a plurality of photodiodes in a substrate of a pixel sensor array (block 1510). For example, one or more semiconductor processing tools may be used to form a plurality of photodiodes 112 in a substrate 246 of a pixel sensor array 222, as described herein.

[0175] As further shown in FIG. 15, process 1500 may include forming a DTI structure around the plurality of photodiodes in the substrate (block 1520). For example, one or more semiconductor processing tools may be used to form a DTI structure 252 around the plurality of photodiodes 112 in the substrate 246, as described herein.

[0176] As further shown in FIG. 15, process 1500 may include forming a metal grid structure above the substrate and over the DTI structure (block 1530). For example, one

or more semiconductor processing tools may be used to form a metal grid structure 224 above the substrate 246 and over the DTI structure 252, as described herein. In some implementations, the DTI structure 252 laterally surrounds the plurality of photodiodes 112.

[0177] As further shown in FIG. 15, process 1500 may include forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure (block 1540). For example, one or more semiconductor processing tools may be used to form an antireflection structure at least one of in a top surface 306 of the metal grid structure 224 or on the top surface 306 of the metal grid structure 224, as described herein. In some implementations, the antireflection structure includes an antireflective layer 404 on the top surface 306 of the metal grid structure 224. In some implementations, the antireflection structure includes a plurality of nanostructures 304 on the top surface 306 of the metal grid structure 224. In some implementations, the antireflection structure includes a plurality of nanostructures 304 formed on an antireflective layer 902 on the top surface 306 of the metal grid structure 224. In some implementations, the antireflection structure includes a plurality of cavities 502 formed in the top surface 306 of the metal grid structure 224.

[0178] Process 1500 may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0179] In a first implementation, forming the antireflection structure includes depositing a layer of dielectric material (e.g., the antireflective layer 402, the antireflective layer 902) on the metal grid structure 224 and on the semiconductor layer 246, and removing portions of the layer of dielectric material such that remaining portions of the layer of dielectric material correspond to the antireflection layer (e.g., the antireflective layer 402, the antireflective layer 902) on the top surface 306 of the metal grid structure 224.

[0180] In a second implementation, alone or in combination with the first implementation, removing the portions of the layer of dielectric material includes removing the portions of the layer of dielectric material from sidewalls 310 of the metal grid structure 224.

[0181] In a third implementation, alone or in combination with one or more of the first and second implementations, process 1500 includes performing a thermal anneal operation to form nanowires (e.g., nanostructures 304) on the antireflection layer.

[0182] In a fourth implementation, alone or in combination with one or more of the first through third implementations, forming the antireflection layer includes forming a pattern (e.g., a pattern 1304, a pattern 1402) in a masking layer 1302 on the metal grid structure 224, and etching the top surface 306 of the metal grid structure 224 based on the pattern to form a plurality of cavities 502 in the top surface 306 of the metal grid structure 224, wherein the plurality of cavities 502 correspond to the antireflection structure in the top surface 306 of the metal grid structure 224.

[0183] Although FIG. 15 shows example blocks of process 1500, in some implementations, process 1500 includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 15. Additionally, or alternatively, two or more of the blocks of process 1500 may be performed in parallel.

[0184] In this way, an image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

[0185] As described in greater detail above, some implementations described herein provide a pixel sensor array. The pixel sensor array includes a plurality of pixel sensors, arranged in a grid, comprising a plurality of photodiodes in a substrate. The pixel sensor array includes a DTI structure, laterally surrounding the plurality of photodiodes, in the substrate. The pixel sensor array includes a metal grid structure above the DTI structure and above the substrate, where the metal grid structure surrounds the photodiodes. The pixel sensor array includes an antireflective layer on a top surface of the metal grid structure. The top surface faces away from the substrate.

[0186] As described in greater detail above, some implementations described herein provide an image sensor device. The image sensor device includes a plurality of pixel sensors, arranged in a pixel sensor array, that include a plurality of photodiodes in a substrate of the image sensor device. The image sensor device includes a DTI structure, around the plurality of photodiodes, in the substrate. The image sensor device includes a metal grid structure above the DTI structure and above the semiconductor layer, where the metal grid structure surrounds the photodiodes, and where the metal grid structure includes a plurality of concave surfaces in a top surface of the metal grid structure.

[0187] As described in greater detail above, some implementations described herein provide a method. The method includes forming a plurality of photodiodes in a substrate of a pixel sensor array. The method includes forming a DTI structure around the plurality of photodiodes in the substrate. The method includes forming a metal grid structure above the substrate and over the DTI structure. The method includes forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure.

[0188] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

[0189] The terms “approximately” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., +1%, +2%, +3%, +4%, +5% of the value). These values are merely examples and are not intended to be limiting. It is to be understood that the terms “approximately” and “substantially” can refer to a percentage of the values of a given quantity in light of this disclosure.

[0190] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes

and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A pixel sensor array, comprising:
 - a plurality of pixel sensors, arranged in a grid, comprising a plurality of photodiodes in a substrate;
 - a deep trench isolation (DTI) structure, laterally surrounding the plurality of photodiodes, in the substrate;
 - a metal grid structure above the DTI structure and above the substrate, wherein the metal grid structure surrounds the photodiodes; and
 - an antireflective layer on a top surface of the metal grid structure, wherein the top surface faces away from the substrate.
2. The pixel sensor array of claim 1, wherein the metal grid structure has a substantially trapezoidal cross-sectional profile.
3. The pixel sensor array of claim 1, wherein antireflective layer is further included on sidewalls of the metal grid structure.
4. The pixel sensor array of claim 1, wherein the metal grid structure has a substantially square-shaped cross-sectional profile.
5. The pixel sensor array of claim 1, wherein the metal grid structure further includes a plurality of nanowires on the antireflective layer.
6. The pixel sensor array of claim 5, wherein the plurality of nanowires comprises a plurality of carbon nanowires.
7. The pixel sensor array of claim 1, wherein the antireflective layer comprises at least one of:
 - zinc oxide (ZnO_x),
 - aluminum oxide (Al_xO_y), or
 - aluminum-doped zinc oxide (AZO).
8. An image sensor device comprising:
 - a plurality of pixel sensors, arranged in a pixel sensor array, comprising a plurality of photodiodes in a substrate of the image sensor device;
 - a deep trench isolation (DTI) structure, around the plurality of photodiodes, in the substrate; and
 - a metal grid structure above the DTI structure and above the substrate, wherein the metal grid structure surrounds the photodiodes, and wherein the metal grid structure comprises a plurality of concave surfaces in a top surface of the metal grid structure.
9. The image sensor device of claim 8, wherein the plurality of concave surfaces have a rounded cross-sectional profile.
10. The image sensor device of claim 8, wherein the metal grid structure has a substantially trapezoidal cross-sectional profile.

11. The image sensor device of claim 8, wherein the plurality of concave surfaces have a substantially V-shaped cross-sectional profile.

12. The image sensor device of claim 8, wherein the metal grid structure comprises another plurality of concave surfaces in sidewalls of the metal grid structure.

13. The image sensor device of claim 12, wherein a cross-sectional profile of the plurality of concave surfaces in the top surface of the metal grid structure, and a cross-sectional profile of the other plurality of concave surfaces in the sidewalls of the metal grid structure, are different cross-sectional profiles.

14. The image sensor device of claim 12, wherein a cross-sectional profile of the plurality of concave surfaces in the top surface of the metal grid structure, and a cross-sectional profile of the other plurality of concave surfaces in the sidewalls of the metal grid structure, are approximately a same cross-sectional profile.

15. The image sensor device of claim 8, wherein the metal grid structure has a substantially square-shaped cross-sectional profile.

16. A method, comprising:

- forming a plurality of photodiodes in a substrate of a pixel sensor array;
- forming a deep trench isolation (DTI) structure around the plurality of photodiodes in the substrate;
- forming a metal grid structure above the substrate and over the DTI structure; and
- forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure.

17. The method of claim 16, wherein forming the antireflection structure comprises:

- depositing a layer of dielectric material on the metal grid structure and on the substrate; and
- removing portions of the layer of dielectric material such that remaining portions of the layer of dielectric material correspond to an antireflection layer on the top surface of the metal grid structure.

18. The method of claim 17, wherein removing the portions of the layer of dielectric material comprises:

- removing the portions of the layer of dielectric material from sidewalls of the metal grid structure.

19. The method of claim 17, further comprising:

- performing a thermal anneal operation to form nanowires on the antireflection layer.

20. The method of claim 16, wherein forming the antireflection structure comprises:

- forming a pattern in a masking layer on the metal grid structure; and
- etching the top surface of the metal grid structure based on the pattern to form a plurality of cavities in the top surface of the metal grid structure, wherein the plurality of cavities correspond to the antireflection structure in the top surface of the metal grid structure.

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